The internal computation can be partially read: a method-pluralist account of LLM interpretability, 2016–2026

Theory KB — Paper 4 of 5

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Abstract

Five interpretability methods — logit/tuned lenses, probes, sparse autoencoders, activation patching, and circuit tracing — make different commitments about what counts as a feature, a circuit, or an explanation. Method-pluralism is principled, not a sign of immaturity. Cross-method evidence converges in some cases (the IOI circuit, factual-recall localization) and diverges in others ("truth direction" across datasets, SAE feature canonicality). We walk the methods, the math each commits to, and the evidence pattern we should expect to see at each scale.

Contents

1 Introduction

Interpretability is method-plural. A modern frontier transformer is, at the level of weights and activations, a stack of attention and MLP sublayers writing into a residual stream of dimension $D \in \{2048, 4096, 8192, 16384\}$ over sequences of length S up to $\sim 10^6$ tokens, with parameter counts from 10^9 to 10^{12} and a vocabulary of $|\mathcal{V}| \in \{32\text{k}, 128\text{k}, 256\text{k}\}$ tokens (Paper 1 §2). The mechanistic-interpretability program asks what that stack is *computing* — not just what it predicts, but what intermediate representations it builds, which of those representations it uses, and how those representations compose into the behaviors we care about. The field has not converged on a single method for answering that question, and this paper argues it should not. The underlying object is plural: the same residual-stream activation $\mathbf{h}^{(\ell,t)} \in \mathbb{R}^D$ at layer ℓ, position t is simultaneously a partial vocabulary distribution (Belrose et al., 2023), a vector of linearly decodable properties (Marks and Tegmark, 2023), a sparse combination of overcomplete features (Gao et al., 2024), a node in a causal graph (Meng et al., 2022), and a participant in a sparse behavior-implementing subgraph (Wang et al., 2022). Each of those is a real fact about $\mathbf{h}^{(\ell,t)}$, observable with a different instrument, and load-bearing for a different question. The methodological pluralism is principled, not transitional.

This paper is a method-by-method comparison of the five interpretability techniques that have stabilized into the field's canon between roughly 2017 and 2026. Method-pluralism is the throughline, but the load-bearing claim is sharper: the underlying transformer's internal computation can be *partially* read — we can extract human-legible structure for *some* behaviors, on *some* models, with cross-method agreement on *some* claims — and the field is honest about which behaviors, which models, and which claims. Complete reverse-engineering of a frontier model is not on the table in 2026; partial, locally-faithful, cross-method-validated explanation is.

1.1 Five methods, four commitments

Five interpretability methods anchor the literature, in roughly the order they crystallized:

Lens techniques (§3). Project an intermediate residual-stream state $\mathbf{h}^{(\ell,t)}$ through the model's own pre-unembed normalization and unembedding W_U to obtain a layer-ℓ "as-if-final" distribution over $\mathcal{V}$. The *logit lens* (nostalgebraist, 2020) is the no-training baseline; the *tuned lens* (Belrose et al., 2023) fixes documented cross-model failures with a per-layer affine translator.

Probes (§4). Train a (typically linear) classifier $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{h}^{(\ell,t)} + b)$ on *frozen* activations to predict an external property y (Alain and Bengio, 2016). High accuracy is the (correlational) signal that y is linearly decodable from $\mathbf{h}^{(\ell,t)}$.

Sparse autoencoders (§7). Decompose activations $\mathbf{x} \in \mathbb{R}^n$ into a sparse linear combination of $M \gg n$ learned dictionary directions $\{\mathbf{d}_i\}$ via an encoder–decoder pair under a joint reconstruction-plus-sparsity loss (Gao et al., 2024). The commitment: features are linear, additive, and identifiable from activations alone, without supervision.

Activation patching (§9). The canonical causal intervention. The three-run protocol of Meng et al. (2022) caches the clean run, records the corrupted run, then substitutes one component's clean activation into the corrupted run; the indirect effect $\text{IE}_C = m_{\text{pt}} - m_*$ is the causal estimator (Zhang and Nanda, 2023). ROME (Meng et al., 2022) extends this from diagnostic to weight-level surgery (§10).

Circuit tracing (§11). Compose the previous methods into a sparse, causal subgraph $C \subseteq M$ of the model's full computational graph that explains an identifiable capability. Wang et al. (2022) hand-reverse-engineered indirect-object identification in GPT-2 small; Conmy et al. (2023) automated the discovery as iterative edge-pruning; Lindsey et al. (2025b) scaled the program to production Claude 3.5 Haiku via cross-layer transcoders.

These five do not partition cleanly along a single axis. They commit along (at least) four:

1. **Correlational vs causal.** Lenses and probes read off activations and report what is *decodable*; activation patching, ROME, and circuit tracing intervene on activations or weights and report what is *used*. The correlation–causation gap (§5) is the methodological seam: a high-accuracy probe is consistent with the decoded property being computed and consumed, computed and discarded (*vestigial information*), or being a confound outside the model entirely (Belinkov, 2022).

2. **Component-level vs feature-level.** Activation patching and circuit-tracing-via-attention-heads identify *components* (heads, MLPs, hidden states) that mediate a behavior; SAEs and feature-level circuit tracing identify *directions* in residual-stream space, which can be smaller or larger than any component basis. The two granularities answer different questions.

3. **Supervised vs unsupervised.** Probes train against labels; SAEs train against the activations themselves. Lenses train against the model's own final-layer output (a form of self-supervision). The supervision signal determines which properties the method can find: a probe finds what it is asked to find; an SAE finds what its sparsity prior considers a feature.

4. **Static readout vs intervention.** Lenses, probes, and SAEs are static — they compute on cached activations without perturbing the forward pass. Activation patching, ROME, and circuit-validation knockouts are interventional. The static methods scale gracefully; the interventional methods carry the causal load.

The methods are not strictly ranked. Lenses are the cheapest readout ($L \cdot (D^2 + D)$ tuned-lens parameters — under 10^9 for Pythia 12B (Belrose et al., 2023)); probes are next; SAEs sit at training-pretraining-fraction compute (Gemma Scope at $\sim 20\%$ of GPT-3 pretraining (DeepMind, 2024)); activation patching costs $O(N_{\text{components}})$ forward passes per input pair; circuit tracing on production models inherits the SAE/transcoder training cost *plus* the per-task graph-pruning cost. The compute envelope is itself a methodological commitment: a method that is too expensive to run does not enter the field's working vocabulary.

Intuition

The five methods are five different instruments aimed at the same residual stream. The lens reads what the model is about to predict; the probe reads what the model has computed; the SAE reads how the model has decomposed what it computed; activation patching tests which of those computations the model uses; circuit tracing composes all four into a sparse, causal explanation of a specific behavior. Returning to canonical form: each method is a different projection of the activation tensor $\mathbf{h}^{(\ell,t)} \in \mathbb{R}^{\mathsf{B} \times \mathsf{S} \times \mathsf{D}}$, and "method-plural" is the recognition that no single projection is sufficient.

1.2 What "partially read" means

The thesis sentence — *the internal computation can be partially read* — has three quantifiers. *Some behaviors*: the GPT-2-small IOI circuit (Wang et al., 2022), GPT-2-XL factual recall (Meng et al., 2022), and selected Claude 3.5 Haiku capabilities (Lindsey et al., 2025b) are reverse-engineered to a mechanism-level account; almost everything else a frontier model does is not. *Some models*: SAEs are fully released for the Gemma 2 family at 2B/9B/27B (DeepMind, 2024); tuned lenses cover the GPT-2 / Pythia / LLaMA-1 lineage (Belrose et al., 2023) but have not all been published for Llama 3/4, Qwen 2.5/3, DeepSeek-V3, or Gemma 3 (§13). *Some claims*: the mid-layer-MLP localization of factual recall, the IOI circuit's 26-head decomposition, and the linear decodability of truth on LLaMA-family models are claims multiple methods agree on; the canonicality of SAE features (Leask et al., 2025), the existence of a single truth direction across datasets (Marks and Tegmark, 2023), and the reproducibility of attribution-graph findings outside the originating lab are claims the methods disagree on.

The closing question of this paper, taken up in §12, is the diagnostic version of that disagreement: *when do two methods have to agree, and what does it mean when they don't?* When two methods answer the same question about the same mechanism — e.g., activation patching and ACDC on IOI — they *have* to agree, and divergence is a falsification signal for at least one method's assumptions. When two methods answer different questions — e.g., a probe (decodable?) and patching (used?) — they *can* disagree without either being wrong, and the disagreement is principled. Distinguishing the two cases is a load-bearing methodological skill that this paper tries to teach.

1.3 Roadmap

The paper is organized to make the method-pluralism legible. §2 fixes the substrate and notation: the residual stream as a linear additive bus, attention as QK+OV, MLPs as candidate key–value associative memories, the unembedding W_U as the canonical readout, and the formal definitions of *feature* (a direction $\mathbf{d} \in \mathbb{R}^D$) and *circuit* (a subgraph satisfying Wang et al. (2022)'s faithfulness/completeness/minimality criteria) used throughout.

§§3–11 present the five methods and their immediate critical literature. §3 covers lenses; §4 covers probes; §5 formalizes the correlation–causation gap and the activation-intervention resolution; §6 works the truth-direction case study; §7 covers SAE architectures; §8 takes up the canonicality question raised by Leask et al. (2025); §9 covers activation patching; §10 covers ROME and the interpretability-to-surgery bridge; §11 covers circuit tracing from IOI to Haiku.

§§12–14 are the cross-method synthesis. §12 catalogs the convergence and divergence cases method-by-method and behavior-by-behavior; §13 surveys what scales (SAEs, probes, attribution graphs in narrow regimes) and what does not yet (manual circuit hunts, full activation patching at frontier scale, lens coverage of MLA / MoE / NSA architectures), with a 2026 frontier snapshot table of which method covers which model; §14 concentrates the live disagreements — SAE canonicality, lens cross-model unreliability, probe correlation vs causation, truth-direction multiplicity, circuit-tracing reproducibility past Anthropic-2025 — and names, for each, the experiment that would resolve it.

The reader who wants the headline of each method in isolation can read §2 and any single §3–§11. The reader who wants the comparative argument should read §2 through §14 in order: the cross-method evidence is what makes "the internal computation can be partially read" a defensible claim rather than a hopeful slogan, and that evidence is built incrementally across the paper's arc.

2 What we mean by "internal computation"

Every interpretability method in this paper reads, intervenes on, or decomposes a transformer's internal computation. Before the methods can disagree productively, the substrate they share must be pinned down. This section codifies four pieces of architectural vocabulary — the residual stream as a linear additive bus, attention as a QK + OV decomposition, the feedforward sublayer as a candidate associative memory, and the LayerNorm + unembedding as the canonical readout — and the load-bearing notation $\mathbf{h}^{(\ell,t)}, \mathbf{d}_y, f_i^{(\ell,t)}, C$ that §§3–11 inherit. Where Paper 1's architecture chapter establishes the substrate as a design object, this section recasts it as a *measurement* object: the quantities the lens reads, the directions the probe projects onto, the sites the patcher swaps, the subgraphs the circuit picks out.

2.1 The residual stream as an additive bus

Fix a decoder-only transformer $\mathcal{M}$ with L layers, model dimension D, vocabulary $\mathcal{V}$ of size $V = |\mathcal{V}|$, H attention heads of head-dimension d_h, and unembedding matrix $W_U \in \mathbb{R}^{D\times V}$ (Paper 1 notation). On an input sequence $\mathbf{x} \in \mathcal{V}^S$ of length S, the model emits a length-S sequence of residual-stream activations $\{\mathbf{h}^{(\ell,t)}\}_{\ell=0,\dots,L;\,t=1,\dots,S}$ with $\mathbf{h}^{(\ell,t)} \in \mathbb{R}^D$. The index $\ell = 0$ denotes the post-embedding state; $\ell = L$ denotes the final pre-unembed state. Batched, the residual stream is a tensor of shape $\mathsf{B}\times\mathsf{S}\times\mathsf{D}$, with B the batch axis. A head-split per-layer attention tensor is $\mathsf{B}\times\mathsf{H}\times\mathsf{S}\times d_h$. The output logits are shape $\mathsf{B}\times\mathsf{S}\times\mathsf{V}$.

The pre-LayerNorm transformer block updates the residual stream additively[1] (Belrose et al., 2023, §2, Eq. 1):

$$\mathbf{h}^{(\ell+1,t)} \;=\; \mathbf{h}^{(\ell,t)} \;+\; F_\ell\Big(\mathbf{h}^{(\ell,t)}\Big)\,, \tag{1}$$

where F_ℓ is the residual output of layer ℓ — attention followed by a feedforward sublayer, each with its own pre-block LayerNorm or RMSNorm. Eq. (7) is the load-bearing equation of this paper: every component reads $\mathbf{h}^{(\ell,t)}$, computes some function of it, and writes the result back into the same residual stream by addition. There is no gating, no replacement, no nonlinearity at the bus level. Recursing Eq. (7) from any intermediate ℓ to the output gives the residual decomposition[2] (Belrose et al., 2023, §2, Eq. 2):

$$\mathcal{M}_{>\ell}\Big(\mathbf{h}^{(\ell,t)}\Big) \;=\; \mathrm{LN}\left[\mathbf{h}^{(\ell,t)} \;+\; \underbrace{\sum_{\ell'=\ell}^{L} F_{\ell'}\Big(\mathbf{h}^{(\ell',t)}\Big)}_{\text{residual update}}\right] W_U. \tag{2}$$

Here LN is the model's own final pre-unembed normalization (LayerNorm or RMSNorm), and $W_U \in \mathbb{R}^{D\times V}$ is the unembedding (often weight-tied with the input embedding). Two identifications follow immediately. Setting the residual update to zero in Eq. (8) gives the *logit lens* of §3 (Belrose et al., 2023, §2, Eq. 3); replacing $\mathbf{h}^{(\ell,t)}$ with $A_\ell\mathbf{h}^{(\ell,t)} + \mathbf{b}_\ell$ before the same readout gives the *tuned lens*. Both are read-offs of the same bus.

Intuition

The residual stream is a linear additive bus. Attention heads, MLPs, and LayerNorms are devices wired into that bus, each reading a projection of $\mathbf{h}^{(\ell,t)}$ and writing back a vector

[1] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-decomposition`.

[2] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-decomposition`.

that *adds* to it; nothing replaces or multiplies the running state. This is what makes lenses cheap (the bus's contents at any layer are already in the basis the unembedding wants, up to a per-layer affine correction); what makes patching causally clean (substituting a single addend leaves all other addends untouched in expectation); and what makes the dictionary view of §7 applicable (a sum of sparse contributions decomposes into a sparse linear combination of feature directions in $\mathbb{R}^D$). Returning to canonical form: Eq. (7) says every layer is $\mathbf{h}_{\text{new}} = \mathbf{h}_{\text{old}} + \Delta$, and every interpretability method is a particular way of reading or perturbing the running sum.

2.2 Attention as QK + OV

A single attention head h at layer ℓ contributes to F_ℓ a term that the *Mathematical Framework for Transformer Circuits* factors into two read–write circuits. The **QK circuit** computes the attention-weight pattern from the residual stream:

$$\alpha_{t,s}^{(\ell,h)} \;=\; \mathrm{softmax}_{s'}\left(\frac{1}{\sqrt{d_h}}\left(W_Q^{(\ell,h)}\,\mathbf{h}^{(\ell,t)}\right)^\top \left(W_K^{(\ell,h)}\,\mathbf{h}^{(\ell,s')}\right)\right)\Bigg|_{s'=s}, \tag{3}$$

with $W_Q^{(\ell,h)}, W_K^{(\ell,h)} \in \mathbb{R}^{d_h \times D}$. The **OV circuit** determines what is read and written along each attended position:

$$\mathrm{Attn}_{\ell,h}\left(\mathbf{h}^{(\ell,t)}\right) \;=\; W_O^{(\ell,h)} \sum_{s=1}^{S} \alpha_{t,s}^{(\ell,h)}\, W_V^{(\ell,h)}\,\mathbf{h}^{(\ell,s)}, \tag{4}$$

with $W_V^{(\ell,h)} \in \mathbb{R}^{d_h \times D}$ and $W_O^{(\ell,h)} \in \mathbb{R}^{D \times d_h}$. The QK pair determines *where* a head looks; the OV pair determines *what* it copies and *which direction* that copy writes into the residual stream. Multi-head attention is a sum over heads of Eq. (4). The decomposition is load-bearing for circuit analysis: when Wang et al. (2022) report (§11) that a Name Mover Head "copies the attended name", the human-interpretable claim is exactly that its OV circuit acts as a near-identity on the residual-stream subspace encoding "name", and its QK circuit attends to the IO position; the attention-pattern–vs–effect split is the QK–vs–OV split.

2.3 Feedforward sublayers as candidate associative memory

The other half of F_ℓ is the feedforward (or MLP) sublayer:

$$\mathrm{MLP}_\ell\left(\mathbf{h}^{(\ell,t)}\right) \;=\; W_{\text{down}}^{(\ell)}\,\sigma\left(W_{\text{up}}^{(\ell)}\,\mathbf{h}^{(\ell,t)}\right), \tag{5}$$

with $W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d_{\text{ff}} \times D}$, $W_{\text{down}}^{(\ell)} \in \mathbb{R}^{D \times d_{\text{ff}}}$, an elementwise nonlinearity σ (GELU / SiLU / SwiGLU per the model; biases and gating dropped here for brevity), and intermediate width $d_{\text{ff}} \approx 4D$ in dense models. The hypothesis that ties this to interpretability is the *Localized Factual Association* reading of Meng et al. (2022): the rows of $W_{\text{up}}^{(\ell)}$ act as *keys* and the columns of $W_{\text{down}}^{(\ell)}$ act as *values* in an associative memory indexed by subject representations, with mid-layer MLPs storing the bulk of factual recall content. This is the framing the ROME rank-one MLP edit of §10 acts on; its empirical warrant comes from the activation-patching localization at $\mathrm{AIE}_{\text{MLP}} = 6.6\%$ vs. $\mathrm{AIE}_{\text{attn}} = 1.6\%$ on GPT-2 XL (Meng et al., 2022, §2.2)[3]. We adopt the key–value framing as the *anchor for §10*, not

[3] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-2`.

as a settled fact: alternative framings (FFNs as feature-direction *detectors* in superposition, the Leask et al. (2025) non-atomic critique of §8) coexist in the 2026 literature and are taken up where they bear on the methods.

2.4 LayerNorm and the canonical readout

Two normalization variants appear in the residual-stream literature covered here. LayerNorm (Ba et al., 2016) centers and rescales $\mathbf{h}^{(\ell,t)}$ along its D-dim axis with a learned affine; RMSNorm drops the centering. Either is denoted $\mathrm{LN}[\,\cdot\,]$ when the distinction does not matter for the argument, and pre-block or post-block placement is folded into the model's specification. The canonical *readout* is the model's own pre-unembed LN followed by W_U; the layer-L logits are

$$\mathrm{logits}(\mathbf{x})_t \;=\; \mathrm{LN}\Big[\mathbf{h}^{(L,t)}\Big]\, W_U \;\in\; \mathbb{R}^V. \tag{6}$$

Equation (8) is exactly Eq. (6) re-expressed at an arbitrary depth ℓ. The lens family of §3 reuses Eq. (6) verbatim on $\mathbf{h}^{(\ell,t)}$ for $\ell < L$; the correlation-vs-causation gap of §5 is precisely the gap between "Eq. (6) *would* produce token r at layer ℓ" and "the model *uses* the ℓ-th-layer information that drives that prediction".

2.5 What we mean by a feature

Throughout this paper a **feature** is a direction $\mathbf{d} \in \mathbb{R}^D$ in residual-stream space (or in some other activation space named explicitly), *not* a neuron and *not* necessarily a basis vector. The motivating commitment is the *linear representation hypothesis* : many human-interpretable properties of the input are linearly readable from the residual stream, so the natural object of interpretability is an axis (or a low-dimensional subspace) of $\mathbb{R}^D$ rather than an individual coordinate. Every method in this paper picks features under a different commitment.

- §3 (lens). The *token-aligned* features are the columns of W_U themselves: the lens projects $\mathbf{h}^{(\ell,t)}$ onto every column simultaneously and reads the arg max.
- §4 (probe). A **property direction** $\mathbf{d}_y \in \mathbb{R}^D$ is a feature associated with an external label y — truth, sentiment, syntactic role, sycophancy — discovered either by supervised fitting (Eq. (14)-equivalent) or in closed form by the mass-mean rule $\mathbf{d}_{\mathrm{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$.
- §7 (SAE). A **dictionary feature** i at layer ℓ, position t is the activation $f_i^{(\ell,t)} \in \mathbb{R}_{\geq 0}$ of the i-th latent in a sparse autoencoder trained on $\mathbf{h}^{(\ell,t)}$, with associated decoder direction $\mathbf{d}_i$. The dictionary $\{\mathbf{d}_i\}_{i=1}^M$ is overcomplete ($M \gg D$); on any given input, $\mathbf{f}^{(\ell,t)} \in \mathbb{R}^M$ is sparse.

A feature is therefore an *axis*, not a *coordinate*: the neuron basis is one possible feature basis, and the production record suggests it is rarely the right one. The choice of basis — W_U's columns, a learned $\mathbf{d}_y$, an SAE's $\mathbf{d}_i$, or some other — is the methodological commitment that distinguishes one interpretability technique from the next.

2.6 What we mean by a circuit

A **circuit** $C \subseteq \mathcal{M}$ is a subgraph of $\mathcal{M}$'s computational graph — nodes are forward-pass components (residual-stream sites, attention heads, MLP sublayers, SAE/transcoder features) and edges are the residual / attention / projection links between them — whose function $C(\mathbf{x})$ agrees with $M(\mathbf{x})$ on the task distribution after *knockout* of $\mathcal{M} \setminus C$ (Wang et al., 2022, §2)[4]. A circuit *claim* commits to three quantitative criteria from Wang et al. (2022, §4)[5]:

Faithfulness. C performs the task as well as the whole model: $C(\mathbf{x}) \approx \mathcal{M}(\mathbf{x})$ on the task distribution under a chosen metric.

Completeness. C contains every node used to perform the task: knocking out $\mathcal{M} \setminus C$ does not change behavior beyond noise.

Minimality. C contains no task-irrelevant nodes: each node in C fails an "ablate-it-and-see-if-it-matters" test.

The substantive content of a circuit claim is therefore threefold: the relevant components are *localized* (a small fraction of the graph), the explanatory edges are *causal* (knockout matters), and the subgraph is *sparse* (most of $\mathcal{M}$ is irrelevant on the task distribution). Wang et al. (2022) note that real circuits routinely pass faithfulness and minimality but fail the strongest completeness tests — the structural fact that §11.2 unpacks via the *backup head* phenomenon, in which ablating Name Mover heads reactivates latent backups that were dormant in the unablated forward pass. The features-in-circuits dichotomy is fundamental: a circuit can route a feature from one position to another without itself *computing* that feature, so the wiring diagram (the circuit) and the encoding (the feature) are independent specifications of an explanation (Wang et al., 2022, §2.1).

2.7 The notation table

Table 1 fixes the symbols inherited by the rest of the paper. Tensor shapes are listed in the $\mathsf{B}\times\mathsf{S}\times\mathsf{D}$ convention of Paper 1.

The four objects in the second half of the table fix what each method *produces*. A lens (§3) maps $\mathbf{h}^{(\ell,t)}$ to a layer-ℓ token distribution via Eq. (8). A probe (§4) reads $\mathbf{h}^{(\ell,t)} \mapsto \sigma(\mathbf{d}_y^\top \mathbf{h}^{(\ell,t)})$ for a learned property direction $\mathbf{d}_y$. An SAE (§7) decomposes $\mathbf{h}^{(\ell,t)}$ into a sparse $\mathbf{f}^{(\ell,t)} \in \mathbb{R}^M$ with components $f_i^{(\ell,t)}$ along directions $\mathbf{d}_i$. Activation patching (§9) intervenes on $\mathbf{h}^{(\ell,t)}$ (or on a finer-grained component) and measures the indirect effect on a downstream metric. Circuit reverse-engineering (§11) returns a subgraph C satisfying the Wang criteria. All four methods read or write the same underlying object.

2.8 Forward link: four reads of one bus

The unifying picture of this section is that the residual stream defined by Eq. (7) is one object, and the five interpretability methods this paper covers are five different ways of reading or perturbing it. §3 reads the bus's running prediction by applying the canonical readout (6) at every layer. §4 reads the bus under supervision by training a classifier that projects $\mathbf{h}^{(\ell,t)}$ along a learned $\mathbf{d}_y$. §7 reads the bus *without* supervision by fitting a sparse dictionary $\{\mathbf{d}_i\}_{i=1}^M$ such that $\mathbf{h}^{(\ell,t)} \approx \sum_i f_i^{(\ell,t)} \mathbf{d}_i$. §9 *intervenes* on the bus, swapping a component's contribution to F_ℓ between a clean and a corrupted forward pass and measuring the downstream metric change. §11 returns a subgraph of $\mathcal{M}$ whose nodes are precisely the bus sites and the components writing into them. The methods commit to different objects — the W_U columns, the property direction $\mathbf{d}_y$, the SAE direction $\mathbf{d}_i$, the component

[4]KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-2-definition`. Wang et al. define knockout via mean ablation on a reference distribution p_{ABC} rather than zero ablation, on the empirical grounds that "zero ablation [...] lead[s] to noisy results".

[5]KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-4-criteria`.

Symbol	Shape / type	Meaning
$\mathbf{h}^{(\ell,t)}$	$\mathbb{R}^D$ ($\mathsf{B}\times\mathsf{S}\times\mathsf{D}$ batched)	residual-stream activation at layer $\ell \in \{0,\ldots,L\}$, position $t \in \{1,\ldots,S\}$
F_ℓ	$\mathbb{R}^D \to \mathbb{R}^D$	residual contribution of layer ℓ: $\mathrm{Attn}_\ell + \mathrm{MLP}_\ell$ (Eq. (7))
$\mathrm{LN}[\cdot]$	$\mathbb{R}^D \to \mathbb{R}^D$	LayerNorm or RMSNorm, model-specific
W_U	$\mathbb{R}^{D\times V}$	unembedding matrix; canonical readout is $\mathrm{LN}[\mathbf{h}]\,W_U$
$\alpha_{t,s}^{(\ell,h)}$	scalar in $[0,1]$	attention weight from query t to key s, head h, layer ℓ (Eq. (3))
$W_Q^{(\ell,h)}, W_K^{(\ell,h)}, W_V^{(\ell,h)}$	$\mathbb{R}^{d_h\times D}$	QK and OV input projections
$W_O^{(\ell,h)}$	$\mathbb{R}^{D\times d_h}$	OV output projection (writes back into the bus)
$\mathbf{d}_y$	$\mathbb{R}^D$ (unit-norm)	property direction associated with label y (probe / steering target)
$f_i^{(\ell,t)}$	$\mathbb{R}_{\geq 0}$	SAE / transcoder feature i's activation at (ℓ,t)
$\mathbf{d}_i$	$\mathbb{R}^D$	decoder column for feature i (the feature's direction in residual space)
C	subgraph of $\mathcal{M}$	circuit, in the Wang-2022 sense
B, S, D, H, d_h, V, L	—	batch, sequence, model-dim, heads, head-dim, vocab, layers (Paper 1)

Table 1: The notation inherited by §3–§11. $\mathbf{h}^{(\ell,t)}$ is the residual-stream activation site every method reads; $\mathbf{d}_y$ is what supervised probes find; $f_i^{(\ell,t)}$ is what unsupervised SAEs find; C is what circuit reverse-engineering returns. The tensor-shape conventions $\mathsf{B}\times\mathsf{S}\times\mathsf{D}$, $\mathsf{B}\times\mathsf{H}\times\mathsf{S}\times d_h$, $\mathsf{B}\times\mathsf{S}\times\mathsf{V}$ follow Paper 1's architecture chapter.

C at a knockout site, the subgraph C as a wiring diagram — but the underlying read site is the same $\mathbf{h}^{(\ell,t)}$ throughout. The convergences and divergences of §12 live exactly at the boundaries between these reads of one bus.

3 Lens techniques

A *lens* takes a transformer's intermediate residual-stream state $\mathbf{h}_\ell$ at some layer ℓ and returns a distribution over the vocabulary $\mathcal{V}$ — the prediction the model would commit to if it were forced to early-exit at ℓ. Lenses are the cheapest interpretability readout: no training data, no auxiliary classifier, no causal intervention; the lens reuses the model's own pre-unembed normalization and unembedding matrix W_U. This section covers three lens techniques that anchor the literature: the *logit lens* (nostalgebraist, 2020), the no-training baseline that introduced the framework on a LessWrong post in 2020; the *tuned lens* (Belrose et al., 2023), which inserts a learned per-layer affine translator and fixes the logit lens's documented cross-model failures; and *DoLa* (Chuang et al., 2023), an inference-time contrastive-decoding strategy that uses the layerwise-readout structure to suppress hallucinations. Each commits to a different question: what *would* the model predict at layer ℓ? what *is* the model already predicting there once basis drift is accounted for? and what content did the *later* layers specifically commit?

3.1 Setup: the iterative-inference view

Belrose et al. (2023) frame the transformer as iterative inference[6]: each block makes a small additive update to the residual stream, and the per-layer prediction tends to converge smoothly toward the final-layer output. Concretely, for a pre-LayerNorm transformer the residual update at layer ℓ is

$$\mathbf{h}_{\ell+1} \;=\; \mathbf{h}_\ell + F_\ell(\mathbf{h}_\ell), \tag{7}$$

where F_ℓ is the residual output of layer ℓ (attention followed by MLP). Splitting the model $\mathcal{M}$ into the layers up to ℓ and the layers after, the function from a hidden state $\mathbf{h}_\ell$ to the final-layer logits is (Belrose et al., 2023, §2, Eq. 2)[7]

$$\mathcal{M}_{>\ell}(\mathbf{h}_\ell) \;=\; \text{LayerNorm}\left[\mathbf{h}_\ell \;+\; \underbrace{\sum_{\ell'=\ell}^{L} F_{\ell'}(\mathbf{h}_{\ell'})}_{\text{residual update}}\right] W_U, \tag{8}$$

with $W_U \in \mathbb{R}^{D\times V}$ the unembedding matrix (often weight-tied with the input embedding) and LayerNorm the model's own final pre-unembed normalization. Tensor shapes for the rest of this section: residual stream $\mathbf{h}_\ell \in \mathbb{R}^D$ at one (B, S) position, with D the model dim, $V = |\mathcal{V}|$ the vocabulary, and L the layer count from Paper 1's notation. A batched residual stream is $\mathsf{B}\times\mathsf{S}\times\mathsf{D}$; the lens output is $\mathsf{B}\times\mathsf{S}\times\mathsf{V}$.

The *prediction trajectory* $\{\text{Lens}_\ell(\mathbf{h}_\ell)\}_{\ell=0}^{L}$ across layers is the diagnostic object: a single lens reading is a snapshot, but the evolution across layers fingerprints the model's iterative computation. Belrose et al. (2023, §1) report that for clean in-distribution prompts the trajectory converges smoothly to the final prediction many layers before L, and that prompt-injected or otherwise anomalous inputs deviate from the smooth trajectory in detectable ways — the basis for a near-perfect prompt-injection detector (Belrose et al., 2023, §5.3)[8].

3.2 The logit lens: no-training baseline

The logit lens drops the post-ℓ residual updates from Eq. (8) entirely — it sets $\sum_{\ell'=\ell}^{L} F_{\ell'} = 0$ and reads (Belrose et al., 2023, §2, Eq. 3)[9]:

$$\boxed{\text{LogitLens}(\mathbf{h}_\ell) \;:=\; \text{LayerNorm}[\mathbf{h}_\ell]\, W_U} \tag{9}$$

The result is a logit vector in $\mathbb{R}^V$; softmax gives the layer-ℓ "as-if-final" distribution and $\arg\max$ gives the top-1 token. Per-(layer, token) cost is one LayerNorm plus one $D \times V$ matmul — exactly the same compute as the model's actual unembed step at the top of the stack. The technique has no training parameters and no hyperparameters beyond the choice of normalization; it is the cheapest interpretability readout of any kind.

[6] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-1-iterative`.
[7] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-decomposition`.
[8] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-5-3`.
[9] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-logit-lens`.

Forum signal

The logit lens was introduced by nostalgebraist on LessWrong in August 2020 (nostalgebraist, 2020) and propagated for nearly three years before any peer-reviewed treatment landed (Belrose et al., 2023, §2). We treat the LessWrong post as the canonical source of the technique, but flag it explicitly as a forum-tier (Tier C) artifact under the source-tier convention of Paper 1: the technique is widely reused (Halawi 2023, Geva 2022, Dar 2022 referenced in Belrose et al., 2023, §2), but the original write-up is a LessWrong essay, not a journal paper. Hard technical claims about lens behavior in this section back to the Belrose et al. peer-reviewed treatment, not to the original post.

Why the logit lens fails on some models

Belrose et al. (2023, §2) document three failure modes of the raw logit lens[10]. **Bias.** The logit lens systematically puts more mass on certain vocabulary items than the final layer does. On Pythia 12B, the bias measured as KL between logit-lens marginal and final-layer marginal is "around 4 to 5 bits for most layers," against 0.0068 bits between Pythia 160M's and Pythia 12B's final-layer distributions (Belrose et al., 2023, §2). A rational agent's beliefs should not update predictably over time, so the prediction trajectory under a biased lens cannot be read as Bayesian belief updating. **Representational drift.** Hidden states at different layers occupy slightly different covariance subspaces; the final-layer covariance often shifts sharply relative to earlier layers (Belrose et al., 2023, §3), so a fixed final-layer unembedding applied to an earlier-layer state may be misapplied to the wrong basis. **Cross-model unreliability.** The logit lens works "reasonably well" on GPT-2 (Radford et al., 2019) but fails on BLOOM and GPT-Neo: its top-1 predictions in the bottom 21 layers of GPT-Neo-2.7B are nonsense (e.g., "`@G:`") rather than coherent intermediate tokens (Belrose et al., 2023, §2 + Fig. 1). For BLOOM and OPT 125M, the top-1 logit-lens prediction is the input token rather than any plausible continuation in more than half the layers (Belrose et al., 2023, §2).

These three failure modes set up the tuned lens as the targeted correction.

3.3 The tuned lens: a per-layer affine translator

Belrose et al. (2023, §3) insert a learned affine *translator* $(A_\ell, \mathbf{b}_\ell)$ at each layer that maps $\mathbf{h}_\ell$ into the basis the unembedding expects:

$$\boxed{\mathrm{TunedLens}_\ell(\mathbf{h}_\ell) \;:=\; \mathrm{LogitLens}(A_\ell \mathbf{h}_\ell + \mathbf{b}_\ell)} \tag{10}$$

(Belrose et al., 2023, §3, Eq. 8)[11], with $A_\ell \in \mathbb{R}^{D\times D}$ and $\mathbf{b}_\ell \in \mathbb{R}^D$. Translators are initialized to the identity, so a freshly initialized tuned lens is identical to the logit lens; training pulls the translator off the identity to absorb basis drift. Crucially, the unembedding W_U is *frozen* — there is exactly one W_U per model, shared across all L tuned lenses (Belrose et al., 2023, §3).

The training objective is a per-layer distillation against the model's own final-layer prediction:

$$\mathcal{L}(\ell) \;=\; \mathbb{E}_{\mathbf{x}}\big[D_{\mathrm{KL}}(\mathcal{M}_{>\ell}(\mathbf{h}_\ell) \,\|\, \mathrm{TunedLens}_\ell(\mathbf{h}_\ell))\big] \tag{11}$$

[10] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-failures`.

[11] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-3-tuned-lens`.

(Belrose et al., 2023, §3, Eq. 9)[12]. Using $\mathcal{M}_{>\ell}$ as the soft target rather than ground-truth tokens is what makes Eq. (11) a distillation loss: the translator is incentivized to recover what the model itself already knows, never to learn extra information beyond it. This is the design choice that distinguishes tuned-lens translators from Alain and Bengio (2016)-style early-exit probes, which retrain a full unembedding per layer and risk learning structure not present in the underlying model — the classic "probe-as-feature-extractor" confound flagged by Hewitt and Liang (2019) and engaged in §4.

Cost. Tuned-lens parameters total $L \cdot (D^2 + D)$ — a fraction of the model's own size, since $D \times D$ is roughly the size of one attention block's W^Q projection. For Pythia 12B with $D = 5120$ and $L = 36$, this works out to roughly 944M extra parameters (Belrose et al., 2023, §3.2)[13] — considerably smaller than per-layer unembeddings of size $V \times D$ would be, since $V \in [50\mathrm{K}, 250\mathrm{K}]$ across the models Belrose et al. study. The cost-per-(layer, token) at inference is one extra $D \times D$ matmul on top of the logit-lens LayerNorm + W_U chain.

Why a single affine map is enough. The tuned lens fixes all three logit-lens failures with one affine map per layer: $\mathbf{b}_\ell$ absorbs the bias term, A_ℓ absorbs the representational drift (basis shift), and the joint training across all layers adapts to the cross-model pathologies that defeated the raw logit lens. Belrose et al. (2023, Fig. 1) report that the tuned lens recovers coherent intermediate predictions throughout the GPT-Neo-2.7B stack where the logit lens produced nonsense, and generalizes across Pythia, GPT-Neo, GPT-NeoX-20B, OPT, BLOOM, and GPT-2. The fact that a *linear* correction suffices is itself informative: representational drift in residual-stream architectures is approximately linear, not learned-feature-dependent.

Intuition

The lens framework exploits one architectural fact: the residual stream is an additive bus. The final-layer hidden state is $\mathbf{h}_L = \mathbf{h}_\ell + \sum_{\ell'=\ell}^{L-1} F_{\ell'}(\mathbf{h}_{\ell'})$ by recursing Eq. (7). If the residual sum is small relative to $\mathbf{h}_\ell$, the unembedding of $\mathbf{h}_\ell$ already approximates the unembedding of $\mathbf{h}_L$ — the logit lens is a zero-order Taylor expansion that the residual is unchanged. The tuned-lens translator $A_\ell \mathbf{h}_\ell + \mathbf{b}_\ell$ in Eq. (10) extends this to a first-order correction that compensates for whatever systematic drift the actual residual contributes at layer ℓ. Returning to canonical form: the tuned lens is the logit lens with a per-layer linear basis correction; the underlying readout is still $\mathrm{LayerNorm}[\,\cdot\,]W_U$.

3.4 DoLa: lens techniques as a decoding strategy

The lens framework is not only diagnostic. Chuang et al. (2023) turn the layerwise readout into an inference-time decoding strategy by contrasting logits across layers. *Decoding by Contrasting Layers* (DoLa) defines, at each generation step,

$$\mathrm{logits}_{\mathrm{DoLa}}(t) \;=\; \mathrm{logit}^{(L)}(t) \;-\; \mathrm{logit}^{(\ell^\star)}(t), \tag{12}$$

where $\ell^\star$ is selected per token as the layer whose logit distribution is maximally divergent (in Jensen-Shannon divergence) from the final-layer logit distribution (Chuang et al., 2023). Tokens whose probability rises sharply between layer $\ell^\star$ and layer L are exactly the tokens the later layers

[12] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-3-loss`.
[13] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-3-tuned-lens`.

specifically committed to; reweighting toward those tokens preserves the late-stage factual signal while suppressing tokens that were already high in early layers (plausibly hallucination-shaped). DoLa requires no fine-tuning, adds only a modest decoding-time overhead, and reports improved TruthfulQA scores on LLaMA-7B/13B/33B (Chuang et al., 2023). The conceptual point: layerwise readout carries decision-relevant information that can be fed back into generation, not just inspected.

Analogy

DoLa is to the logit lens what a numerical *difference* is to a function evaluation. The logit lens reads $\mathrm{LogitLens}(\mathbf{h}_\ell)$ as the model's belief at layer ℓ; DoLa reads $\mathrm{LogitLens}(\mathbf{h}_L) - \mathrm{LogitLens}(\mathbf{h}_{\ell^\star})$ as the *change* in belief that the later layers contributed. The analogy returns to canonical form: Eq. (12) is exactly the difference of two logit-lens evaluations (Eq. (9)) on the same token, so the underlying object is still the layerwise unembed — read differentially across depth instead of pointwise at one layer.

3.5 Diagnostic: the prediction trajectory

The empirical signature of a working lens is the smoothness of the prediction trajectory $\{\mathrm{Lens}_\ell(\mathbf{h}_\ell)\}_{\ell=0}^{L}$. Belrose et al. (2023, §1) document three regimes. On *clean in-distribution prompts*, the tuned-lens trajectory converges smoothly to the final prediction, with each layer typically reducing perplexity. On *prompt-injected* or otherwise anomalous inputs, the trajectory deviates from the smooth manifold of normal trajectories detectably enough that off-the-shelf isolation-forest / local-outlier-factor classifiers achieve near-perfect prompt-injection detection accuracy (Belrose et al., 2023, §5.3)[14]. On *hard examples* — data points the model required many training steps to learn — the trajectory's commit-layer extends deeper into the stack: harder examples are classified later (Belrose et al., 2023, §5.4). The trajectory is therefore not just a richer readout than the layer-pointwise prediction; it is also a fingerprint that distinguishes input regimes the model treats differently.

Contradiction

Layer-of-prediction vs. layer-of-decision. The trajectory view assumes the lens reading is monotonically improving in ℓ: later layers reflect the model's better-considered prediction. Halawi et al. (2023), however, find that on certain tasks predictions extracted from *earlier* layers are *more* robust to incorrect demonstrations than the final-layer prediction. Belrose et al. (2023, §5.2) reproduce this on three models (BLOOM 560M, GPT-Neo 1.3B, GPT-Neo 2.7B), but flag that without peeking at ground-truth labels the lens user has no a-priori recipe for choosing the right layer[a]. So "commit later = commit better" is sometimes false, and the field lacks a principled criterion for layer selection. The strongest form of the claim is that lens techniques diagnose what *would* happen at ℓ, not what the model *should* commit to; closing that gap requires the causal interventions of §4 and the patching machinery of §9.

[a] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-5`.

3.6 Natural-language autoencoders: lens output as text

The three lenses above project $\mathbf{h}_\ell$ through the model's own unembedding to produce a vocabulary distribution. Fraser-Taliente et al. (2026) (Anthropic 2026-05-07, Transformer Circuits Thread)

[14] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-5-3`.

keep the read-out-the-residual-stream goal but swap the projection target: a Natural Language Autoencoder (NLA) routes the activation through a free-form natural-language explanation. Two LLM modules form the pair: an activation verbalizer (AV) that maps $\mathbf{h}_\ell \in \mathbb{R}^{d_{\text{model}}}$ to an explanation token sequence z, and an activation reconstructor (AR) that maps z back to a reconstruction $\hat{\mathbf{h}}_\ell$. Both are jointly trained with reinforcement learning on the round-trip MSE,

$$\mathcal{L} = \mathbb{E}_{\mathbf{h}_\ell \sim \mathcal{H}}\, \mathbb{E}_{z \sim \text{AV}(\cdot|\mathbf{h}_\ell)} \left[\|\mathbf{h}_\ell - \text{AR}(z)\|_2^2 \right], \tag{13}$$

reported as a Fraction of Variance Explained (FVE) score normalised by the residual variance around the mean activation (Fraser-Taliente et al., 2026, §method; kb/excerpts/anthropic2026-nla#sec-method).

The bottleneck is the explanation: whatever information the AV omits, the AR cannot recover. Unlike the logit / tuned lens which output a deterministic distribution over $\mathcal{V}$, an NLA output is a stochastic free-form text — more expressive, more computationally expensive (two RL-trained LLM modules instead of one W_U projection), and unlike lenses can *confabulate*: the paper explicitly notes that NLA explanations "can also be noisy . . . sometimes include contradictory information or verifiably false claims about the context" (Fraser-Taliente et al., 2026, §confabulation; kb/excerpts/anthropic2026-nla#sec-confabulation). The round-trip FVE provides a calibration signal — confabulated explanations score badly — but a single sample taken at face value can still mislead.

The headline application is auditing. NLAs were used in the Anthropic pre-deployment audit of Claude Opus 4.6 to surface unverbalized evaluation awareness — cases where the model internally suspects it is being tested without saying so in its visible output (Fraser-Taliente et al., 2026, §auditing; kb/excerpts/anthropic2026-nla#sec-auditing). On the intentionally-misaligned-model auditing game introduced in prior Anthropic work by Marks et al., NLA-equipped auditing agents "more effectively discover the root cause of the model's misalignment than baselines" (Fraser-Taliente et al., 2026, §auditing) .

NLAs sit on a different point of the lens-techniques frontier than DoLa: same goal (use intermediate state for safety-relevant readout), opposite output modality (free-form natural language vs. a contrastive adjustment to the final-layer logits). Whether NLAs displace lenses or coexist depends on whether the FVE score is a sufficient gate against confabulation in workflows where the reader cannot run the round-trip themselves.

3.7 Frontier coverage and forward link

The lens framework as published covers the open-source generation through 2023: GPT-2, Pythia (70M–12B), GPT-Neo, GPT-NeoX-20B, OPT, BLOOM, and (via lens transfer) Vicuna-13B (Belrose et al., 2023, §3, §5). The 2025–2026 frontier — Llama 3 / 4, Qwen 2.5 / 3, DeepSeek-V3, Gemma 3 — is largely uncovered by published tuned lenses as of writing. The `logitlens4llms-2025` extension targets Qwen-2.5 and Llama-3.1; broader open-lens releases for the rest of the frontier-2026 lineup are an outstanding need. Whether the lens framework extends cleanly to architectures with substantially modified attention or FFN substrate — DeepSeek-V3's MLA-compressed-KV plus fine-grained-MoE, NSA-sparse attention, sliding-window layers — is partially empirical and not yet documented at frontier scale. The residual stream is still well-defined in all of these, but the layer structure is more irregular, and the assumption that a single affine translator suffices to absorb basis drift may need re-examination on per-architecture basis.

The lens is correlational by construction: it tells you what the model *would* predict if it committed at ℓ, not whether the model *uses* the layer-ℓ information that supports that prediction. Belrose et al.

(2023, §4) partially close this gap with *Causal Basis Extraction*, which finds a tuned-lens-derived orthonormal basis ordered by mean-ablation influence and validates that the directions important to the lens are also important to the underlying model (Spearman $\rho = 0.89$ on Pythia 410M, layer 18)[15]. The next two sections take up this correlational/causal split directly: §4 introduces probes as the supervised correlational analog of lenses — a trained classifier on frozen activations rather than the model's own unembedding — and §5 shows how the field couples probes with activation interventions to upgrade correlational readouts into causal claims. The lens results in this section reappear in §12: the tuned-lens prediction trajectory on IOI prompts is one of the three independent methodologies that converge on the GPT-2-small IOI-circuit picture, and the lens cross-model unreliability documented in §3.2 is one of the field's open contradictions catalogued there.

4 Probing

A *probe* is the supervised correlational readout: a (typically linear) classifier trained on *frozen* activations of a pre-trained transformer to predict an external property — truth, sentiment, syntactic role, factuality, sycophancy. Where the lens of §3 reuses the model's own unembedding W_U to ask *what would the model predict at layer ℓ*, a probe trains a fresh classifier on $\mathbf{h}^{(\ell,t)}$ to ask a different question: *is the property y linearly decodable from the activation at layer ℓ, position t?* The implicit claim that motivates the methodology is sharper than the question: if a high-accuracy probe exists for property y in $\mathbf{h}^{(\ell,t)}$, then the model *linearly represents* y at that layer–position. Whether the model also *uses* that representation is the correlation–causation gap that §5 takes up; this section sets up the methodology, the canonical four probe families, and the layer-position story that lets a probe-accuracy curve diagnose *where* a property is encoded.

4.1 Setup: the linear probe

Let $\mathcal{M}$ be a frozen pre-trained transformer with L layers and hidden size D (Paper 1 notation). For an input sequence $\mathbf{x}$, write $\mathbf{h}^{(\ell,t)}(\mathbf{x}) \in \mathbb{R}^D$ for the residual-stream activation at layer ℓ, position t. A *linear probe* for a binary property y is a classifier

$$\hat{y} \;=\; g(\mathbf{h}^{(\ell,t)}) \;=\; \sigma\Big(\mathbf{w}^\top \mathbf{h}^{(\ell,t)} + b\Big)\,, \tag{14}$$

with $\mathbf{w} \in \mathbb{R}^D$, $b \in \mathbb{R}$, and σ the sigmoid (Alain and Bengio, 2016)[16]. For a multiclass property $y \in \{1, \ldots, K\}$ the probe is a softmax classifier with $\mathbf{W} \in \mathbb{R}^{K \times D}$. The probe is fit on a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$ by minimizing a regularized cross-entropy loss

$$\mathcal{L}(\mathbf{w}, b) \;=\; \frac{1}{|\mathcal{D}|} \sum_i \mathcal{L}_{\mathrm{CE}}\Big(g(\mathbf{h}^{(\ell,t)}(\mathbf{x}_i)),\, y_i\Big) \;+\; \lambda \|\mathbf{w}\|_2^2, \tag{15}$$

with $\mathcal{M}$'s parameters held fixed throughout. The layer ℓ and the token position t are hyperparameters on equal footing with λ. Probe accuracy on a held-out test split is the headline diagnostic; per the control-task convention of Hewitt and Liang (2019), it must be reported alongside an *accuracy delta* against a randomly-initialized-model baseline so that the probe's expressivity does not masquerade as the model's representation. Tensor-shape note: a batched cache of activations at layer ℓ is $\mathsf{B}{\times}\mathsf{S}{\times}\mathsf{D}$; the probe output is $\mathsf{B}{\times}\mathsf{S}{\times}1$ (binary) or $\mathsf{B}{\times}\mathsf{S}{\times}K$ (multiclass).

[15] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-4-cbe`.
[16] KB excerpt: `kb/excerpts/alain-bengio-2017.md#abstract`.

Alain and Bengio (2016) introduced this framework on Inception v3 and ResNet-50 and reported that “the linear separability of features increase[s] monotonically along the depth of the model”[17]: a class-label probe at layer ℓ does at least as well as the same probe at any earlier layer. The result anchored the “hierarchical-abstraction” picture that subsequent NLP probing work inherits — though, as we will see in §4.5, the LLM picture is non-monotonic and property-dependent in ways the vision result does not anticipate.

4.2 The structural probe: from binary to geometric

Linear logistic regression in Eq. (14) treats y as a single class label. Many properties of interest — syntactic dependency trees, coreference chains, semantic role frames — are higher-arity: they are *relations between* tokens, not labels *on* tokens. Hewitt and Manning (2019) generalize the framework with a *structural probe*: a learned linear transform $\mathbf{B} \in \mathbb{R}^{k \times D}$ such that, in $\mathbf{B}$-space, the squared L_2 distance between two tokens' activations approximates the parse-tree distance between those tokens (Hewitt and Manning, 2019)[18]:

$$\boxed{d_{\mathbf{B}}(\mathbf{h}_i, \mathbf{h}_j)^2 \;:=\; \|\mathbf{B}(\mathbf{h}_i - \mathbf{h}_j)\|_2^2 \;\approx\; d_T(w_i, w_j)} \tag{16}$$

where $d_T(w_i, w_j)$ is the number of edges on the parse-tree path between tokens w_i and w_j. A companion variant probes scalar *tree depth* via $\|\mathbf{B}\mathbf{h}_i\|_2^2 \approx \mathrm{depth}_T(w_i)$. The rank k is a hyperparameter; the entire probe is a single $k \times D$ matrix per (model, layer). Hewitt and Manning report that such $\mathbf{B}$ exists for ELMo and BERT but *not* for randomly-initialized baselines, providing evidence that “entire syntax trees are embedded implicitly in deep models' vector geometry” (Hewitt and Manning, 2019). The methodological point: a structural probe shifts the question from “does the model represent y?” to “does the model represent the *structure* of y?”, and the answer is geometric rather than classificatory.

4.3 Mass-mean probing: closed-form, parameter-free

The 2023+ truth-direction literature found that a probe with no learned parameters at all often suffices, and is in fact *more* causally implicated than its trained counterparts. Given a balanced labeled set $\{(\mathbf{h}_i, y_i \in \{+, -\})\}$ at a chosen (ℓ, t), the *mass-mean* (or *difference-in-means*) probe direction is (Marks and Tegmark, 2023)[19]:

$$\boxed{\mathbf{d}_{\mathrm{MM}} \;:=\; \boldsymbol{\mu}_+ - \boldsymbol{\mu}_- \;=\; \mathbb{E}_{i:\, y_i=+}[\mathbf{h}_i] \;-\; \mathbb{E}_{i:\, y_i=-}[\mathbf{h}_i]} \tag{17}$$

The classifier is the projection $\mathbf{h}^\top \mathbf{d}_{\mathrm{MM}}$ thresholded at the midpoint $\frac{1}{2}(\boldsymbol{\mu}_+ + \boldsymbol{\mu}_-)^\top \mathbf{d}_{\mathrm{MM}}$. There is no optimization, no regularization, no hyperparameter tuning beyond (ℓ, t), and no stochasticity beyond the sample split. Marks and Tegmark (2023) argue that despite this simplicity, mass-mean “performs as well as other probing techniques while identifying directions which are more causally implicated in model outputs” (Marks and Tegmark, 2023, abstract): the absence of moving parts is the *point*, since the discovered direction is then a property of the data, not a property of an optimizer. The strong form of the methodological claim — that the mass-mean direction is the direction the model actually *uses* — is the load-bearing contribution of §5, and the geometry-of-truth case study of §6 unpacks the evidence and the caveats.

[17] KB excerpt: `kb/excerpts/alain-bengio-2017.md#abstract` (verbatim).
[18] KB excerpt: `kb/excerpts/hewitt2019-structural-probe.md#abstract`.
[19] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#abstract`, `#sec-1-contributions`.

4.4 Burns CCS: probing without labels

A fourth probe family drops the label requirement entirely. Contrast-Consistent Search (CCS) fits a linear probe by minimizing a self-consistency loss across paired contrast prompts (e.g., "X is true." / "X is false."), enforcing that the probe's outputs on the two completions of a contrast pair should sum to one and be confident. The headline claim is that CCS recovers a *truth-like* direction in LLM activations without ever observing a truth label, and that the recovered direction's classification accuracy is competitive with supervised probes on in-distribution examples. Marks and Tegmark (2023) situate CCS alongside logistic-regression and contrastive probes (Marks and Tegmark, 2023, §1.1)[20] and report that subsequent work has documented generalization failures under negation — CCS-discovered directions on 'cities' do not transfer cleanly to 'neg_cities'. The unsupervised commitment is methodologically appealing (it removes the curated-label confound) but inherits a strong form of the spurious-correlate failure: any consistent latent feature that distinguishes the two halves of the contrast set, not just truth, is a candidate. The deeper treatment of CCS's geometry — why it works, where it fails, what it shares with mass-mean — is in §6.

Intuition

A probe is a *point-spread function* on the model's representation space. Like the optical PSF of a microscope, it is the linear filter $\mathbf{w}^\top \mathbf{h}$ in Eq. (14) through which we observe the model's activation at (ℓ, t) — and the property y must be *in focus along that filter axis* for the probe to detect it. The mass-mean direction $\mathbf{d}_{\mathrm{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$ in Eq. (17) is a closed-form choice of PSF: the axis along which the two classes are maximally separated in expectation. The structural probe in Eq. (16) replaces the scalar projection with a k-dimensional one, so the PSF becomes a k-dimensional subspace rather than a single axis. Returning to canonical form: in every case the probe is a linear map $\mathbb{R}^D \to \mathbb{R}^k$ (with $k = 1$ for binary, $k > 1$ for structural) followed by a fixed decision rule, and probe accuracy quantifies how well the chosen k-dimensional subspace separates the labeled data.

4.5 Where to probe: layer and token position

Probe accuracy is a function of (ℓ, t), and the shape of that function tells a structural story about *where* a property is represented. Three regimes recur across the LLM probing literature.

Surface features early. Token identity, basic part-of-speech role, and other surface-syntactic properties probe highest in *early* layers; the probe accuracy curve rises steeply through the embedding and first few transformer blocks and then plateaus or decays (Hewitt and Manning, 2019). The structural probe of §4.2 is the canonical demonstration: parse-tree distances are recovered with $\mathbf{B}$ trained on early-to-mid BERT layers.

Syntax mid. Tree-structure properties — dependency-parse distances, constituent boundaries, agreement features — peak in *middle* layers. Alain and Bengio (2016) report monotonic linear separability with depth on vision class labels, but the LLM generalization is non-monotonic: information about syntactic structure is most accessible mid-stack and degrades thereafter as later layers recombine it for downstream prediction.

[20] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-1-related`.

Semantic and pragmatic features late-mid. Truth, factuality, sentiment, speaker stance — properties that require the model to integrate information across a clause — peak in *late-mid* layers and stay accurate through the end of the stack. The Marks and Tegmark (2023) finding on LLaMA-2-13B is the canonical worked example: the truth direction localizes around layers 12–15 and remains salient through layer 35, with PCA visualizations showing the linear true/false separation emerging in early-middle layers and "emerg[ing] later for datasets of more structurally complex statements (e.g. conjunctive statements)" (Marks and Tegmark, 2023, §4)[21].

Position selection: the period-token trick. For free-form text, the most-informative probe position is often the *end-of-statement punctuation token* (typically the period). The intuition is summarization: the period sits one position past the last content token, and the model's attention pattern at that position has had the opportunity to summarize the preceding clause into a single hidden-state vector. Marks and Tegmark (2023, §3) document this on LLaMA-2 family: causal-tracing localizes truth representations to a group of hidden states "over the final token of the statement and end-of-sentence punctuation," which they hypothesize "store a representation of the statement's truth"[22]. The period-token convention is now standard in the truth-direction literature; Marks and Tegmark (2023) cite Tigges et al. 2023 as the prior empirical observation of the same summarization behavior on sentiment.

4.6 Probe lineage: eight families on one timeline

Table 2 traces the eight families that anchor the LLM probing literature, in the form lifted from `kb/notes/interpretability/probing.md` §4.1. The lineage tells a coherent story. The 2017–2019 wave (rows 1–4) establishes the methodology and the first methodology critique: the control-task baseline of Hewitt and Liang (2019) is the explicit recognition that probe accuracy alone is insufficient, since a sufficiently expressive probe can learn y from features the model itself does not use. The 2022 review of Belinkov (2022)[23] consolidates the critique into three confounders — probe expressivity, spurious correlation, vestigial information — and surveys the "advances" that pair probes with causal interventions (Belinkov, 2022, §abstract; advances). Rows 5–8 are the 2022+ post-critique wave: closed-form probes that minimize moving parts (mass-mean), unsupervised probes that remove the curated-label confound (CCS), and the activation-steering family that turns a probe direction into a behavioral intervention. §5 takes up the methodological turn that connects rows 1–4 and rows 5–8; §6 unpacks the row-5–6 case study at the level of cross-dataset alignment.

4.7 The probing protocol in six steps

Synthesizing across the four probe families, the practical recipe for probing a property y at layer ℓ, position t is the following six-step pipeline:

1. **Curate** a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$ in *minimal-pair* form: inputs that vary only on y and on as few confounders as possible. Marks and Tegmark (2023, §2) construct `cities`, `neg_cities`, `larger_than`, `cities_cities_conj`, and the `likely` control dataset (10 000 nonfactual completions) precisely to control for surface confounders such as text probability[24].

[21] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-3-layers`, `#sec-4-misalignment`.
[22] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-3-layers`.
[23] KB excerpt: `kb/excerpts/belinkov2022-probing-survey.md#abstract`.
[24] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-2-datasets`.

2. **Cache** activations $\{\mathbf{h}_i^{(\ell,t)}\}_i$ via one forward pass per example. Cost is $|\mathcal{D}|$ forward passes plus storage for $|\mathcal{D}| \cdot D$ floats per (ℓ, t) slice.
3. **Fit** a probe — Eq. (14) via logistic regression, Eq. (17) closed-form, or Eq. (16) via least-squares regression — on a train split. Multi-(ℓ, t) runs are common.
4. **Evaluate** held-out accuracy on a same-distribution test split, paired with a control-task accuracy from Hewitt and Liang (2019) on randomly-permuted labels.
5. **Generalization tests** (optional but increasingly standard). Train on $\mathcal{D}_1$, evaluate on $\mathcal{D}_2$ at a structurally distinct distribution; failures here reveal that the probe was reading a surface confounder rather than the abstract property. Marks and Tegmark (2023, §1, §4) make cross-dataset transfer the cornerstone of their analysis.
6. **Causal validation** (optional, but the *closing* step for any claim that the model *uses* the probe direction). Ablate $\mathbf{d}_y$ from the residual stream during a forward pass and measure behavioral change; or add $\alpha\hat{\mathbf{d}}_y$ to steer the model. The ablation operator and the steering rule are spelled out in §5.

Steps 1–4 produce a correlational claim of the form "y is linearly decodable from $\mathbf{h}^{(\ell,t)}$ with accuracy a above a control-task baseline". Step 5 sharpens the correlational claim by testing whether the decoded direction generalizes across surface distributions. Step 6 converts a correlational claim into a causal one, and is the methodological centerpiece of the next section.

4.8 Cross-paper thread and forward link

Probes are read from *training-time activations*: the property y is whatever the model has acquired by the end of pre-training (or post-training, depending on which checkpoint the activations are cached from), and the probe-accuracy curve is a snapshot of the model's representational competence at that checkpoint. The training pipeline of Paper 2 is therefore the upstream substrate of every claim made in this section: probe-accuracy results read off a checkpoint at post-pre-training time inherit whatever pre-training data, mixture, and optimizer choices produced that checkpoint, and a probe-accuracy *shift* between two checkpoints (e.g., pre-SFT vs. post-SFT) is itself a probing measurement of how a training stage moves the model's internal representation of y.

The methodological gap that probing alone leaves open — that a high-accuracy probe is correlational, not causal — is the load-bearing critique that §5 closes by pairing probes with activation interventions: direction ablation $\mathbf{h} \mapsto \mathbf{h} - (\mathbf{h}^\top \hat{\mathbf{d}}_y)\hat{\mathbf{d}}_y$ tests whether the direction is causally used, and direction addition $\mathbf{h} \mapsto \mathbf{h} + \alpha\hat{\mathbf{d}}_y$ is the activation-steering family that turns a probe direction into an inference-time control surface. §6 then returns to the geometry-of-truth case study at full depth: when probes across datasets agree at late layers but diverge at early layers, what does the divergence tell us about whether "truth" is a single linear axis or a multi-axis subspace?

5 The correlation–causation gap

A high-accuracy probe at layer ℓ tells us that property y is *linearly decodable* from $\mathbf{h}^{(\ell,t)}$. It does not tell us that the model *uses* the decoded direction to compute its output (Belinkov, 2022). The 2017–2019 wave of probing methodology summarized in §4 put the correlational claim on firm footing; the 2019–2022 wave, anchored by Hewitt and Liang (2019) and consolidated by Belinkov (2022), made the gap between *decodable* and *used* into a methodological problem the field had to confront. The 2023+ wave closes the gap by pairing probes with two activation-level interventions: *direction ablation* along the probe direction $\hat{\mathbf{d}}_y$, which tests whether removing $\mathbf{d}_y$ from the residual stream changes behavior, and *direction addition* $\mathbf{h} \mapsto \mathbf{h} + \alpha\hat{\mathbf{d}}_y$, the activation-steering family that turns a probe direction into an inference-time control surface. This section states the critique formally, gives the two intervention operators in canonical form, walks through the Marks–Tegmark 2023 truth-direction worked example, and shows how the resulting "patch-then-probe" pipeline composes patching (§9) with probing (§4) into a single causal recipe.

5.1 The Hewitt–Liang control-task critique

Hewitt and Liang (2019) surface the load-bearing methodology problem: a sufficiently expressive probe can achieve high accuracy even when the underlying model does not represent the property in any useful way, because the probe itself learns the property from low-information features in $\mathbf{h}$. The proposed fix is the *control task*: an alternative probing target with the same input–output type as the real task but with the labels permuted to be informationally meaningless. A part-of-speech control task, for instance, assigns each word type a single random label drawn once and held fixed across the dataset; the probe is then asked to learn this nonsense mapping. The diagnostic is the *selectivity*, defined as the gap

$$\text{selectivity}(g, \ell) \;:=\; \text{acc}(g, \ell, \text{real task}) \;-\; \text{acc}(g, \ell, \text{control task}). \tag{18}$$

(Hewitt and Liang, 2019). A probe that scores high on both real and control tasks has low selectivity: its accuracy is a property of the probe's expressivity, not of the model's representation. A probe that scores high on the real task but low on the control task has high selectivity, and the real-task accuracy can be attributed to information the model itself encoded in $\mathbf{h}^{(\ell,t)}$. Hewitt and Liang (2019) argue this should be a non-negotiable companion measurement to any probing accuracy reported in the literature, and Eq. (18) is now the standard control alongside the held-out test split of the probing protocol of §4.7.

The control-task critique is sharper than "be careful with deep probes." Even a linear probe can fail selectivity on a poorly initialized model, on noisy activations, or on a label set whose distribution is dominated by frequency effects rather than the property of interest. Conversely, a high-selectivity probe is not yet proof of *causal* use, only of *informational* encoding; selectivity rules out *probe-expressivity* confounds, but two further confounds remain.

5.2 Belinkov's three confounders

Belinkov (2022) consolidates the post-control-task critique into a three-confounder taxonomy that has become the methodological vocabulary of the field[25]. The first is Hewitt and Liang (2019)'s **probe-expressivity** confound, addressed by Eq. (18). The second is the **spurious-correlate** confound: even a selective probe may be reading a feature z that correlates with y in the probing dataset, not y itself. The probe direction $\hat{\mathbf{d}}_y$ is then a direction for z, and any downstream intervention along $\hat{\mathbf{d}}_y$ is an intervention on z masquerading as one on y. The standard mitigation is *cross-distribution generalization*: train the probe on dataset $\mathcal{D}_1$, evaluate on dataset $\mathcal{D}_2$ chosen to break the z–y correlation. Marks and Tegmark (2023, §1, §4) make exactly this move the cornerstone of their truth-direction analysis, training on `cities` and evaluating on `neg_cities`, `cities_cities_conj`, and the `likely` control of nonfactual completions designed to match text probability across the true/false split.

[25]KB excerpt: `kb/excerpts/belinkov2022-probing-survey.md#abstract`.

The third is **vestigial information**: the model may have computed y at some intermediate stage, but later layers may not use the resulting direction; the probe reads the trace, not the driver. This is the most insidious confound because it is invisible to both selectivity and cross-distribution generalization — a probe of a truly vestigial feature can be high-selectivity *and* robust across distributions, because the feature really is encoded by the model on those distributions. The only way to distinguish a vestigial encoding from a used one is intervention: change the activation, hold the rest of the model fixed, observe whether the output behavior moves. Belinkov (2022)'s "advances" section catalogs the three intervention families that close this gap — amnesic probing (Elazar et al. 2021, surveyed in Belinkov, 2022), causal-mediation analysis (the ancestor of the activation-patching protocol of §9), and interchange interventions — and the 2023+ probing literature is the inheritor of all three.

Contradiction

Decodable does not entail used. A probe with selectivity gap of $\Delta = 0.9$ on a held-out test split, generalizing across five datasets that break the obvious surface confounders, is still *correlational.* The model may compute the property at some intermediate layer and discard it; later layers may use a different representation; the probe direction may correlate near-perfectly with the used representation while not being it. The strongest form of the critique (Belinkov, 2022, §abstract): probing methodology has historically conflated "y is in $\mathbf{h}$" with "the model uses y in $\mathbf{h}$," and the field's record of behavioral interventions that did *not* work after a probe-only validation is precisely what motivated the 2023+ pivot to intervention-paired probing in §5.3.

5.3 Direction ablation and direction addition

The 2023+ resolution is to compose probing with one of two activation-level interventions, both of which act linearly on the residual stream along a unit-norm probe direction $\hat{\mathbf{d}}_y \in \mathbb{R}^D$.

Direction ablation. Project the residual-stream activation onto the orthogonal complement of $\hat{\mathbf{d}}_y$:

$$\boxed{\mathbf{h} \;\mapsto\; \mathbf{h} \;-\; (\mathbf{h}^\top \hat{\mathbf{d}}_y)\,\hat{\mathbf{d}}_y \;=\; (\mathbf{I} - \hat{\mathbf{d}}_y \hat{\mathbf{d}}_y^\top)\,\mathbf{h}.} \tag{19}$$

The operator on the right is the orthogonal projector $\mathbf{P}_{\hat{\mathbf{d}}_y^\perp} := \mathbf{I} - \hat{\mathbf{d}}_y \hat{\mathbf{d}}_y^\top$ onto the $(D-1)$-dimensional subspace orthogonal to $\hat{\mathbf{d}}_y$ (Marks and Tegmark, 2023). Inserting Eq. (19) into the residual stream at one or more layers and re-running the rest of the model removes whatever signal $\hat{\mathbf{d}}_y$ carries, leaving every other direction unchanged in the activation. The diagnostic question is then simple: if the model's output behavior on property y changes substantially under ablation, $\hat{\mathbf{d}}_y$ is causally implicated; if behavior is unchanged, the direction is vestigial or correlated with a separate driver. Eq. (19) is the formal closing move on the vestigial-information confound of §5.2: it does not just check whether information is encoded along $\hat{\mathbf{d}}_y$, it checks whether the model's downstream computation *depends* on it.

The shape ledger: at one batched residual-stream slice the activation is $\mathbf{h} \in \mathbb{R}^{\mathsf{B}\times\mathsf{S}\times\mathsf{D}}$, the direction is $\hat{\mathbf{d}}_y \in \mathbb{R}^D$, the projection $\mathbf{h}^\top \hat{\mathbf{d}}_y$ contracts the D-axis to a scalar per (B, S) position, and the ablation rewrites $\mathbf{h}$ in place. The intervention is one matrix subtraction per layer where it is applied; cost is dominated by re-running the suffix $\mathcal{M}_{>\ell}$ from the modified state.

Direction addition (activation steering). Add a scaled copy of $\hat{\mathbf{d}}_y$ to the residual stream:

$$\boxed{\mathbf{h} \;\mapsto\; \mathbf{h} \;+\; \alpha\,\hat{\mathbf{d}}_y,} \tag{20}$$

with $\alpha \in \mathbb{R}$ a steering coefficient that can be positive (push the model toward $y = +$) or negative (push toward $y = -$) (Marks and Tegmark, 2023). Direction addition is the method underneath the activation-steering family of (representation engineering, which casts steering as a control-theoretic intervention on the residual stream's representational subspace) and (Contrastive Activation Addition, which derives $\hat{\mathbf{d}}_y$ from a difference-of-means over contrastive prompt pairs and applies it at a chosen layer). The 2025 generalization to behaviorally-singular properties that decompose into multi-axis subspaces appears in , who report that sycophancy in current chat models factorizes into three independently steerable directions (agreement, praise, genuine-agreement), each detected by mass-mean probing and controllable by Eq. (20) at the single-direction level.

The relationship between the two operators: ablation in Eq. (19) is the limit of subtraction along $\hat{\mathbf{d}}_y$ that takes the projection coefficient to zero, while addition in Eq. (20) is a free shift along $\hat{\mathbf{d}}_y$ leaving the projection of the original $\mathbf{h}$ in place. Ablation tests *necessity* (does removing the direction break behavior?); addition tests *sufficiency* (does adding the direction induce behavior?). A causally-implicated direction passes both tests; a spurious or vestigial direction passes neither, or passes only one in a way that fails to compose.

Intuition

The ablation operator $\mathbf{P}_{\hat{\mathbf{d}}_y^{\perp}} = \mathbf{I} - \hat{\mathbf{d}}_y\hat{\mathbf{d}}_y^{\top}$ in Eq. (19) is the same orthogonal projector that appears in Gram–Schmidt orthogonalization: subtracting from $\mathbf{h}$ its component along $\hat{\mathbf{d}}_y$ leaves a vector in the hyperplane normal to $\hat{\mathbf{d}}_y$. Causally probing along $\hat{\mathbf{d}}_y$ is, in this sense, asking whether the model's downstream computation lives *entirely in that hyperplane*: if so, the projection is a no-op and behavior is unchanged; if the computation steps out of the hyperplane along $\hat{\mathbf{d}}_y$ to produce its answer, the projection blocks that step. Returning to canonical form: Eq. (19) is rank-$(D-1)$, the lost rank is exactly $\operatorname{span}(\hat{\mathbf{d}}_y)$, and the experiment is whether the model's output changes when this rank-1 subspace is zeroed out at the residual stream.

5.4 The Marks–Tegmark worked example: flipping TRUE/FALSE

Marks and Tegmark (2023) provide the canonical worked example of the resolution. The mass-mean direction $\mathbf{d}_{\mathrm{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$ introduced in §4.3 (Eq. 4 of §4) is fit on the `cities` dataset of LLaMA-2-13B residual-stream activations at the period-token of layer 12. Probe accuracy on the held-out test split is high; cross-dataset transfer to `neg_cities`, `larger_than`, and `cities_cities_conj` is sufficient to argue that $\mathbf{d}_{\mathrm{MM}}$ is not a surface artifact of the prompt template (Marks and Tegmark, 2023, §1.1, §4)[26]. The closing causal step is the ablation experiment from Eq. (19): the authors apply $\mathbf{P}_{\hat{\mathbf{d}}_{\mathrm{MM}}^{\perp}}$ at the localized hidden states identified by the patching pipeline of §9 and report:

> Causal evidence obtained by surgically intervening in a LLM's forward pass, causing it to treat false statements as true and vice versa.(Marks and Tegmark, 2023, abstract)[27]

[26]KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-1-related`, `#sec-4-misalignment`.
[27]KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#abstract`.

The TRUE/FALSE flip is the load-bearing observation. The probe alone identifies a direction that linearly separates the two classes (§4.1); cross-dataset generalization rules out the obvious spurious-correlate failure modes (§5.2); the ablation in Eq. (19) demonstrates that removing $\mathbf{d}_{\mathrm{MM}}$ from the localized hidden states causes the model to systematically misclassify factual statements that the unmodified model gets right. By the necessity criterion of §5.3, the mass-mean direction is causally implicated; the model is not just *representing* truth, it is *using* the representation. The same paper reports that direction-addition (Eq. (20)) along $\mathbf{d}_{\mathrm{MM}}$ steers the model's outputs in the predicted direction, providing the matching sufficiency test[28].

The strongest form of the conclusion — "$\mathbf{d}_{\mathrm{MM}}$ is the truth direction" — is, however, qualified by the misalignment finding of §6 (forward link): even the mass-mean direction is dataset-dependent in early-mid layers, with `cities` and `neg_cities` truth directions *antipodal* in early LLaMA-2-13B layers and only aligned by late layers (Marks and Tegmark, 2023, §4)[29]. The causal-validation argument lifts the load-bearing claim from "decodable" to "used" but does not, by itself, resolve whether "truth" is a single linear axis or a multi-axis subspace; the next section takes up that geometry directly.

5.5 The patch-then-probe pipeline

The Marks–Tegmark causal pipeline is more than a probe with an ablation appended; it is a two-stage composition that uses the patching machinery of §9 to choose *where* to probe. The first stage is activation patching: sweep the (component, token, layer) grid of AIE_C values (Eq. 34 of §9.1) on a true/false counterfactual benchmark, identify the small group of hidden states with high causal mediation, and restrict the probing domain to those states. Marks and Tegmark (2023, §3) report three groups of causally-implicated hidden states for LLaMA-2-13B on the `cities` dataset; the third (group c) reads directly as TRUE/FALSE under the model's own decoder head, while the second (group b) sits over the final token of the statement and end-of-sentence punctuation and "store[s] a representation of the statement's truth"[30].

The second stage is probing within those states. Mass-mean (Eq. 17 of §4) is fit on the group-b activations rather than on the full grid, and the ablation of Eq. (19) is applied at the same locations. The composition is what makes the causal argument load-bearing: patching localizes the components that carry the behavior, probing localizes the direction *within* those components, and ablation tests whether that direction is the one the model uses. Without the patching stage, the probing direction is fit on activations that may or may not be causally relevant; with it, the probe's domain is restricted to states that patching has already shown to mediate the behavior.

Analogy

Patching localizes the *room*; probing localizes the *shelf within the room*; ablation tests whether the book on that shelf is the one being read aloud. A probe alone tells us a book about y is on *some* shelf in the library; patching alone tells us a book about y is in *this* room; the composition tells us *this book on this shelf in this room* is the one whose absence makes the narrator stop reading. Returning to canonical form: patching's C is the room (a component subset of the residual-stream grid), the probe's $\hat{\mathbf{d}}_y$ is the shelf (a one-dimensional subspace of $\mathbb{R}^D$ at that component), and the ablation $\mathbf{I} - \hat{\mathbf{d}}_y \hat{\mathbf{d}}_y^\top$ in Eq. (19) tests the causal entailment by zeroing the shelf and re-running the suffix $\mathcal{M}_{>\ell}$.

[28] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-1-contributions`.

[29] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-4-misalignment`.

[30] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-3-layers`.

The patch-then-probe pipeline is now the de facto causal-probing recipe in the truth-direction literature (Marks and Tegmark, 2023), and the dual-method recipe behind many of the cross-method convergence cases catalogued in §12. It also exposes the dependencies between methods: patching's STR-vs-GN corruption sensitivity (§9.3, especially the `[CONTRADICTION]` block on GN mis-localization) and the metric-choice pitfalls (§9.2) propagate into the probe domain, since the localized hidden states are the *output* of a patching pass whose recommendations matter.

5.6 Cross-paper threads and forward link

Two cross-paper threads sit in the load-bearing path of this section. The chain-of-thought-faithfulness question of Paper 3 §11 is, in the framing of this section, a probing question: a behavioral probe of "the model is using its written reasoning" is correlational unless paired with an intervention — delete or rewrite the intermediate reasoning trace and observe whether the final answer changes — in exactly the necessity-and-sufficiency form of §5.3. The probe-on-activations and the probe-on-CoT-text are different observation modalities, but the methodological pattern is the same one this section formalizes, and the same set of confounders (probe expressivity, spurious correlate, vestigial information) translates across modalities. The alignment chapter of Paper 5 consumes behavioral probes as the detection methodology for deception, refusal-circumvention, sycophancy, and scheming; every claim there of the form "the model is being y-deceptive" depends on a probe that has passed both the selectivity and the ablation tests of this section, or it inherits the correlational caveat in full.

The forward links within this paper are immediate. §6 picks up the truth-direction case study at the geometry-of-truth depth, asking when the same model's mass-mean direction is dataset-stable and when it is not, and what the misalignment in early-mid layers tells us about whether "truth" is a single linear axis or a multi-axis subspace. §12 returns to the patch-then-probe pipeline as one of the field's clearest cross-method convergence stories, with §9's machinery on the activation-patching side and §4's machinery on the probing side composing into a causal pipeline that neither method supports alone. The correlation–causation gap is, in this paper's framing, the cleanest example of why method-pluralism is principled: the gap is closed not by a better single method, but by composing two methods whose individual commitments are insufficient and whose joint commitments are not.

6 Truth directions and the geometry of truth

If §4 sets up the probing methodology and §5 closes the correlation–causation gap with activation interventions, this section is the worked case study that exercises both. "Truth" — whether the model's residual stream encodes the truth value of a factual statement as a linearly decodable direction — is the most-studied probing target since 2022, and the empirical record is rich enough to separate what the field has actually converged on from what remains open. The convergence claim is narrow: on the LLaMA-family on simple factual true/false statements, a linear direction in the residual stream at late-mid layers carries the truth value with high probe accuracy and is causally implicated in the model's TRUE/FALSE prediction (Marks and Tegmark, 2023). The open claims are the load-bearing ones: whether the *same* direction is recovered across prompt formats, whether the unsupervised CCS direction recovers truth or a near-truth confound, whether "truth" is a single linear axis or a multi-axis subspace, and whether any of this transfers cross-model-family. We work through each in turn.

6.1 The Marks–Tegmark convergence claim

Marks and Tegmark (2023) make the strongest published case for a linear truth representation, with three lines of evidence assembled on LLaMA-2 7B/13B/70B and the curated `cities` / `neg_cities` / `larger_than` / `cities_conj` datasets of §4.7[31]: *visualization* (PCA of layer-ℓ activations shows clear linear true/false separation in the top two PCs); *transfer* (probes trained on one dataset classify a structurally distinct dataset above chance and ablation-flip its predictions); and *causal intervention* (ablating the probe-derived direction in the residual stream during the forward pass causes the model to "treat false statements as true and vice versa" (Marks and Tegmark, 2023, abstract)). The headline methodological recommendation is that, among the four probe families of §4, the parameter-free *mass-mean* direction $\mathbf{d}_{\text{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$ of Eq. (17) is the one "most causally implicated in model outputs" (Marks and Tegmark, 2023, §1) — generalizes across datasets at least as well as logistic regression, while intervening on it produces larger behavioral effects.

The localization picture is sharp. Causal patching localizes truth representations to a small group of hidden states "over the final token of the statement and end-of-sentence punctuation," which the authors hypothesize "store a representation of the statement's truth" (Marks and Tegmark, 2023, §3)[32]. On LLaMA-2-13B the separation emerges in early-middle layers (around $\ell = 12$), is salient through the rest of the stack, and "emerges later for datasets of more structurally complex statements (e.g. conjunctive statements)" (Marks and Tegmark, 2023, §4). The pattern — linear separation, localized to a layer band, summarized at the period-token — is the picture the field now defaults to when saying "the LLM has a truth direction."

Intuition

Mass-mean turns a probing claim into a claim about the data's class geometry: $\mathbf{d}_{\text{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$ is the axis along which the per-class means are separated, and the projection $\mathbf{h}^\top \mathbf{d}_{\text{MM}}$ is the "how far does this activation lean toward the true class" scalar. Why does the parameter-free direction beat trained logistic regression on *causal* mediation? Because logistic regression fits the discriminative direction — the axis that minimizes classification error — and that axis can read a confound z that linearly separates the data without the model using z. Mass-mean fits no boundary; it points along the class-difference of the data itself, so the recovered axis depends only on $\mathcal{D}$, not on an optimizer. Returning to canonical math: with isotropic Gaussian classes of equal covariance, mass-mean and logistic regression recover the *same* direction up to scale (Fisher LDA); under heteroscedastic or non-Gaussian residuals they diverge, and the divergence is exactly the discriminative flexibility that mass-mean chooses to forgo.

6.2 The Burns CCS direction — truth-like, but unsupervised

Contrast-Consistent Search (CCS) attacks the same question from the unsupervised side. Given an unlabeled set of contrast pairs $(\mathbf{x}^+, \mathbf{x}^-)$ — e.g., a statement appended with "yes" versus "no" — CCS fits a linear probe $p_\theta(\mathbf{h}) = \sigma(\mathbf{w}^\top \mathbf{h} + b)$ by minimizing a self-consistency loss

$$\mathcal{L}_{\text{CCS}}(\theta) \;=\; \mathbb{E}_{(\mathbf{x}^+,\mathbf{x}^-)}\Big[\big(p_\theta(\mathbf{h}^+) + p_\theta(\mathbf{h}^-) - 1\big)^2 \;+\; \min\big(p_\theta(\mathbf{h}^+),\, p_\theta(\mathbf{h}^-)\big)^2\Big], \tag{21}$$

which jointly enforces (i) that the probe assigns complementary probabilities to the two halves of the pair — a constraint any *truth* feature must satisfy, since a statement and its negation have

[31] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#abstract`.

[32] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-3-layers`.

opposite truth values — and (ii) that the probe is confident on at least one side. The discovered $\mathbf{w}$ is reported to be "truth-like": probe accuracy on held-out labeled true/false data is competitive with supervised logistic regression on in-distribution splits . The contribution is conceptual: CCS removes the curated-label confound that motivates Step 1 of the probing protocol of §4.7; the discovered direction is a property of the model's representation *and* of the contrast-pair templates, not of any human-curated label.

The empirical record on CCS, however, is more cautious than the in-distribution accuracy headline suggests. Marks and Tegmark (2023) situate CCS alongside other probing techniques in their related work and note that *transfer to negated statements* is the load-bearing test[33]: a direction that separates "X is true" from "X is false" but fails on "X is not false" versus "X is not true" is reading something correlated with truth (sentiment, statement plausibility, surface template features) rather than truth itself. The pattern in their related-work discussion is that prior unsupervised and contrastive probing work has documented "failures of these probes to generalize in basic ways" (Marks and Tegmark, 2023, abstract) — and the `cities` versus `neg_cities` contrast is exactly the form of generalization test on which the Marks and Tegmark mass-mean direction does pass and on which CCS-style directions are documented to struggle. We treat this as a strong empirical signal that the unsupervised contrast-pair objective of Eq. (21) can latch onto a near-truth confound rather than truth proper, and we flag the underlying claim as anchored to in-distribution accuracy until a peer-reviewed cross-dataset reproduction is in hand.

6.3 Stark misalignment: the dataset-dependent direction

The cleanest open question is the one Marks and Tegmark (2023) surface in their own §4 visualization analysis. After establishing that each of `cities`, `neg_cities`, `larger_than`, and `cities_conj` admits a linearly separating direction, the authors ask whether those directions *align* across datasets. The answer is: they align *at late layers*, but not at early/mid layers[34]. The verbatim characterization is worth quoting in full:

> *The axes of separation for various true/false datasets align often, but not always. . . . Fig. 3(c) shows stark failures of alignment, with the axes of separation for datasets of statements and their negations being approximately orthogonal. (Marks and Tegmark, 2023, §4)*

and the layer-wise dynamics:

> *In early layers, `cities` and `neg_cities` separate antipodally, before rotating to lie orthogonally. . . , and finally aligning in later layers. (Marks and Tegmark, 2023, §4 / App. C)*

The implication for the "does the model have a truth direction" question is direct. The probe direction recovered from `cities` alone is *not* the probe direction recovered from `neg_cities`, in early or middle layers; it would be a mistake to treat either single-dataset-derived $\hat{\mathbf{d}}_y$ as a canonical truth axis at those depths. Only at late layers do the per-dataset directions begin to share a common subspace, and the authors flag a scale dimension on top of the layer dimension: `larger_than` and `smaller_than` separate *antipodally* in LLaMA-2-13B but along a *common* direction in LLaMA-2-70B (Marks and Tegmark, 2023, §4), suggesting that larger models compress what 13B encodes as multiple per-template axes into a single more-abstract axis.

[33]KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-1-related`.

[34]KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-4-misalignment`.

6.4 Vennemeyer 2025: the multi-component generalization

Vennemeyer et al. (2025) extend the multi-component finding from truth to behavioral phenomena. Applying difference-in-means probing to sycophancy on LLaMA-family models, they report that the behavior decomposes into *three independently-steerable directions*: agreement (do I match the user's stated position?), praise (do I compliment the user?), and genuine-agreement (do I give the answer I would have given without the user's lead?). Activation-steering along each direction independently changes a different facet of the sycophantic behavior; combining them at chosen scales reproduces the full behavioral profile. The methodological generalization is direct: where Marks and Tegmark document *representational plurality* for "truth" *across datasets within a model*, Vennemeyer et al. document representational plurality for *a single behavior at a single distribution* — both observations point at the same underlying claim: **behaviorally-singular properties may be representationally-plural**, and a single direction is a *projection* of a richer subspace rather than the full representation.

Contradiction

Is there a canonical truth direction? Three observations strain the strong-form claim that "the LLM linearly represents truth" as a single axis $\mathbf{d}_{\text{truth}} \in \mathbb{R}^D$. (1) Within a model, Marks and Tegmark (2023, §4) document early-/mid-layer "stark misalignment" — per-dataset truth directions are antipodal, rotate to orthogonal, then align only late: a single-dataset-derived $\hat{\mathbf{d}}_y$ is not a depth-uniform truth axis. (2) CCS, the unsupervised analog of Eq. (21), is documented to suffer generalization failures under negation (Marks and Tegmark, 2023, §1.1) — a putative truth direction that fails the most basic logical operation on truth is better understood as a near-truth confound than as truth. (3) The Vennemeyer et al. (2025) sycophancy decomposition shows that an alignment-relevant phenomenon thought to be representationally-singular admits a multi-axis decomposition; the generalization to truth is direct (and tested informally in their follow-up). **What evidence would resolve it.** A multi-axis basis recovery procedure that, applied independently to `cities`, `neg_cities`, `larger_than`, and `cities_conj` at matched layers, returns a low-dimensional subspace whose per-dataset projections recover the single-dataset directions modulo a known equivalence (rotation, sign, scale). Until then, "the truth direction" is a useful abstraction at late layers in fixed model + dataset family — a *representational subspace* treated as a one-dimensional projection — not a basis-free property of the model.

6.5 What the field has converged on, and what remains open

The convergence picture, restricted to load-bearing claims that multiple labs and probing methodologies have triangulated:

- **Late-mid layer linear decodability.** On the LLaMA-2 family, simple factual true/false statements admit a linear probe with high held-out accuracy in residual-stream activations at layers $\ell \approx 12-15$ (LLaMA-2-13B) and equivalent relative depths on 7B/70B; the direction remains salient through the rest of the stack (Marks and Tegmark, 2023, §3, §4).
- **Period-token summarization.** The most-informative probe position is the end-of-statement punctuation token; this summarization behavior was observed prior on sentiment (Marks and Tegmark, 2023, §3, citing Tigges et al. 2023) and the truth result reproduces the geometric pattern.

- **Mass-mean as the direction with strongest causal mediation.** Among the probe families of §4, $\mathbf{d}_{\mathrm{MM}}$ flips TRUE/FALSE under direction-ablation more strongly than logistic-regression directions on matched datasets (Marks and Tegmark, 2023, §1, §abstract).
- **Larger models' truth axes are more abstract.** Per-template direction differences shrink as model size grows; LLaMA-2-70B compresses what 13B represents as multiple per-template axes into a more shared axis (Marks and Tegmark, 2023, §4.1).

The open claims, in falsification-priority order:

- **Cross-prompt-format alignment.** Whether the early-/mid-layer per-dataset misalignment is a deficiency of the *probe* (single-direction summary of a multi-axis subspace) or a property of the *model* (no shared early-layer truth representation across prompt formats) is unresolved (Marks and Tegmark, 2023, §4).
- **CCS under negation.** The empirical record on whether Eq. (21) recovers truth versus a confound on negated-statement transfer is fragmented across follow-ups; a matched-protocol reproduction at LLaMA-2-13B/70B scale is missing .
- **Multi-class generalization of mass-mean.** Eq. (17) is binary by construction; the analog for K-class truthfulness gradations (e.g., model-graded confidence buckets) is not yet a settled methodological recipe.
- **Cross-model-family transfer.** Truth directions in LLaMA-2-13B and Qwen-2.5-14B almost certainly do not have the same coordinates (the bases differ); whether there is a canonical truth direction up to model-specific basis or whether truth is fundamentally model-dependent is the question that Paper 5 needs answered to sharpen behavioral probes for deception, sycophancy, scheming, and alignment-faking into cross-model evaluation tools.

6.6 Forward link

Behavioral probes for alignment-relevant phenomena ride on the methodology this section assembles. Paper 5 §9 (sycophancy) consumes the Vennemeyer et al. multi-component result directly; Paper 5 §10 (scheming) and §11 (alignment-faking) use the same probe-then-causally-validate pipeline of §5 with behavioral targets in place of factual ones. The single methodological lesson the truth case study exports is the contradiction box of §6.3: treat the recovered direction as a single-dataset summary, not as a basis-free property; report cross-dataset transfer in *both* accuracy *and* ablation effect; and prefer multi-axis subspace recovery over single-axis fits when the underlying behavior admits a multi-component decomposition.

The same canonicality question generalizes one floor down, to the *unsupervised* feature framing. Sparse autoencoders (§7, written) decompose residual-stream activations into a learned dictionary of $M \gg D$ directions, each plausibly a "feature" candidate; whether two SAE training runs at different sparsity targets recover the *same* feature inventory is the structural cousin of "do two probes on different truth datasets recover the same direction." §8 takes that question up at the SAE level, and the cross-method synthesis of §12 returns to both: probe-derived truth directions and SAE-derived truth-related features partially overlap, partially do not, and the cases of disagreement are exactly the cases where a single-method claim should not be taken as canonical.

7 Sparse autoencoders

A sparse autoencoder (SAE) makes a specific commitment about what a feature *is*: a direction $\mathbf{d}_i \in \mathbb{R}^n$ in activation space such that, on any given input, the activation $\mathbf{x}$ is well-approximated by a sparse linear combination of a small subset of such directions drawn from an overcomplete dictionary $\{\mathbf{d}_i\}_{i=1}^M$ with $M \gg n$. The commitment is strong — features are linear, additive, and identifiable from activations alone without supervision — and the methodological payoff is that activations acquire an addressable, named basis on which downstream interpretability work (probing, patching, circuit tracing) can land. This section covers the two production-grade architectures (Top-K (Gao et al., 2024), JumpReLU (Rajamanoharan et al., 2024)), the open scale demonstration (Gemma Scope (DeepMind, 2024)), and the structural cousin (transcoders (Dunefsky et al., 2024)) that carries the framing into the circuit-tracing pipeline of §11. The canonicality question raised by these commitments is taken up in §8.

7.1 The framing: dictionary learning on activations

Given an activation $\mathbf{x} \in \mathbb{R}^n$ — typically the residual stream at some layer ℓ, or an MLP/attention-output vector — an SAE is a pair of an encoder and a decoder (Rajamanoharan et al., 2024, §2, Eq. 1–2)[35]:

$$\mathbf{f}(\mathbf{x}) := \sigma(\mathbf{W}_{\text{enc}}\,\mathbf{x} + \mathbf{b}_{\text{enc}})\,, \tag{22}$$

$$\hat{\mathbf{x}}(\mathbf{f}) := \mathbf{W}_{\text{dec}}\,\mathbf{f} + \mathbf{b}_{\text{dec}}, \tag{23}$$

where $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{M\times n}$, $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{n\times M}$, and the encoder activation σ is the architectural choice that defines the SAE family. The columns of $\mathbf{W}_{\text{dec}}$, written $\mathbf{d}_i$, are the *decoder feature directions*; the expansion factor M/n is typically 8–1000. The training objective always combines reconstruction and sparsity:

$$\mathcal{L}(\mathbf{x}) \;=\; \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{f}(\mathbf{x}))\|_2^2}_{\mathcal{L}_{\text{reconstruct}}} + \mathcal{L}_{\text{sparsity}}. \tag{24}$$

The sparsity term and the activation σ jointly determine the position the SAE occupies on the L_0-vs-reconstruction Pareto frontier, and that frontier is, empirically, the only training axis that matters: every modern SAE family is an attempt to push the Pareto curve down-and-left at fixed compute (Rajamanoharan et al., 2024, §1).

The original recipe of Templeton et al. (2024) following Gao et al. (2024)'s baseline uses ReLU encoder with an ℓ_1 sparsity penalty $\lambda\|\mathbf{f}\|_1$ (Gao et al., 2024, §2.2, Eq. 1)[36]. Two known issues motivate the TopK and JumpReLU successors. First, *shrinkage*: ℓ_1 biases all positive activations toward zero, so reconstructions systematically underestimate true feature magnitudes (Gao et al., 2024, §2.3). Second, *reparameterization non-invariance*: scaling encoder rows up and decoder columns down preserves $\mathbf{W}_{\text{dec}}\mathbf{f}$ but changes $\|\mathbf{f}\|_1$, so the ℓ_1 penalty is degenerate without a unit-norm constraint on $\mathbf{d}_i$ (Rajamanoharan et al., 2024, §3).

7.2 Top-K SAEs (Gao et al. 2024)

Gao et al. (2024) replace the ReLU+ℓ_1 recipe with a k-sparse activation (after Gao et al., 2024): keep the top-K pre-activations, zero the rest. The encoder becomes (Gao et al., 2024, §2.3, Eq. 2)[37]

$$\mathbf{f}(\mathbf{x}) \;=\; \text{TopK}(\mathbf{W}_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}))\,, \tag{25}$$

[35] KB excerpt: `kb/excerpts/rajamanoharan2024-jumprelu.md#sec-2`.

[36] KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-2-2-baseline`.

[37] KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-2-3`.

and because the sparsity is enforced exactly — $L_0 = K$ at every token by construction — the loss reduces to $\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$ with the ℓ_1 penalty removed entirely. The four claimed benefits are direct consequences of this design choice (Gao et al., 2024, §2.3): (i) no shrinkage bias from ℓ_1; (ii) L_0 is set directly rather than tuned via λ, simplifying sweeps and model comparison; (iii) the empirical sparsity-reconstruction frontier improves over ReLU+ℓ_1, with the gap widening at scale (Gao et al., 2024, Fig. 2b); (iv) clamping small activations to zero increases the monosemanticity of random activating examples (Gao et al., 2024, §4.3).

The cost is the *dead-latent problem*: at large M, latents that fail to fire early in training never recover. Gao et al. (2024) report that without mitigation, "up to 90% dead latents" (Gao et al., 2024, §2.4) appear at the largest scales[38]. Two ingredients fix it: encoder rows are initialized as the transpose of decoder columns; and an auxiliary "AuxK" loss models the residual reconstruction error using the top-k_{aux} *dead* latents, forcing them to reactivate. With these, even a 16M-latent SAE on GPT-4 residual-stream activations trained for 40B tokens keeps only 7% of latents dead (Gao et al., 2024, §1, §2.4)[39]. Optimal compute follows a clean power law $L(C) = 0.09 + 0.056\, C^{-0.084}$ on the MSE-vs-compute frontier (Gao et al., 2024, Fig. 1, §3), so the top-K family scales predictably to GPT-4-class models.

7.3 JumpReLU SAEs (Rajamanoharan et al. 2024)

Rajamanoharan et al. (2024) take a different route to the same goal: replace ReLU with a learned-threshold step,

$$\text{JumpReLU}_{\boldsymbol{\theta}}(z) \;:=\; z\, H(z - \theta), \tag{26}$$

where H is the Heaviside step function and $\boldsymbol{\theta} \in \mathbb{R}_+^M$ is a *per-feature learned threshold* below which encoder pre-activations are set to zero (Rajamanoharan et al., 2024, §2, Eq. 4)[40]. The full SAE encoder is $\mathbf{f}(\mathbf{x}) := \text{JumpReLU}_{\boldsymbol{\theta}}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}})$ (Rajamanoharan et al., 2024, §3, Eq. 8), and sparsity is trained directly via an L_0 penalty rather than its ℓ_1 proxy:

$$\mathcal{L}(\mathbf{x}) \;=\; \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{f}(\mathbf{x}))\|_2^2 \;+\; \lambda\, \|\mathbf{f}(\mathbf{x})\|_0. \tag{27}$$

Equation (27) is non-differentiable both in $\boldsymbol{\theta}$ (because of H) and in $\|\cdot\|_0$, which Rajamanoharan et al. (2024) resolve with *straight-through estimators*: the backward pass replaces the discontinuous derivative with a kernel-density-estimated surrogate of the derivative of the expected loss, separately for $\boldsymbol{\theta}$ and for the L_0 term (Rajamanoharan et al., 2024, §3). The key design point is that this avoids the shrinkage bias of ℓ_1 entirely while keeping the encoder elementwise — no partial sort, no batchwise communication — so a JumpReLU SAE trains as cheaply as a plain ReLU SAE.

The empirical headline: at any fixed sparsity level on Gemma 2 9B residual-stream, MLP-output, and attention-output activations, JumpReLU SAEs "provide reconstructions that are at least as good as, and often slightly better than, TopK SAEs" and consistently dominate Gated SAEs (Rajamanoharan et al., 2024, §1)[41]. The caveat is interpretability, not reconstruction: both TopK and JumpReLU produce a small fraction of *high-frequency* latents (firing on $> 10\%$ of tokens, fewer than 0.06% of a 131K-width SAE) which manual ratings find systematically less interpretable (Rajamanoharan et al., 2024, §1)[42]. Pareto-winning on reconstruction-at-fixed-sparsity does not imply Pareto-winning on per-feature interpretability.

[38] KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-2-4`.
[39] KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-1-headline`.
[40] KB excerpt: `kb/excerpts/rajamanoharan2024-jumprelu.md#sec-2-activations`.
[41] KB excerpt: `kb/excerpts/rajamanoharan2024-jumprelu.md#sec-1-pareto`.
[42] KB excerpt: `kb/excerpts/rajamanoharan2024-jumprelu.md#sec-1-caveat`.

Intuition

A sparse autoencoder is a search for the directions in activation space such that most points use only K of them. The encoder asks "which K dictionary atoms best explain this activation?"; the decoder reassembles the activation as a K-sparse linear combination $\hat{\mathbf{x}} = \sum_{i \in S(\mathbf{x})} f_i \, \mathbf{d}_i$ with $|S(\mathbf{x})| = K$. The choice of activation σ in Eq. (22) is the choice of *how* to enforce that K: TopK does it by argmax (exactly K active per token); JumpReLU does it by a learned per-feature threshold (approximately K on average, learned). Both reduce to the same canonical form $\hat{\mathbf{x}} = \sum_i f_i \, \mathbf{d}_i$ with $\mathbf{f}$ sparse — the dictionary-learning math of Eq. (22)–(23).

7.4 Gemma Scope: the open scale demonstration

Until 2024, large-scale SAE work was largely Anthropic-internal (Templeton et al., 2024 on Claude 3 Sonnet) or OpenAI-internal (Gao et al., 2024 on GPT-4). *Gemma Scope* (DeepMind, 2024) is the open-release inflection: a suite of JumpReLU SAEs trained on *all* layers and sublayers — residual stream, MLP output, attention output — of Gemma 2 2B and 9B base models, plus selected layers of Gemma 2 27B. The release contains "more than 400 sparse autoencoders in the main release, with more than 30 million learned features in total" (DeepMind, 2024, §1)[43], trained on 4–16B tokens each at widths $\{16.4\text{K}, 65.5\text{K}, 131\text{K}, 1\text{M}\}$. The compute headline:

> *We used over 20% of the training compute of GPT-3 (Brown et al., 2020), saved about 20 Pebibytes (PiB) of activations to disk, and produced hundreds of billions of sparse autoencoder parameters in total (DeepMind, 2024, §1).*

The justification for breadth is methodological: DeepMind (2024, §1) argue that "a comprehensive suite is essential for enabling more ambitious applications of interpretability, such as circuit analysis," which require feature dictionaries at every site through which causal information flows.

A load-bearing implementation detail for the cross-method work in §11: DeepMind (2024, §4.1) report that delta-loss is consistently higher for residual-stream SAEs than for MLP- or attention-sublayer SAEs at matched FVU, because "even small errors in reconstructing the residual stream can have a significant impact on LM loss, whereas relatively large errors in reconstructing a single MLP or attention sub-layer's output have a limited impact on the LM loss" (DeepMind, 2024, §4.1). Residual SAEs sit on the model's communication bus and any reconstruction error compounds; sublayer SAEs sit on additive contributions and absorb error locally. Circuit-analysis work that re-inserts SAE reconstructions in place of the original activations must account for this: residual-SAE reconstructions are the high-leverage failure mode.

The release explicitly disclaims a converged evaluation framework: "as of now there is no consensus on what constitutes a reliable metric for the quality of a sparse autoencoder or its learned latents and [...] this is an ongoing area of research and debate" (DeepMind, 2024, §4). Gemma Scope reports several metrics (delta-loss, FVU, downstream-task evaluation) precisely because no single one is canonical — a frontier point we develop in §8.

7.5 Transcoders: from feature decomposition to circuit primitive

A *transcoder* (Dunefsky et al., 2024) is structurally an SAE but trained on a different objective: rather than reconstructing an activation, it learns to mimic an MLP sublayer's input-to-output

[43] KB excerpt: `kb/excerpts/gemma-scope-2024.md#sec-1`.

mapping with a sparse hidden representation in between. The architecture (Dunefsky et al., 2024, §3.1, Eq. 3–5)[44] is:

$$\mathbf{z}_{\mathrm{TC}}(\mathbf{x}) = \mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\,\mathbf{x} + \mathbf{b}_{\mathrm{enc}})\,, \tag{28}$$
$$\mathrm{TC}(\mathbf{x}) = \mathbf{W}_{\mathrm{dec}}\,\mathbf{z}_{\mathrm{TC}}(\mathbf{x}) + \mathbf{b}_{\mathrm{dec}}, \tag{29}$$

trained against the MLP function the transcoder is meant to replace:

$$\mathcal{L}_{\mathrm{TC}}(\mathbf{x}) \;=\; \underbrace{\|\mathrm{MLP}(\mathbf{x}) - \mathrm{TC}(\mathbf{x})\|_2^2}_{\text{faithfulness}} + \lambda_1\,\|\mathbf{z}_{\mathrm{TC}}(\mathbf{x})\|_1. \tag{30}$$

The structural difference from an SAE is one substitution in the faithfulness term: an SAE-on-MLP-output minimizes $\|\mathrm{MLP}(\mathbf{x}) - \hat{\mathbf{x}}\|_2^2$ where $\hat{\mathbf{x}}$ depends on the *post*-MLP activation, and so the SAE's feature dictionary is a decomposition of that *specific* activation. A transcoder's $\mathrm{TC}(\mathbf{x})$ depends on the *pre*-MLP input $\mathbf{x}$, and so its feature dictionary is a decomposition of the MLP *function*, not its particular output (Dunefsky et al., 2024, §1)[45].

The methodological consequence (Dunefsky et al., 2024, §1): the connection from a pre-MLP feature f_i (associated with decoder column $\mathbf{d}_i$) to a post-transcoder feature f_j (associated with encoder row $\mathbf{e}_j$) factorizes into an *input-dependent* term (the activations $f_i(\mathbf{x})$ and $f_j(\mathbf{x})$ at the input of interest) and an *input-invariant* term (the bilinear weight product $\mathbf{e}_j^\top \mathbf{d}_i$ that holds for any input). This is the factorization that the cross-layer transcoder of Lindsey et al. (2025b) generalizes to skip-layer feature flow, and it is the architectural primitive on which the attribution-graph construction of §11.4 is built. As an analytical tool, transcoders bridge two questions: SAEs ask "what features are present in an activation?"; transcoders ask "what features does this MLP *compute*, and how do they depend on its inputs?". The latter is the question circuit analysis needs answered.

7.6 Forward link: are these features canonical?

The architectures of this section — ReLU+ℓ_1, Top-K, JumpReLU, and transcoders — give us a working pipeline for converting polysemantic neuron-basis activations into a sparse, named feature basis at scale. What they do not give us, on their own, is a guarantee that the basis is unique. Two independent SAE training runs at different widths M on the same activations may produce different feature inventories (Leask et al., 2025); a feature in a wider SAE may decompose into a combination of features in a narrower one (*non-atomicity*); features present in a wider SAE may have no counterparts in the narrower (*incompleteness*). The "true features" framing implicit in Templeton et al. (2024) treats SAE features as the model's underlying units of computation; the Leask et al. (2025) critique argues this is unwarranted. Section 8 takes up the canonicality question directly, both for SAEs and for the transcoders that inherit it.

8 Are SAE features canonical?

The architectures of §7 give us a working pipeline for converting polysemantic neuron-basis activations into a sparse, named feature basis at scale. They do not, on their own, tell us whether that basis is *the* basis. The implicit framing of the production-scale SAE program — that scaling M and pushing further down the L_0-vs-reconstruction Pareto curve converges toward a canonical decomposition of model activations into atomic, interpretable features — is a load-bearing assumption: if features are

[44] KB excerpt: `kb/excerpts/dunefsky2024-transcoders.md#sec-3-1`.
[45] KB excerpt: `kb/excerpts/dunefsky2024-transcoders.md#sec-1-comparison`.

canonical, then SAE features are the units of the model's internal computation, and downstream interpretability claims (probes cite them, patching targets them, circuits compose them) inherit their canonicality. Leask et al. (2025) make a direct empirical argument that this assumption is unwarranted, with two techniques that probe atomicity and completeness separately. We work through the argument, then take up *transcoders* (Dunefsky et al., 2024) — structural cousins of SAEs that sit at the input-output boundary of an MLP rather than on the activation itself — and show that the additional analytical leverage they provide for circuit work in §11 comes with the same canonicality concern inherited unchanged.

8.1 What "canonical" would mean

Leask et al. (2025, §1) fix the terminology before attacking the claim[46]. A set of units of analysis is **canonical** when it is simultaneously: **unique** — the same set is recovered (modulo a known equivalence such as rotation or permutation) by independent training runs and across reasonable choices of dictionary size M; **complete** — every feature the model uses is represented; and **atomic** — each unit is irreducible, not decomposable into smaller units in the same dictionary family. The Bricken et al. (2023) "Towards Monosemanticity" paper and the Templeton et al. (2024) scaling follow-up both lean on a strong-form version of this picture: SAE features are the *true* units, and a sufficiently scaled SAE recovers them. Both citations are HTML-only on `transformer-circuits.pub`; we cite them by paper-key with attribution pending Phase 2 transcription .

The empirical content of the canonicality claim is testable. If features are canonical, then (i) a wider SAE on the same activations should not contain features absent from a sufficiently wide narrower SAE — it should refine, not enlarge, the inventory; and (ii) a feature in a wide SAE should not decompose into a sparse combination of features from a narrower SAE — it should be a single unit. The two techniques of Leask et al. (2025) test exactly these two predictions, and report failure on both.

8.2 Meta-SAEs: large-SAE features are not atomic

The first technique trains an SAE on the *decoder matrix* of a target SAE rather than on model activations. Each row of the meta-SAE's input is a single decoder column $\mathbf{d}_i \in \mathbb{R}^n$ from the target SAE; the meta-SAE learns a sparse dictionary of *meta-latents* into which the target's feature directions decompose. The reported finding[47] is the now-canonical example:

> *[Figure 1] Decomposition of an SAE latent representing "Einstein" into a set of interpretable meta-latents. . . . revealing how a complex concept can be represented as a sparse combination of meta-latents. (Leask et al., 2025, §1)*

The Einstein latent in the target SAE decomposes into meta-latents corresponding to "scientist", "Germany", "famous person", and a "starts with E-" feature. Atomicity — the claim that the Einstein direction is irreducible within the SAE's feature family — fails on inspection: it is a sparse combination of more elementary directions discovered without looking at any new model activations, just at the target SAE's own decoder. The summary characterization is direct[48]:

[46] KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#sec-1-canonical`.
[47] KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#sec-1-einstein`.
[48] KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#abstract`.

> *Using meta-SAEs — SAEs trained on the decoder matrix of another SAE — we find that latents in SAEs often decompose into combinations of latents from a smaller SAE, showing that larger SAE latents are not atomic.* (Leask et al., 2025, abstract)

The decoder-matrix substrate is the load-bearing methodological choice: meta-SAEs do not introduce new activation data, so the decomposition of $\mathbf{d}_{\text{Einstein}}$ into meta-latent directions is a property of the target SAE's learned geometry alone. Whatever "Einstein" meant inside the target SAE, it meant it as a sparse combination of more-elementary directions in $\mathbb{R}^n$ — and the meta-SAE recovers them.

8.3 SAE stitching: smaller SAEs are incomplete

The second technique tests completeness by inserting or swapping latents from a larger SAE into a smaller one and measuring reconstruction performance on held-out activations. The paper's classification[49] sorts larger-SAE latents into two disjoint sets:

- **Reconstruction latents.** When inserted into the smaller SAE, they can replace — with comparable behavior — one or more latents already present. They are coordinate refinements of features the smaller SAE already had.
- **Novel latents.** When inserted, they *improve* the smaller SAE's reconstruction. They are features the smaller SAE did not have; absent in its dictionary; inaccessible to its forward pass.

The empirical finding is that novel latents exist in non-trivial numbers: scaling M does not just refine, it discovers. The methodological consequence (Leask et al., 2025, §1) is that any single-width SAE is provably incomplete relative to a sufficiently wider one — and since the field has no a-priori stopping criterion for M, the "complete dictionary" premise of the canonicality claim has no operational test that any finite SAE can pass. The contrast with Bricken et al. (2023)'s earlier *feature-splitting* framing is sharp: Leask et al. note that under feature splitting alone, "latents from smaller SAEs 'split' into multiple, more fine-grained latents in larger SAEs" (Leask et al., 2025, §1)[50] — a relationship in which the smaller dictionary is *coarser* but still complete. Stitching's novel latents fail this picture: some larger-SAE features have no coarse-grained counterpart in the smaller dictionary at all.

Contradiction

Are SAE features canonical? Two lines of evidence strain the strong-form "true features" framing implicit in Bricken et al. (2023) and Templeton et al. (2024). (1) Non-atomicity. Meta-SAEs trained on the decoder matrix $\mathbf{W}_{\text{dec}}$ of a target SAE recover a sparse decomposition of the target's feature directions; an "Einstein" latent decomposes into "scientist", "Germany", "famous person" (Leask et al., 2025, §1). The target's atoms are not atoms. (2) Incompleteness. SAE stitching shows that larger SAEs contain *novel latents* — features absent from smaller SAEs that improve reconstruction when transplanted in (Leask et al., 2025, abstract). No fixed-width SAE qualifies as complete relative to a sufficiently wider trained counterpart. **What evidence would close it.** A basis-finding criterion under which two independent SAE training runs — at different widths M_1, M_2, different sparsity targets, different random seeds, and on the same activations — converge on the same feature inventory modulo a

[49] KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#abstract`.
[50] KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#sec-1-feature-splitting`.

known equivalence (rotation, sign, permutation, scale). Until that criterion is exhibited and shown to hold empirically, the field's working assumption is the pragmatic one Leask et al. (2025, §1) themselves recommend[a]: "the choice of dictionary size is subjective. We suggest taking a pragmatic approach to applying SAEs to mechanistic interpretability tasks, trying SAEs of several widths to see which is best suited." (Leask et al., 2025, §1)

[a]KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#sec-1-still-useful`.

8.4 Transcoders: function approximation, not activation decomposition

The architectural cousin of an SAE relevant to circuit analysis — introduced in §7.5 and reused as the primitive of §11 — is the *transcoder* (Dunefsky et al., 2024). We restate the loss for reference, because the canonicality picture rides on its faithfulness term being defined against the MLP *function* rather than against its activation[51]:

$$\mathcal{L}_{\mathrm{TC}}(\mathbf{x}) \;=\; \underbrace{\|\mathrm{MLP}(\mathbf{x}) - \mathrm{TC}(\mathbf{x})\|_2^2}_{\text{faithfulness}} + \lambda_1 \, \|\mathbf{z}_{\mathrm{TC}}(\mathbf{x})\|_1 \tag{31}$$

with $\mathrm{TC}(\mathbf{x}) = \mathbf{W}_{\mathrm{dec}}\,\mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\mathbf{x} + \mathbf{b}_{\mathrm{enc}}) + \mathbf{b}_{\mathrm{dec}}$ from Dunefsky et al. (2024, §3.1, Eq. 3–5). The contrast with an SAE on the MLP output is one substitution: an SAE minimizes $\|\mathrm{MLP}(\mathbf{x}) - \hat{\mathbf{x}}\|_2^2$ where $\hat{\mathbf{x}}$ depends on the *post*-MLP activation; a transcoder's $\mathrm{TC}(\mathbf{x})$ depends only on the *pre*-MLP input $\mathbf{x}$. The transcoder's feature dictionary is a decomposition of the MLP *function*, not of any particular output (Dunefsky et al., 2024, §1)[52].

The methodological payoff (Dunefsky et al., 2024, §1)[53] is the **input-invariance factorization**. The connection from a pre-MLP feature f_i (associated with decoder column $\mathbf{d}_i$) to a post-MLP transcoder feature f_j (associated with encoder row $\mathbf{e}_j$) factors into an input-dependent term — the activations $f_i(\mathbf{x})$ and $f_j(\mathbf{x})$ at the input of interest — and an input-invariant term, the bilinear product

$$W_{ji} \;:=\; \mathbf{e}_j^\top \mathbf{d}_i, \tag{32}$$

which holds for any input. The motivating example Dunefsky et al. (2024, §1) give for why this matters is worth quoting in full:

> *SAEs cannot tell us about the general input-output behavior of an MLP across all inputs. . . . Doing e.g. patching on one input shows that a pre-MLP feature for Polish last names is important for causing the post-MLP feature to activate. But on other inputs, would features other than the Polish last name feature also cause the post-MLP feature to fire (e.g. an English last names feature)? Could there be other inputs where the Polish last names feature fires but the post-MLP feature does not? We can see that without input-invariance, it is difficult to make general claims about model behavior. (Dunefsky et al., 2024, §1)*

An SAE-on-output answers "which features explain *this* activation?"; a transcoder answers "which features does this MLP *compute*, and how do they depend on its inputs?". The latter is the question circuit analysis needs answered, and the cross-layer transcoder of Lindsey et al. (2025b) generalizes the bilinear factorization (32) to skip-layer feature flow. That generalization is the architectural primitive on which the attribution-graph construction of §11.4 is built.

[51]KB excerpt: `kb/excerpts/dunefsky2024-transcoders.md#sec-3-1`.
[52]KB excerpt: `kb/excerpts/dunefsky2024-transcoders.md#sec-1-comparison`.
[53]KB excerpt: `kb/excerpts/dunefsky2024-transcoders.md#sec-1-input-invariance`.

Intuition

A transcoder is to an MLP what a sparse Taylor expansion is to a smooth function: the original is a dense, hard-to-read black box; the transcoder is a sparse linear combination of basis functions $\{\mathbf{d}_i\}$ that approximates it everywhere. The input-invariance factorization is exactly the statement that "coefficients of the expansion" (the bilinear products $\mathbf{e}_j^\top \mathbf{d}_i$) are properties of the function, while "where the expansion is being evaluated" (the activations $f_i(\mathbf{x}), f_j(\mathbf{x})$) are properties of the input. Returning to canonical math: the Jacobian of a post-transcoder feature f_j with respect to a pre-transcoder feature f_i separates into a bilinear weight term and a pointwise activation derivative — the structural condition that makes attribution graphs computable in closed form rather than by sampling.

8.5 Inheritance: transcoders carry the SAE canonicality problem

The architectural difference between Eq. (31) and the SAE loss of Eq. (24) does not, on its own, resolve the canonicality concerns of §§8.2–8.3. A transcoder is still a sparse-dictionary architecture; its feature directions $\{\mathbf{d}_i\}$ are still trained against a reconstruction objective with a sparsity penalty; nothing in Eq. (31) privileges any particular basis modulo the rotational/scale freedoms of dictionary learning. Two independent transcoders trained on the same MLP at different widths d_{features} should be expected — by the same arguments Leask et al. (2025) make for SAEs — to exhibit incompleteness (the wider transcoder finds function-features the narrower one missed) and non-atomicity (a wider feature $\mathbf{d}_i^{\text{wide}}$ admits a sparse decomposition into function-features of a narrower transcoder). The Leask et al. (2025) stitching protocol applied between transcoders rather than between SAEs has not, as of 2026, been published as a separate experiment, and we flag the inheritance as empirically unconfirmed but theoretically expected .

The pragmatic consequence for circuit work is real. The bilinear factorization of Eq. (32) is well-defined *within* a fixed transcoder: given the $(\mathbf{W}_{\text{enc}}, \mathbf{W}_{\text{dec}})$ trained for a specific MLP, the input-invariant attribution weights W_{ji} are deterministic. But the *interpretation* of the post-transcoder feature f_j — what it "represents" — is identified at the basis level, and the basis is not canonical. Two attribution graphs computed against two different transcoders trained on the same MLP will produce different node sets, different edge weights, and potentially different qualitative narratives about the underlying circuit. This is the *transcoder-canonicality* concern inherited from the SAE setting, and the cross-method synthesis of §12 returns to it: when activation patching at the head/MLP level and attribution-graph tracing at the transcoder-feature level disagree, one possibility is that the patching result is wrong; another is that the attribution graph reads features as canonical that the model does not treat as canonical.

8.6 Where the canonicality question stands

The 2026 community position, as registered by Leask et al. (2025, §1) and consistent with Anthropic's own move from "true features" to circuit-tracing's pragmatic transcoder usage , is *pragmatic*: SAEs and transcoders are useful, basis-dependent tools for converting polysemantic activations into legible sparse representations, and the right working response to the canonicality critique is to vary M, retrain, and check that load-bearing interpretability claims survive across widths and seeds. This is the recommendation Leask et al. themselves make: "trying SAEs of several widths to see which is best suited" (Leask et al., 2025, §1).

The *theoretical* question — whether canonical units exist at all, and whether some other dictionary-learning architecture (one explicitly designed to converge across widths) could find them

— is open. The shape of an answer would be: a basis-finding criterion that two independent runs satisfy modulo a known equivalence, and an architecture for which the criterion is provably (or at least robustly empirically) achievable. Neither is in hand as of 2026.

8.7 Forward link

The transcoder framing of §8.4 is the architectural primitive of §11, where Lindsey et al. (2025b)'s cross-layer transcoders provide the bilinear-attribution substrate (32) on which Anthropic's 2025 attribution graphs on Claude 3.5 Haiku are built . The canonicality concern propagates forward: any circuit claim made against a transcoder-derived attribution graph is a claim about *this* transcoder's features, not about a model-level canonical basis. §12 takes up the cross-method synthesis question — when two interpretability methods agree on a phenomenon (the IOI circuit, factual recall localization), and when they disagree (the truth direction across datasets, SAE feature inventories across widths) — and uses the canonicality picture from this section to explain why some disagreements are method-internal artifacts (one method reads a basis as canonical that another method does not) rather than disagreements about the underlying model computation.

9 Activation patching

The lens and probe of §3–§4 read the residual stream; they tell us what intermediate states *contain*. They do not tell us what the model *uses*. A high-accuracy probe at layer ℓ is consistent with the property being computed at ℓ and consumed by the unembed, with the property being computed at ℓ and ignored by the unembed (*vestigial information*), or with a confound entirely outside the model (Belinkov, 2022). Closing that gap requires an *intervention*: change the activation at ℓ, hold everything else fixed, observe the output. Activation patching is the canonical form of that intervention. The three-run protocol of Meng et al. (2022), generalized and named by Zhang and Nanda (2023), gives the field its workhorse causal estimator and the heatmap that ROME made famous; the variant zoo of path patching (Wang et al., 2022), attribution patching, and ACDC (Conmy et al., 2023) are all refinements of the same protocol along three methodological axes that this section names explicitly.

9.1 The three-run protocol

Meng et al. (2022) fix the protocol as three forward passes[54], restated by Zhang and Nanda (2023, §2.1) as the now-standard recipe[55]. Fix a clean prompt X_{cl} with target answer r (e.g., "*The Eiffel Tower is in*" → "*Paris*") and a corrupted prompt X_* that changes the answer (e.g., "*The Colosseum is in*" → "*Rome*"). Let $\mathbf{h}_i^{(\ell)}$ denote the model's residual-stream activation at sequence position i and layer ℓ, and C a candidate *component* indexing some collection of these activations (a single hidden state, an MLP output, an attention-head output, or a subspace; see §9.4).

1. **Clean run.** Run $\mathcal{M}$ on X_{cl} and cache every internal activation $\{\mathbf{h}_i^{(\ell)}\}_{i,\ell}$. Record the model's output distribution $\mathbb{P}_{\text{cl}}$.

2. **Corrupted run.** Run $\mathcal{M}$ on X_* and record the corrupted output distribution $\mathbb{P}_*$ and a target metric m_* (probability, logit difference, or KL — see §9.2).

[54] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2`.
[55] KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-2-1`.

3. **Patched run.** Run $\mathcal{M}$ on X_*, but at component C overwrite the activation with the cached clean value from step (1). Record the patched output distribution $\mathbb{P}_{\rm pt}$ and the metric $m_{\rm pt}$.

The *indirect effect* of component C on this prompt pair is the metric difference between the patched and corrupted runs (Meng et al., 2022, §2)[56]:

$$\mathrm{IE}_C(X_{\rm cl}, X_*) \;:=\; m_{\rm pt} \;-\; m_*. \tag{33}$$

Averaging over a dataset $\{(X_{\rm cl}^{(n)}, X_*^{(n)})\}_{n=1}^N$ of paired counterfactuals gives the *average indirect effect* (Meng et al., 2022, §2),

$$\mathrm{AIE}_C \;:=\; \frac{1}{N}\sum_{n=1}^{N} \mathrm{IE}_C(X_{\rm cl}^{(n)}, X_*^{(n)}), \tag{34}$$

which is the statistic the field actually reports: a single IE_C on one prompt is noisy, AIE_C on a counterfactual benchmark is the population-level localization signal. A large AIE_C means substituting C's clean activation back into the corrupted run *recovers* the clean behavior, so C is a sufficient causal mediator for the metric on this task.

Iterating C over every (component, token, layer) cell of the residual-stream grid produces the *causal trace heatmap* of Meng et al. (2022, Fig. 1): rows are token positions, columns are layers, cells are AIE_C. Bright cells localize the behavior; dark cells are causally inert. The heatmap is the visual signature of patching, and the substantive output of §10 is what its peaks revealed for factual recall.

Tensor-shape ledger. A residual-stream activation at one (B, S) position is $\mathbf{h}^{(\ell,t)} \in \mathbb{R}^D$; the batched tensor is $\mathsf{B}\times\mathsf{S}\times\mathsf{D}$. A component C identifies a subset of this grid: "MLP output at (ℓ, t)" is one $\mathbb{R}^D$ vector; "attention-head h output at (ℓ, t)" is one $\mathbb{R}^{d_h}$ vector before the W^O projection; "head h contribution to the residual stream" is the $\mathbb{R}^D$ vector after W^O. A subspace patch replaces only the projection $\mathbf{h}^{(\ell,t)} \mapsto \mathbf{h}^{(\ell,t)} - (\mathbf{h}^{(\ell,t)})^\top \hat{\mathbf{d}}\,\hat{\mathbf{d}} + (\mathbf{h}_{\rm cl}^{(\ell,t)})^\top \hat{\mathbf{d}}\,\hat{\mathbf{d}}$ along a probe direction $\hat{\mathbf{d}}$, leaving the orthogonal complement unchanged — the natural granularity for "patch the truth direction" or "patch the Paris feature."

9.2 The metric: logit difference is the default

The metric m in Eq. (33) is a free choice with real consequences. Zhang and Nanda (2023, §2.1) catalog three options[57]. Probability, $m^{\rm P} = \mathbb{P}(r)$, gives a unitless patching effect $\mathbb{P}_{\rm pt}(r) - \mathbb{P}_*(r)$; it is the metric of Meng et al. (2022). Logit difference, with r' the corrupted-prompt answer, is

$$\mathrm{LD}(r, r') \;:=\; \mathrm{logit}(r) - \mathrm{logit}(r'), \tag{35}$$

and the canonical normalized patching effect is (Zhang and Nanda, 2023, §2.1)

$$m_{\rm pt}^{\rm LD} \;:=\; \frac{\mathrm{LD}_{\rm pt}(r, r') - \mathrm{LD}_*(r, r')}{\mathrm{LD}_{\rm cl}(r, r') - \mathrm{LD}_*(r, r')}, \tag{36}$$

typically in $[0, 1]$ (where 1 is fully restored, 0 is corrupted). KL is $D_{\rm KL}(\mathbb{P}_{\rm cl} \,\|\, \mathbb{P}_*) - D_{\rm KL}(\mathbb{P}_{\rm cl} \,\|\, \mathbb{P}_{\rm pt})$.

The methodological case for logit difference as the default is made by Heimersheim and Nanda (2024, §4.1)[58]: it is a "mostly linear function of the residual stream (unlike probability-based metrics) which makes it easy to directly attribute logit difference to individual components," and it is "a 'softer' metric, allowing us to see partial effects on the model even if they don't change the rank of the output tokens." Heimersheim and Nanda (2024, §4.2) catalog the failure modes of the alternatives[59]:

[56] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-effects`.
[57] KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-2-1-metrics`.
[58] KB excerpt: `kb/excerpts/heimersheim2024.md#sec-4-1`.
[59] KB excerpt: `kb/excerpts/heimersheim2024.md#sec-4-2`.

probability is "non-linear, in the sense that [it] track[s] the logits exponentially. For example, a model component adding +2 to a given logit can create a 1 or 40 percentage point probability increase, just depending on what the baseline was"; log-probabilities saturate ("once the correct answer becomes the model's top guess, the logprob stops increasing meaningfully"); rank- and accuracy-based metrics "round off each input to a discrete metric." The single most important consequence of choosing probability over logit difference is invisibility of negative components: the IOI Negative Name Mover heads (Wang et al., 2022, §3)[60] push *against* the correct answer, and a probability metric near $\mathbb{P}(r) \in \{0, 1\}$ saturates exactly the regime in which their effect is detectable. Logit difference does not.

9.3 The corruption: STR over GN

The second methodological axis is how X_* is generated from X_{cl}. Zhang and Nanda (2023, §2.1) fix two canonical choices[61]. **Gaussian Noising** (GN), used by Meng et al. (2022), "adds Gaussian noise $\mathcal{N}(0, \nu)$ to the embeddings of certain key tokens, where ν is 3 times the standard deviation of the token embeddings from the textset." **Symmetric Token Replacement** (STR), used by Wang et al. (2022), "replaces the key tokens by similar ones with equal sequence length"; "Eiffel Tower" becomes "Colosseum", "Mary" becomes "Alice". STR's defining property is that "X_{corrupt} is identically distributed as a fresh draw of a clean prompt" (Zhang and Nanda, 2023, §2.1).

Contradiction

GN can mis-localize relative to STR. Zhang and Nanda (2023, §1) report that "GN and STR can lead to inconsistent localization and circuit discovery outcomes (Section 3.1)" on the same task. The conjectured mechanism: "GN breaks model's internal mechanisms by putting it off distribution"[a] — in-distribution prompts that the model knows how to process produce clean signals about which components matter; out-of-distribution noise produces signals about which components fail under noise, which need not coincide. Zhang and Nanda (2023, §1) "advocate for STR, as it supplies in-distribution corrupted prompts that help to preserve consistent model behavior." Concrete consequence: Meng et al. (2022)'s GN-derived AIE peak at GPT-2-XL layer 15 broadly survives STR re-validation (Zhang and Nanda, 2023, §3), but the precise per-layer peak shifts, and downstream uses of ROME's exact layer-of-edit recommendation inherit that sensitivity. The strongest form of the methodology claim — "GN-derived localizations should be re-validated under STR before being treated as canonical for a new task" — is the field's current default.

[a] KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-findings`.

A practical refinement, due to Heimersheim and Nanda (2024, §2.3), is the *denoising vs. noising* distinction[62]. The protocol above is *denoising* (clean → corrupted): the patched run inserts a clean activation into a corrupted run and asks *which patches were sufficient to restore the behavior*. The inverse protocol, *noising* (corrupted → clean), inserts a corrupted activation into a clean run and asks *which patches were necessary to maintain the behavior*. Heimersheim and Nanda (2024, §2.4) show with a worked AND/OR-gate example that the two directions are not symmetric: an AND circuit ($C = A \wedge B$) has both A and B as necessary, so noising flags both; denoising of either alone leaves the other at the corrupted baseline and finds nothing. An OR circuit ($C = A \vee B$) has either alone as sufficient, so denoising of either restores behavior; noising of either alone leaves the other

[60] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-3-circuit`.
[61] KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-2-1-corruption`.
[62] KB excerpt: `kb/excerpts/heimersheim2024.md#sec-2-3`.

operative. The practical recommendation is to choose direction by the suspected circuit topology, or to run both.

9.4 Granularity: from hidden states to subspaces

The third axis is what counts as C. The granularities used in the literature (Zhang and Nanda, 2023, §2.1) form a coarse-to-fine ladder, summarized in Table 3.

The headline trade-off: granularity governs the strength of the claim the experiment can support. Hidden-state patching localizes *layers* and *token positions*; MLP-vs-attention patching splits the contribution by sublayer; head-level patching localizes specific attention heads and is the granularity needed for circuit claims; subspace patching localizes specific representational directions. Coarse patching that finds "MLPs at layer 15 matter" does not yet justify "the `Paris` direction in those MLPs matters," even if both are true — the latter requires either finer-grained patching at the subspace level or a downstream probe-on-patched-states design (Zhang and Nanda, 2023, §2.3).

9.5 Path patching: isolating direct effects

Standard activation patching gives the *total* effect of a component on the metric: any influence routed through C is captured, including effects mediated by downstream components. Circuits with multiple parallel paths — which is most circuits — need a finer estimate. Wang et al. (2022, §3.1)'s *path patching*[63] is the partial-derivative analogue: it isolates the direct effect of a head h on a downstream component R by patching only along the residual and MLP paths from h to R, holding indirect routes through other attention heads at their corrupted values. Verbatim from Wang et al. (2022, §3.1):

> Given inputs x_{orig} and x_{new}, and a set of paths $\mathcal{P}$ emanating from a node h, path patching runs a forward pass on x_{orig}, but for the paths in $\mathcal{P}$ it replaces the activations for h with those from x_{new}. In our case, h will be a fixed attention head and $\mathcal{P}$ consists of all direct paths from h to a set of components R, i.e. paths through residual connections and MLPs (but not through other attention heads); this measures the counterfactual effect of h on the members of R.

The methodological gain is exactly what makes the seven-class IOI decomposition (§11) legible: full activation patching attributes "Name Mover Heads matter," but only path patching can attribute "S-Inhibition Heads write into the queries of the Name Mover Heads," i.e., a directed effect $h \to R$ via a specific projection. Without path patching the IOI circuit collapses into a flat ranked list of important heads; with it, it is a wiring diagram.

Intuition

Activation patching is counterfactual surgery on the residual-stream graph. The clean run is the patient who knows the answer; the corrupted run is the patient who lost it; the patched run grafts one specific tissue from the clean patient into the corrupted patient. If the corrupted patient recovers, that tissue carried the answer. Returning to canonical math: the recovery measure is exactly $\text{IE}_C = m_{\text{pt}} - m_*$ from Eq. (33), with C the grafted tissue and m a behaviorally meaningful metric (logit difference, by default). Path patching extends the surgery to grafting along specific *conduits* between tissues rather than the tissues themselves; the canonical math is the same Eq. (33) restricted to a path set $\mathcal{P}$.

[63] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-3-1-path-patching`.

9.6 Attribution patching: the linearization

The cost of full activation patching is one forward pass per component: sweeping every (component, token, layer) cell of the heatmap costs $O(N_{\text{components}})$ forward passes per (clean, corrupted) pair, and path patching at depth-k costs $O(N^k_{\text{components}})$. On GPT-2 small this is feasible; on production-scale models it is not. *Attribution patching* replaces the substitution by its first-order Taylor approximation: write $\Delta\mathbf{h}_C := \mathbf{h}_{C,\text{cl}} - \mathbf{h}_{C,*}$ for the clean-minus-corrupted activation difference at component C, and approximate the indirect effect of patching C as

$$\text{IE}_C \;\approx\; \langle \nabla_{\mathbf{h}_C} m \,\big|_{\mathbf{h}_C=\mathbf{h}_{C,*}},\; \Delta\mathbf{h}_C \rangle, \tag{37}$$

where the gradient is taken at the corrupted activation. The cost is one forward and one backward pass on the corrupted prompt to obtain $\nabla_{\mathbf{h}} m$ at *every* component simultaneously, plus one cached clean forward pass for $\Delta\mathbf{h}$; this collapses $O(N_{\text{components}})$ patch-forward-passes into $O(1)$ forward+backward passes, an asymptotic win that scales the technique to production models.

Forum signal

The attribution-patching approximation in Eq. (37) was popularized by Nanda's "Attribution Patching: Activation Patching at Industrial Scale" LessWrong post (2023), which is the canonical reference for the technique . Subsequent peer-reviewed venues (notably Conmy et al. 2023 and the Anthropic attribution-graph line of §11) build on the linearization, but the original write-up is a forum post and should be flagged as such. Hard claims about attribution-patching's *accuracy* relative to full patching back to Zhang and Nanda (2023, §1), who note the approximation "is known to fail in non-linear regions"[a] — attention-softmax saturation and ReLU/GeLU dead zones being the canonical failure cases.

[a]KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-1`.

The mediation-analysis lineage of activation patching, predating its mech-interp adoption, traces to causal-mediation work in NLP . Meng et al. (2022, §1) cite this lineage explicitly when introducing the three-run causal-tracing protocol; Zhang and Nanda (2023, §1) list "causal mediation analysis" as one of the synonyms under which activation patching appears in the literature. The genealogy matters because it locates patching in a pre-existing statistical-causal vocabulary (total/indirect effect, mediation, intervention) that the mech-interp community then specialized to neural-network computational graphs.

9.7 Recommendations and pitfalls

The methodology of Zhang and Nanda (2023) and Heimersheim and Nanda (2024) converges on a default recipe. Table 4 summarizes.

Three pitfalls deserve naming, lifted from Heimersheim and Nanda (2024, §4.2) and Zhang and Nanda (2023, §2.1). *Logit-difference false-positives*: because logit difference is a difference, "it may be driven by either getting better at the correct answer or worse at the incorrect answer"[64] — a patch that indiscriminately damages the model toward uniform output reduces the incorrect-answer logit and inflates the apparent patching effect, without localizing anything specific. The fix is sanity-checking against an absolute-probability metric and against unrelated control metrics, not abandoning logit difference. *Probability saturation*: when $\mathbb{P}(r) \in \{0,1\}$ at either endpoint, the

[64]KB excerpt: `kb/excerpts/heimersheim2024.md#sec-4-2`.

patching effect on probability is artificially compressed; logit difference is the explicit fix. *Coarse-granularity sleights of hand*: a heatmap showing AIE concentrated at "MLP layer 15, last subject token" is consistent with the model storing the fact in that MLP, with the model storing the fact upstream and reading it at that MLP, with the model storing the fact in a feature subspace within that MLP, and with the model storing the fact partly there and partly in a redundant location whose AIE is masked by backup (Wang et al., 2022, §3). Each of these claims requires a finer-grained or causally complementary experiment to distinguish.

9.8 Forward link

Activation patching is the method that turns the lens-and-probe correlational reading of the residual stream (§3–§4) into causal claims the field can adjudicate. The next two sections deploy it. Section 10 follows the heatmap of Meng et al. (2022, Fig. 1) from a localization claim ("mid-layer MLPs at the last subject token mediate factual recall") to a weight-level intervention (the rank-one MLP edit) that turns patching's diagnostic output into capability surgery; this is the bridge from interpretability-method to model-editing methodology. Section 11 consumes patching as one of two methods that produce the IOI circuit and the ACDC subgraph: Wang et al.'s manual path-patching reverse-engineering and Conmy et al.'s KL-thresholded automated edge-pruning are both refinements of the three-run protocol, and the cross-method evidence with SAE features and the tuned lens (§12) is what gives the IOI picture its load-bearing status. Activation patching is, in this sense, the spine: every causal interpretability claim downstream of this section either uses it directly or refines it.

10 ROME and direct edits

The activation-patching protocol of §9 is a diagnostic: its output is a heatmap of which (component, token, layer) cells are causal mediators for a behavior. ROME (Meng et al., 2022) is the first paper to take that diagnostic and turn it into *capability surgery* on the model's weights. The argument has two halves. The first half *localizes* factual recall in GPT-2 XL with the three-run protocol of §9.1 and reports a sharp pair of peaks: an early site at mid-layer MLPs at the last subject token, and a late site at the final-layer attention. The second half *intervenes*: treating the localized MLP as a linear associative memory, ROME computes a closed-form rank-one weight update that inserts a new (s, r, o) triple — "the Eiffel Tower is in Paris" becomes "the Eiffel Tower is in Rome" — with 99.8% efficacy and 88% paraphrase robustness on the zsRE editing benchmark. The methodological move is to use causal interpretability to identify the locus, then weight-level surgery to validate the localization by *constructing* the predicted edit. The follow-up critique (, Hase et al. 2023, "Does Localization Inform Editing," forthcoming) complicates that move: the layers ROME's tracing identifies are not always the layers where editing performs best.

10.1 Causal tracing localizes factual recall

Meng et al. (2022) apply the three-run protocol of §9.1 to factual prompts on GPT-2 XL, sweeping the patched component C over every (token, layer) cell of the residual-stream grid and computing the average indirect effect AIE_C from Eq. (34)[65]. On 1000 factual statements, two peaks dominate the heatmap[66]: a *late site* near the final-layer prediction (unsurprising; the model commits to the answer there), and an *early site* "at the last token of the subject name" (Meng et al., 2022, §2.2), which the authors flag as "a new discovery." Decomposing the early-site cells into sublayer

[65] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-effects`.

[66] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-2`.

contributions sharpens the picture further: at the last-subject-token early site, $\mathrm{AIE}_{\mathrm{MLP}} = 6.6\%$ versus $\mathrm{AIE}_{\mathrm{attn}} = 1.6\%$, with the AIE peak across hidden states at 8.7% at layer 15 (Meng et al., 2022, §2.2). The hidden-state grid's strongest mediators are MLPs in the middle of the stack, processing the last subject token, well before any attention head has copied the answer forward.

That localization motivates the paper's central conjecture, the *Localized Factual Association Hypothesis* (Meng et al., 2022, §2.3)[67]:

> *each midlayer MLP module accepts inputs that encode a subject, then produces outputs that recall memorized properties about that subject. Middle layer MLP outputs accumulate information, then the summed information is copied to the last token by attention at high layers.*

The hypothesis localizes factual recall along three axes simultaneously (Meng et al., 2022, §2.3): (i) in the MLP sublayers, not the attention sublayers; (ii) at specific middle layers; (iii) at the processing of the subject's last token. It is the strongest-form factual-localization claim the field has fielded, and the second half of the paper validates it by intervention.

10.2 The associative-memory view of MLPs

ROME's intervention rests on a specific reading of the MLP sublayer (Meng et al., 2022, §3.1)[68]: the second linear layer of an MLP, $W_{\mathrm{proj}}^{(\ell)}$, is a *linear associative memory* in the sense of Kohonen (1972) and Anderson 1972. Any linear map W can act as a key–value store for keys $K = [k_1 \mid k_2 \mid \ldots]$ and values $V = [v_1 \mid v_2 \mid \ldots]$ by solving $WK \approx V$, with the Moore–Penrose pseudoinverse minimizing squared error: $\hat{W} = VK^{+}$. In the MLP context, k is the gated activation produced by the first linear layer (so k is the MLP's encoded query for "what subject is this"), and $v = W_{\mathrm{proj}}^{(\ell)} k$ is the MLP output written into the residual stream (so v is the recalled information about the subject).

Inserting a new association (k_*, v_*) then has a closed-form solution. Constrain the new map $\tilde{W}$ to satisfy $\tilde{W}k_* = v_*$ exactly while leaving other key–value pairs minimally perturbed. The closed-form rank-one update (Meng et al., 2022, §3.1, Eq. 2) is

$$\min_{\tilde{W}} \|\tilde{W}K - V\| \quad \text{such that} \quad \tilde{W}k_* = v_*, \quad \hat{W} = W + \Lambda(C^{-1}k_*)^{\top}, \tag{38}$$

where $C = KK^{\top}$ is the uncentered covariance of keys (pre-cached on a Wikipedia sample), and $\Lambda = (v_* - Wk_*)/(C^{-1}k_*)^{\top}k_*$ is a vector proportional to the residual error of the new pair on the original memory (Meng et al., 2022, §3.1). The update is rank one because $\Lambda(C^{-1}k_*)^{\top}$ is the outer product of two vectors; it is closed-form because C is a fixed data-side statistic and Λ is determined by the constraint. Inserting any new fact reduces to (i) computing the key k_* for the target subject by running the model up to the chosen MLP layer, (ii) computing the value v_* that would produce the target object o at the unembed by gradient-descending against the LM loss with the rest of the model frozen, and (iii) applying Eq. (38).

Intuition

The associative-memory view says: the MLP at the localized layer is running the matrix multiply $v = Wk$ as a literal lookup, with the key k being the model's internal representation of the subject and the value v being the residual-stream contribution that pushes the unembed toward the correct object. ROME's edit replaces one row of the lookup table without touching

[67] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-3`.
[68] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-3-1`.

the rest. Returning to canonical math: the constraint $\tilde{W}k_* = v_*$ in Eq. (38) *is* "write a new row," and the rank-one structure $\Lambda(C^{-1}k_*)^\top$ is the cheapest perturbation to W that satisfies that constraint without smashing the other rows — the C^{-1} term reweights the update by which directions in key-space are already crowded with existing associations, so the perturbation prefers under-used directions.

10.3 Empirical headline: editing zsRE

The closed-form edit of Eq. (38), applied at GPT-2 XL's localized layer $\ell^* = 15$ at the last subject token, achieves the editing results in Table 5 (Meng et al., 2022, §3.2)[69].

The methodological reading is what matters. ROME beats fine-tuning at both efficacy (99.8 vs. 99.6) and paraphrase robustness (88.1 vs. 82.1) while preserving specificity, with no gradient steps on the broader model and no held-out training data — only one rank-one matrix update at the localized layer. That a closed-form, parameter-frugal edit at the layer the patching diagnostic flagged beats general-purpose fine-tuning is the substantive validation of the Localized Factual Association Hypothesis: if the fact is stored as a key–value pair at MLP layer 15, you should be able to rewrite it by rewriting that pair, and you can.

10.4 Hase 2023: locating is not editing-target

The headline that "causal tracing localizes the layer to edit" collapses in a more careful follow-up. *Does Localization Inform Editing* runs ROME-style edits at every layer of GPT-J and GPT-2 XL, not only at the layer the tracing diagnostic peaks at, and reports that the editing performance is *not monotonic* in the AIE peak: layers with substantially lower $\mathrm{AIE}_{\mathrm{MLP}}$ frequently match or exceed the editing performance at the tracing-localized layer, and the layer of best edit is task-dependent in ways the tracing heatmap does not predict. The methodological consequence is that *causal tracing identifies layers that are causal mediators of the behavior*, but *editing exploits a different property of the layer* — something closer to "which layer's MLP, when perturbed, produces a residual-stream change that the rest of the network propagates faithfully to the unembed for that fact." The two need not coincide, and on Hase et al.'s benchmarks they often don't.

This is a falsification of the strong-form coupling claim — causal locus equals best edit-target — but not of either half taken alone. The localization claim survives: the tracing peak at GPT-2-XL layer 15 at the last subject token is robust, and we discuss its survival under STR re-validation in §10. The editing claim survives: rank-one edits at *some* layer near the localized band insert facts at 99.8% efficacy. What does not survive is the prescriptive coupling: "compute the AIE heatmap, edit at the peak." A more careful protocol selects the editing layer by end-to-end editing performance on a held-out set, with the tracing heatmap as a strong prior rather than a determinant.

Contradiction

The ROME locus is robust; the layer-of-edit recommendation is corruption-sensitive. Two distinct claims travel together in the ROME paper. The *locus* claim — "mid-layer MLPs at the last subject token are causal mediators of factual recall in GPT-2 XL" — survives STR re-validation: Zhang and Nanda (2023, §3)[a] report that GN- and STR-derived heatmaps disagree on per-layer details but agree on the broad early-site / late-site decomposition that the hypothesis (Meng et al., 2022, §2.3) rests on. The *layer-of-edit*

[69] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-3-2`.

claim — "edit at exactly the AIE peak, layer 15, for GPT-2 XL" — is more fragile. Zhang and Nanda (2023, §1) document that GN-derived per-layer peaks shift under STR, and document that editing performance does not track AIE peaks across layers. The strongest defensible form of ROME's prescription, after both critiques, is: *causal tracing identifies a layer band where factual recall is mediated; within that band, layer-of-edit should be selected on a held-out editing benchmark, not by AIE peak alone.* The locus is robust; the prescription is methodology-sensitive. Returning to canonical math: the AIE statistic of Eq. (34) is the correct estimator for "which components mediate the behavior" (Zhang and Nanda, 2023, §2.1), but it is not, by itself, an estimator for "which components, when perturbed by Eq. (38), produce edits that propagate faithfully"—those are different objects and the field's experience shows they decouple.

[a] KB excerpt: `kb/excerpts/zhang2023-apatching.md#sec-findings`.

10.5 Frontier note: ROME at frontier-2026 architectures

The ROME causal-tracing diagnostic and the rank-one edit of Eq. (38) were developed on dense transformer MLPs in GPT-2 / GPT-J. Whether the Localized Factual Association Hypothesis extends to frontier-2026 architectures is partially open. Two architectural shifts complicate the picture. First, MLA-compressed-KV attention (the MLA mechanism of Paper 1 §5; DeepSeek-V2/V3) and grouped-query attention (Llama 3 family) change the attention sublayer's computational structure, but the MLP sublayers remain dense linear maps in those architectures, so the associative-memory framing of §10.2 should carry over by direct analogy. Second, fine-grained mixture-of-experts MLPs (DeepSeek-V3 with 256 routed experts; Llama 4 MoE variants; Qwen 2.5 MoE) replace each MLP sublayer with a routing function over expert modules, each of which is itself a dense linear map. The associative-memory view applies per-expert, but the localization claim splits: a fact is now stored across some subset of experts at the localized layer, indexed by which experts the subject token routes to. The closed-form rank-one edit of Eq. (38) would have to be applied to the expert(s) the localized subject-token routes through, with the routing distribution itself becoming a methodology parameter. : a systematic causal-tracing study of factual recall on Llama 3 / DeepSeek-V3 / Llama 4 with comparable benchmarks to Meng et al. (2022) is, to our knowledge, not yet published; whether the hypothesis survives the MoE / MLA architectural shifts is open empirical.

10.6 Cross-paper thread: editing as an alternative to fine-tuning

ROME's success makes weight-level editing a methodologically distinct tool from fine-tuning for surgical knowledge updates. Paper 2's SFT and preference-RL sections treat gradient-based fine-tuning (SFT, RLHF, DPO) as the standard recipe for installing new behaviors and knowledge; both update many or all parameters under a loss. ROME-style editing is closed-form and updates a rank-one slice of one matrix under a constraint. The trade-off is sharp: fine-tuning generalizes to multi-fact updates and behavioral distributions; ROME is cheap and surgical but, in its 2022 form, restricted to factual (s, r, o) triples and inherits the layer-of-edit fragility of §10.4. LoRA/QLoRA adapter merging (, Paper 2 §parameter-efficient-finetuning) is methodologically adjacent — a rank-k update across many layers, gradient-trained against a loss, where ROME is rank-one at one layer, closed-form against a constraint. A unified axis (gradient-based vs. closed-form; many-layer vs. one-layer; loss-driven vs. constraint-driven) is the cross-paper synthesis; the substantive choice is task-dependent and ROME-vs-fine-tuning at production scale is an open 2026 question.

10.7 Forward link

ROME is the bridge between activation patching's diagnostic output (§9) and the model-editing literature; it is also a case study in the broader cross-method convergence picture (§12). Section 11 takes up the generalization from a single localized component (mid-layer MLPs at the last subject token) to a full circuit subgraph: where ROME's factual-recall picture is essentially component-level (which sublayer, at which token, at which layer), the circuit picture is edge-level (which downstream component reads from which upstream component, along which projection). Section 12 returns to factual recall as a four-method convergence case — patching localizes the MLP; the tuned lens shows the predicted-object probability rising across the localized layer band; SAE features on those MLPs include subject-entity features; ROME's rank-one edit succeeds at the localized layer — treating it as the second canonical *positive* cross-method evidence after the IOI circuit.

11 Circuits

A *circuit* is the strongest unit of mechanistic explanation the field has committed to: a sparse subgraph of the model's computational graph whose nodes (attention heads, MLPs, transcoder features) and edges (residual / attention / projection links) jointly implement an identifiable capability. Three reference points anchor the literature. Wang et al. (2022) reverse-engineer indirect-object identification in GPT-2 small to a 26-head subgraph in seven functional classes (Wang et al., 2022, §3)[70]; Conmy et al. (2023) systematize the discovery as iterative edge-pruning under a KL-divergence criterion[71]; Lindsey et al. (2025b) extend the program to production Claude 3.5 Haiku via cross-layer transcoders and Jacobian attribution. The thesis of this section is that these are not three separate methodologies but one: a commitment to *sparse, causal, localizable explanation*, refined and scaled across three computational regimes.

11.1 What a "circuit" claims

The operational definition is due to Wang et al. (2022). Let M be the model's full computational graph and $C \subseteq M$ a candidate circuit. The function $C(x)$ is defined by *knockout*: every node in $M \setminus C$ is replaced by its mean activation on a reference distribution p_{ABC}, leaving C's computations intact (Wang et al., 2022, §2)[72]. A circuit claim then commits to three quantitative criteria, also from Wang et al. (2022, §4)[73]:

- **Faithfulness:** C performs the task as well as the whole model. Tested by metric matching on the task distribution.
- **Completeness:** C contains every node used to perform the task. Tested by knocking out $M \setminus C$ and confirming the behavior is preserved beyond noise.
- **Minimality:** C contains no task-irrelevant nodes. Tested by ablating each node in C individually and verifying the behavior degrades.

[70] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-3-circuit`.

[71] KB excerpt: `kb/excerpts/conmy2023-acdc.md#sec-3`.

[72] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-2-definition`. Wang et al. explicitly prefer mean ablation over zero ablation: "zero ablation [. . .] lead[s] to noisy results in practice" because forced-zero activations are off-distribution.

[73] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-4-criteria`.

The substantive content of a circuit claim, then, is threefold: that the relevant features are *localized* (a small fraction of the graph), that the explanatory edges are *causal* (knockout matters), and that the subgraph is *sparse* (most of M is irrelevant on the task distribution). Wang et al. (2022) note that real circuits typically pass faithfulness and minimality but fail the strongest completeness tests (Wang et al., 2022, §4); this is the crack the *backup head* phenomenon (§11.2) widens.

Intuition

A circuit is a wiring diagram of which components matter, not a content description of what they encode. A diagram saying "Name Mover heads at layer 9 attend to previous-name positions and copy them to the unembed" fixes the topology of the computation but leaves the question of *which direction in the residual stream* encodes "name" to SAEs and probes (§7, §4). The canonical math: $C(x) = M(x)$ on a task distribution after knocking out $M \setminus C$, with C minimal under ablation.

11.2 The IOI circuit (Wang et al. 2022)

The canonical end-to-end mechanistic case study. Wang et al. (2022) analyze indirect-object identification on prompts of the form "When *Mary* and *John* went to the store, *John* gave a drink to" → "*Mary*". The reverse-engineered algorithm is human-interpretable (Wang et al., 2022, §1): identify all previous names; remove duplicates; output the remainder. The circuit that implements it consists of **26 attention heads grouped into seven functional classes**, comprising 1.1% of the (head, token-position) pairs in GPT-2 small (Wang et al., 2022, Abstract). The three major classes follow the algorithm's three steps (Wang et al., 2022, §3):

- **Duplicate Token Heads.** Active at the second-subject token `S2`, attend primarily to the first-subject token `S1`, and write the position of the duplicate to the residual stream. They detect that "John" has been seen.
- **S-Inhibition Heads.** Active at the `END` token, attend to `S2`, and write into the *query* of the Name Mover Heads, suppressing their attention to `S1` and `S2` positions.
- **Name Mover Heads.** Active at `END`, attend to previous names (now restricted to `IO` by the suppression), and copy the attended name to the output. The three Name Mover Heads have "copy score above 95%, compared to less than 20% for an average head" (Wang et al., 2022, §3.1).

Four minor classes complete the picture: **Previous Token Heads** copy information from `S` to `S1+1` as input to the induction route; **Induction Heads** reuse the Olsson et al. (2022) `[A][B]...[A]` → `[B]` mechanism for duplicate-token detection rather than for in-context learning; **Backup Name Mover Heads** replicate the Name Mover function only when the regular Name Mover Heads are ablated (Wang et al., 2022, §3); and **Negative Name Mover Heads** "write in the *opposite* direction of the Name Mover Heads, thus decreasing the confidence of the predictions" (Wang et al., 2022, §3), which the authors speculate is loss-hedging behavior.

The discovery method is *path patching* (Wang et al., 2022, §3.1). For inputs x_{orig} and x_{new} and a candidate head h, path patching "runs a forward pass on x_{orig}, but for the paths in $\mathcal{P}$ it replaces the activations for h with those from x_{new}" (Wang et al., 2022, §3.1), where $\mathcal{P}$ is the set of direct paths from h to the unembedding through residual connections and MLPs but *not* through other attention heads. This isolates the direct effect of h on the logits from indirect routing through other heads — the partial-derivative analogue of full activation patching, and the methodological refinement that makes the seven-class decomposition legible.

The IOI circuit complicated three field expectations simultaneously (Wang et al., 2022, §3): (i) *redundancy* — ablating Name Movers does not kill the behavior, because Backup Name Movers latently reactivate; circuits are not rigid pipelines. (ii) *repurposed mechanism* — induction heads, originally identified for in-context learning, are reused here for duplicate detection; mechanism does not equal function. (iii) *negative components* — a head that consistently writes *against* the correct answer is a structural feature of the trained model, not a bug. Each finding shows up again at scale in the attribution-graph era (§11.4).

11.3 Algorithmic circuit discovery (Conmy et al. 2023)

Hand-discovering the IOI circuit took multiple researcher-months. Conmy et al. (2023) systematize this work as an algorithm (**ACDC**, Automatic Circuit DisCovery) over an explicitly constructed computational DAG. The graph G represents "interactions between attention heads and MLPs" (Conmy et al., 2023, §2.2) — or, at finer granularity, between separate query/key/value activations, individual neurons, or per-token-position nodes[74]. The *interchange-intervention* corruption convention — preferred by Conmy et al. (2023) over zero or mean ablation because both "take the model too far away from actually possible activation distributions" (Conmy et al., 2023, §2.3) — fixes the corrupted input x'_i as a same-distribution counterfactual drawn from a related prompt template.

The metric for edge importance is *KL-divergence* between the full graph's output distribution $G(x_i)$ and the candidate subgraph's output distribution $H(x_i, x'_i)$, where $H(x_i, x'_i)$ is computed by "overwrit[ing] all edges in G that are not present in H to their activation on x'_i" (Conmy et al., 2023, §3). ACDC then iterates the graph in reverse-topological order (output-to-input), at each node greedily attempting to remove every incoming edge whose removal does not increase the KL divergence by more than threshold τ. Verbatim from Conmy et al. (2023, Algorithm 1)[75]:

Listing 1: ACDC, the Automatic Circuit DisCovery algorithm.

```
Algorithm 1: ACDC
Data:   Computational graph G; dataset (x_i)_{i=1..n};
        corrupted datapoints (x'_i)_{i=1..n}; threshold tau > 0.
Result: Subgraph H subset of G.

1  H <- G                              // initialize H to the full graph
2  H <- H.reverse_topological_sort()   // sort H so output node first
3  for v in H do
4    for w parent of v do
5      H_new <- H \ {w -> v}           // tentatively remove candidate
   edge
6      if D_KL(G || H_new) - D_KL(G || H) < tau then
7        H <- H_new                    // edge unimportant, prune
   permanently
8      end
9    end
10 end
11 return H
```

[74]KB excerpt: `kb/excerpts/conmy2023-acdc.md#sec-2-2`.
[75]KB excerpt: `kb/excerpts/conmy2023-acdc.md#sec-3`.

The algorithm's output is a sparse subgraph $H \subseteq G$ that preserves the KL-divergence to the full model up to τ summed over prunings. On GPT-2 small Greater-Than, ACDC "rediscovered 5/5 of the component types in [the manually-found] circuit" and "selected 68 of the 32,000 edges in GPT-2 Small, all of which were manually found by previous work" (Conmy et al., 2023, Abstract)[76]. ACDC is evaluated against two baselines: *Subnetwork Probing* (a learned mask over components, with the linear probe removed and KL substituted for the training metric) and *Head Importance Score for Pruning* (a top-k ranking by head importance score). Both are generalized in Conmy et al. (2023) to use corrupted-activation patching rather than zero patching, for fair comparison. ROC-curve analysis against canonical circuits answers the two evaluation questions (Conmy et al., 2023, §4): does the method find the right edges (true-positive rate) and does it avoid spurious ones (false-positive rate)?

ACDC is the algorithmic spine of automated circuit discovery, but it inherits the cost of activation patching: each candidate edge requires a forward pass with one substituted activation, so the per-task cost scales with the size of the computational graph. On GPT-2-small that is feasible; on production models it is not.

11.4 Attribution graphs (Anthropic 2025)

Lindsey et al. (2025b) synthesize the program for production-scale models. The construction has three pieces. First, *cross-layer transcoders* replace each MLP sublayer with a sparse factorization whose decoder writes into the residual stream of every *subsequent* layer, not just the immediate next layer; this makes skip-layer feature flow explicit in the graph (Dunefsky et al., 2024). Second, a single forward pass on the replaced model records the activations of all transcoder features $\{f_i^{(\ell,t)}\}$. Third, an *attribution graph* is constructed by computing the linear contribution of every active feature to a target node (typically the unembedding direction for the target token), via the Jacobian of the cross-layer transcoder system.

Concretely, the attribution edge from feature i at layer ℓ_1, position t_1 to feature j at layer ℓ_2, position t_2 is

$$a\Big(f_i^{(\ell_1,t_1)} \to f_j^{(\ell_2,t_2)}\Big) \;=\; \frac{\partial f_j^{(\ell_2,t_2)}}{\partial f_i^{(\ell_1,t_1)}}\bigg|_x \;\approx\; \langle \mathbf{e}_j^{(\ell_2)}, \mathbf{A}_{\ell_1\to\ell_2}\, \mathbf{d}_i^{(\ell_1)} \rangle, \tag{39}$$

where $\mathbf{d}_i^{(\ell_1)}$ is the decoder column of feature i in the layer-ℓ_1 transcoder, $\mathbf{e}_j^{(\ell_2)}$ is the encoder row of feature j in the layer-ℓ_2 transcoder, and $\mathbf{A}_{\ell_1\to\ell_2}$ accumulates the linearized residual / attention transport between the two layers (identity through the residual stream; attention-weight-modulated through any intervening attention layers). The bilinear factorization is what makes Eq. (39) input-invariant in its endpoints and input-dependent only through the attention weights and the active set $\{f_i^{(\ell,t)} > 0\}$. Edges are pruned below threshold; the resulting graph is iterated backward from the target node to the embedding layer; the surviving subgraph is the attribution graph (Lindsey et al., 2025b).

Forum signal

Single-source methodology (2026-Q2). The cross-layer-transcoder + Jacobian-attribution pipeline above rests on Lindsey et al. (2025b) as its sole canonical reference; no external lab has published an independent reproduction at frontier scale at this writing. Treat the productionised-methodology framing as Anthropic-internal evidence pending external

[76] KB excerpt: `kb/excerpts/conmy2023-acdc.md#abstract`.

replication. The closing contradiction of this section develops the attendant transcoder-seed-stability and external-reproduction concerns.

The compute regime shifts from manual to automated. Per-task cost on the replaced model is one forward pass plus one backward pass; the amortized cost is the cross-layer transcoder training, which Anthropic estimates at roughly comparable order to SAE training (Gemma Scope reports SAE training at ∼20% of the underlying model's pretraining compute (DeepMind, 2024)). Reported findings on Claude 3.5 Haiku, transferred from Lindsey et al. (2025a): forward-planning features that activate many tokens before a poetic rhyme is produced; multi-step factual chains in which a "Texas" feature in middle layers licenses an "Austin" feature in later layers for the prompt "the capital of the state Dallas is in"; refusal-circuit features on harmful-request prompt classes; and partial cross-lingual feature overlap with language-routing through a partly language-agnostic middle.

Intuition

ACDC and attribution graphs answer the same question — which edges of the computational graph causally implement a behavior — in two compute regimes. ACDC pays $O(|E|)$ forward passes per task and returns components; attribution graphs amortize transcoder training once and pay one forward+backward per task, returning features. The canonical math: ACDC prunes by $D_{\mathrm{KL}}(G\|H_{\mathrm{new}}) - D_{\mathrm{KL}}(G\|H) < \tau$ (§11.3); attribution graphs prune by the linearized Jacobian in Eq. (39).

11.5 When circuits agree across methods

The IOI circuit is the canonical convergence story. Wang et al. (2022)'s manual path-patching reverse-engineering and Conmy et al. (2023)'s ACDC recover overlapping subgraphs — ACDC "rediscovered 5/5 of the component types" (Conmy et al., 2023, Abstract) on related GPT-2-small tasks, and on IOI itself recovers the seven functional classes Wang et al. describe. SAE-feature analysis on the same prompts identifies features playing the QK and OV roles of the discovered heads (§7). Tuned-lens prediction trajectories on IOI prompts are consistent with the circuit's mid-late-layer commit to the IO token (§3). Three independent methodologies converge on one picture.

Factual recall is the second convergence. Meng et al. (2022)'s causal tracing on GPT-2 XL localizes factual recall to mid-layer MLPs at the last subject token (AIE peaks at layer 15; $\mathrm{AIE}_{\mathrm{MLP}} = 6.6\%$ vs. $\mathrm{AIE}_{\mathrm{attn}} = 1.6\%$ (Meng et al., 2022, §2.2), with rank-one weight edits at exactly that locus achieving 99.8% efficacy on zsRE (Meng et al., 2022, §3.2)). SAE features on those layers include subject-entity features (Templeton et al., 2024); the tuned lens shows the predicted-object token's probability rising across the localized layer band; ROME's rank-one MLP edit succeeds at exactly that locus. Four methods, one converging picture.

The induction-head story (Olsson et al. (2022)) is a third. The same two-attention-head pattern is identified by ablation (removing the heads abolishes in-context learning), by phase-change correlation (induction-head formation aligns with the in-context learning loss-curve discontinuity), by direct circuit reverse-engineering, and is then *reused* as a substructure of the IOI circuit (Wang et al., 2022, §3). The cross-method evidence is strong enough that "induction head" is now a load-bearing primitive across the rest of the literature.

11.6 When circuits disagree

Convergence is not universal. The truth-direction circuit (§6) is the cleanest divergence case. The mass-mean direction $\mathbf{d}_{\mathrm{MM}} = \boldsymbol{\mu}_{+} - \boldsymbol{\mu}_{-}$ found by Marks and Tegmark (2023) is dataset-dependent in early-to-mid layers — antipodal between `cities` and `neg_cities`, then orthogonal, only aligned across formats in late layers (Marks and Tegmark, 2023, §4)[77]. There is no single "truth circuit" in this picture: there is a *family* of dataset-conditional truth representations, partially aligned across formats and partially not, with the alignment depth itself varying across model layers.

Sycophancy is a parallel non-convergence. Probing-derived steering directions for sycophancy are reported across multiple labs, but no two reports agree on a single "sycophancy circuit": the proposed mechanisms span steering-vector additions to refusal-feature ablations to fine-tuning interventions, and the published evidence does not yet localize a stable subgraph. The 2025 finding that sycophancy decomposes into agreement, praise, and genuine-agreement components individually suggests *behaviorally singular, representationally plural* — if confirmed, the question "what is the sycophancy circuit?" is malformed: there are several.

Contradiction

Circuit-tracing scalability past Anthropic-2025. The attribution-graph methodology of Lindsey et al. (2025b) and the Haiku findings of Lindsey et al. (2025a) have not been independently reproduced as of 2026. The compute cost ($\sim$20% of pretraining for transcoder training) is a real adoption barrier outside frontier labs. The substantive claims — forward planning, multi-step factual chains, refusal circuits — rest on internal Anthropic evidence and on transcoders whose canonicality inherits the Leask et al. (2025) critique: a different transcoder training run may factor the same MLP into a different sparse basis, and the stability of attribution-graph conclusions across transcoder seeds is open empirical. The strongest form of the claim — that automated circuit discovery scales to Claude-class models with results that converge across replications — is asserted but not yet externally validated.

11.7 Forward link

The cases above set up §12, where we make the convergence/divergence pattern the object of analysis: when methods should agree (because they answer the same question about the same mechanism), divergence falsifies at least one method's assumptions; when methods answer different questions (correlational vs. causal, component-level vs. feature-level), apparent divergence may be principled. The IOI circuit and factual-recall localization are the positive cases; the truth-direction geometry and SAE canonicality are the negative cases; and the open question — whether attribution-graph findings on Haiku will reproduce outside Anthropic — is the load-bearing test for whether circuit-tracing extends from a method to a paradigm.

12 Cross-method convergence and divergence

The five methods of §§3–11 make different commitments. Lenses are correlational and project the residual stream through the model's own unembedding. Probes are correlational and project it through a trained classifier. Sparse autoencoders are unsupervised and project it into a learned $M \gg D$-dimensional sparse basis. Activation patching is causal at the component level. Circuit tracing is causal at the feature level. The methodological question this section asks is: when methods

[77] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-4-misalignment`.

should agree — because the underlying mechanism is the same and they answer the same question about it — what do their points of convergence and divergence tell us about the model and about the methods?

The answer is two-sided. §§12.1–12.2 take up the convergence cases (IOI on GPT-2 small; factual recall on GPT-2 XL), where three or four methods triangulate on the same picture and the joint claim is more robust than any single-method claim. §§12.3–12.5 take up the divergence cases (truth direction across datasets; SAE feature canonicality; lens cross-model unreliability), where the methods disagree and the disagreement is itself diagnostic. §12.6 states the closing argument and forward-links to §13 and §14.

12.1 Convergence: the IOI circuit on GPT-2 small

The most-replicated cross-method picture in the field is indirect-object identification on GPT-2 small. The substrate is $\sim$ 124M parameters; the task is the canonical Wang–prompt template "When Mary and John went to the store, John gave a drink to ___"; the target is the indirect-object name (Wang et al., 2022)[78]. Three methodologically independent readouts converge on the same circuit picture.

Wang et al. (2022) identify a circuit of 26 attention heads in 7 functional classes (Duplicate Token, Previous Token, S-Inhibition, Name Mover, Negative Name Mover, Backup Name Mover, Induction Heads), 1.1% of all (head, token-position) pairs, by manual path-patching. Conmy et al. (2023) recover a substantially overlapping subgraph by automating the discovery: the ACDC algorithm of §11.3 iterates over computational-graph edges, KL-thresholds the indirect effect of each, and prunes below threshold; the surviving subgraph on GPT-2 small recovers the Wang circuit's load-bearing edges without human-supplied prior structure. A third method enters from the lens side: the tuned-lens prediction trajectory of §3.5 on IOI prompts shows the indirect-object name's probability rising in the mid-late layer band of GPT-2 small, consistent with the Name-Mover-and-S-Inhibition commit identified by Wang et al. at the head level. SAE-feature analysis on the same prompts identifies features matching the QK and OV roles of the discovered heads .

Intuition

Three methods, three different commitments (causal-component, causal-edge, correlational-readout), one converging picture: IOI on GPT-2 small uses a sparse subset of attention heads in 7 functional classes, distributed across mid-late layers, and the predicted indirect-object probability emerges in exactly the layer band the heads operate in. Returning to canonical form: the intersection of patching's component-level localization, ACDC's edge-level discovery, and the lens's prediction trajectory is the circuit C of §2.6, satisfying faithfulness, completeness, and minimality at the GPT-2-small scale.

12.2 Convergence: factual recall on GPT-2 XL

The second positive cross-method case is factual recall on GPT-2 XL, where four methods converge on the same localization. The substrate is 1.5B parameters; the task is the (s, r, o)-triple cloze of §10.1 (e.g., "The Eiffel Tower is in ___"); the target is the object completion.

Meng et al. (2022) run causal tracing across (component, token, layer) cells and report two effects: the average indirect effect peaks at MLP layers around layer 15–20 at the last subject token, and the AIE for MLP perturbations (6.6%) substantially exceeds the AIE for attention perturbations

[78] KB excerpt: `kb/excerpts/wang2022-ioi.md#sec-2-task-and-circuit`.

(1.6%) at that locus (Meng et al., 2022, §2.3)[79]. The first method — activation patching with Gaussian noise as the corruption — localizes the mediating component class to mid-layer MLPs at the last subject token.

The second method enters as the weight-level intervention. Following the localized factual association hypothesis, Meng et al. construct a rank-one MLP-weight edit at the localized layer that inserts a new (s, r, o) triple under a key–value constraint $(W + \Delta W)\,\mathbf{k}_* = \mathbf{v}_*$ minimizing $\|\Delta W \mathbf{k}_*\|^2$ (Meng et al., 2022, §3.2). The edit inserts new facts with 99.8% efficacy and 88% paraphrase generalization on zsRE. The patching localization survives the strongest possible test: an edit at the localized site succeeds at inserting facts; an edit elsewhere does not. (See §10.4 for the Hase et al. 2023 "does localization inform editing" caveat : the *precise* layer-of-edit is not always the AIE peak, but the locus is robust.)

The third method enters from the lens side. The tuned-lens prediction trajectory of §3.5 on factual recall prompts shows the predicted-object token's logit rising across exactly the localized layer band: the object is not yet predicted at early layers, becomes predictable across the mid-layer MLPs, and is committed by the late layers. The lens trajectory and the patching heatmap are reading the same phenomenon through two different windows.

The fourth method enters from the SAE side. SAE features trained on the residual stream and MLP outputs of GPT-2 XL recover subject-entity features at exactly the layer band patching localizes . The features are not fully catalogued at the GPT-2 XL scale, but the structural pattern — subject-entity features are sparse in the localized band and not elsewhere — holds.

Four methods, one converging picture: factual recall in GPT-2 XL is mediated by mid-layer MLPs at the last subject token; the effect can be localized causally (patching), validated intervention-ally (rank-one edit), seen correlationally (lens trajectory), and decomposed at the feature level (SAE features). The joint claim is more robust than any single-method claim because each method's confound is different and the four do not align: a lens-only artifact would not produce a successful edit; a patching-only artifact (e.g., Gaussian-noise off-distribution effects (Zhang and Nanda, 2023)) would not align with SAE-discovered features that are training-distribution by construction.

12.3 Divergence: truth direction across datasets

The negative cases begin with the most-studied probing target. The Marks and Tegmark (2023) mass-mean direction on LLaMA-2-13B, $\mathbf{d}_{\mathrm{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$, is the canonical truth probe of §4.3; the same §4 of the paper documents "stark misalignment" across datasets (Marks and Tegmark, 2023, §4)[80]. The truth direction recovered from the `cities` dataset ("The capital of France is Paris.") and the truth direction recovered from `neg_cities` ("The capital of France is not Berlin.") are antipodal in early layers, orthogonal in mid-layers, and aligned only in late layers (around layers 12–15). Whether the recovered direction is a property of the *model*'s truth representation or of the *dataset*'s truth-pattern is layer-dependent.

A second method on the same target gives a different answer. Contrast-Consistent Search (CCS) finds an unsupervised direction satisfying a probabilistic-consistency loss on contrast pairs; Marks and Tegmark (2023, §1 related work)[81] documents that the recovered direction is not the mass-mean direction, and CCS suffers documented failures under negation that mass-mean does not share. A third method on the same target enters from the SAE side. SAE-discovered "truthfulness"-related features (Templeton et al., 2024, §4)[82] overlap partially but not fully with mass-mean directions:

[79] KB excerpt: `kb/excerpts/meng2022-rome.md#sec-2-effects`.
[80] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-4-misalignment`.
[81] KB excerpt: `kb/excerpts/marks-tegmark-2023-truth.md#sec-1-related`.
[82] KB excerpt: `kb/excerpts/templeton2024-scaling-monosemanticity.md#sec-4` (HTML, deepen-cite pending).

the cosine between the probe direction and the SAE feature decoder column on matched concepts is, to the best of our knowledge, not yet systematically reported, and the few hand-spot-check writeups report partial alignment rather than identity.

Contradiction

There may not be a single truth direction. Mass-mean probing on LLaMA-2-13B finds dataset-dependent directions that align only in late layers (Marks and Tegmark, 2023, §4); CCS finds a different direction that fails under negation; SAE-discovered truthfulness-related features overlap partially but not fully with either. Three methods, no single converging direction. The Vennemeyer et al. (2025) multi-component result on sycophancy generalizes the pattern: behaviorally-singular properties may be representationally plural, and the field's single-axis truth-probe assumption may be the wrong shape of model for the underlying representation. The strongest form of the open question — "does the model represent truth along a single linear axis, or along a multi-axis subspace whose components depend on prompt format?" — is unresolved as of 2026, and the experiment that would close it is a cross-method, cross-prompt-format, cross-model-family study reporting probe direction, SAE feature direction, and patching-validated causal direction in matched coordinates.

This is principled divergence in part. A probe asks "what direction linearly decodes property y from this dataset's activations?"; an SAE asks "what sparse basis decomposes activations under reconstruction-and-sparsity loss?". Different answers need not falsify each other — but the divergence becomes load-bearing once the field makes single-axis truth-probe claims ("ablating $\mathbf{d}_y$ flips the model's TRUE/FALSE behavior"): if there is no single $\mathbf{d}_y$, the claim is dataset-conditional. The divergence is the diagnostic.

12.4 Divergence: SAE feature canonicality

A second negative case is the SAE feature inventory itself. Leask et al. (2025) show that two independent SAE training runs at different widths M may produce different feature inventories on the same activations (Leask et al., 2025, §1)[83]: meta-SAEs decompose larger-SAE latents into combinations of smaller-SAE latents (non-atomicity), and SAE stitching reveals novel latents present in larger SAEs but absent from smaller ones (incompleteness). The Bricken et al. (2023)/Templeton et al. (2024) "true features" framing assumes a sufficiently large SAE recovers the model's underlying units of computation modulo training noise; the Leask et al. critique argues that "sufficiently large" is not well-defined. §8 treats the contradiction at depth; the cross-method consequence is that any claim grounding in SAE features ("the SAE feature for truth has cosine ρ with the probe direction") is a claim about *this* SAE's inventory, not about a model-canonical basis. The pragmatic 2026 position — training SAEs at multiple widths and reporting which claims survive across widths — reduces the single-run-artifact risk; the open theoretical question (a basis-finding criterion under which independent runs converge modulo a known equivalence) is documented in §8.6.

12.5 Divergence: lens cross-model unreliability

The third negative case is internal to the lens family. The logit lens of nostalgebraist (2020) works on GPT-2 — the trajectory $\{\text{LogitLens}(\mathbf{h}_\ell)\}_{\ell=0}^{L}$ converges smoothly toward the final-layer prediction — but on BLOOM and GPT-Neo of equivalent scale fails: intermediate distributions are systematically biased and the no-training baseline is no longer a useful diagnostic (Belrose et al.,

[83]KB excerpt: `kb/excerpts/leask2025-sae-not-canonical.md#sec-1-canonical`.

2023, §2.2)[84]. Belrose et al. (2023) fix the bias with the per-layer affine translator $\text{TunedLens}_\ell(\mathbf{h}_\ell) = \text{LogitLens}(A_\ell\, \mathbf{h}_\ell + \mathbf{b}_\ell)$, at the cost of $L \cdot (D^2 + D)$ trained parameters. Chuang et al. (2023)'s DoLa contrastive decoding works on the LLaMA-7B/13B/33B family; whether it generalizes to Llama 3 / 4, Qwen 2.5 / 3, DeepSeek-V3, and Gemma 3 has not been systematically reproduced, and the `logitlens4llms-2025` extension is the partial cover for the 2025–2026 frontier. The cross-method consequence: any synthesis using a lens trajectory (the IOI and factual-recall convergence cases above) is implicitly model-conditional, and the four-method factual-recall convergence of §12.2 is therefore a GPT-2-XL-specific result until a tuned lens for the frontier-2026 architecture exists.

12.6 When divergence is principled, and when it is falsification

The closing argument is structural. Two methods can disagree for two reasons: they answer the same question with different commitments and one is wrong (*falsification divergence*, and the diagnostic action is to identify which method's assumption breaks); or they answer different questions whose answers happen to disagree on a target (*principled divergence*, and the diagnostic action is to be careful about which method's answer one cites for which claim). The convergence cases above are the regime where methods are answering compatible questions and the answers cross-validate. The truth-direction divergence is on the boundary: probes and SAE features ask different questions, but the single-axis truth claim assumes the questions have a common answer, and the divergence falsifies that assumption (or exposes that the representation is multi-axis). The SAE-canonicality and lens cross-model divergences are internal to one method family: two runs of the same procedure return different answers, and the divergence falsifies a stability assumption in each case.

Intuition

Method-pluralism is principled because the methods commit differently and answer different questions. But the convergence cases above show that, for sufficiently constrained behaviors on sufficiently small models, the methods *can* be made to answer the same question, and when they do, agreement is the load-bearing form of evidence. Returning to canonical form: the joint posterior over circuits, given evidence from $\{\text{patching}, \text{lens}, \text{SAEs}, \text{ROME-edit}\}$, is sharply peaked when all four methods agree (the IOI and factual-recall cases) and diffusely supported when they do not (the truth-direction case). The next paper-level methodological move is not better single methods — it is principled cross-method protocols that fix what each method is asked to answer, so that disagreement is interpretable as falsification rather than as a question mismatch.

This synthesis feeds two downstream sections. §13 consumes the convergence cases as the regime where every method has scaled enough to participate, and the divergence cases as the regime where single-method claims are still the only evidence on production-scale models. §14 consolidates the divergence cases of §§12.3–12.5 alongside the §§11.6, 8.6, and 10.4 contradictions surfaced earlier in the paper, and pairs each with the experiment that would close the gap. The thesis holds: "*partially read*" is the right phrase because on the well-constrained behaviors where the methods' angles coincide (IOI, factual recall) the model is read tightly, while on the open behaviors where they do not (truth, sycophancy, the SAE feature inventory) the field has identified the disagreement and is working on the resolution.

[84] KB excerpt: `kb/excerpts/belrose2023-tuned-lens.md#sec-2-failures`.

13 Scaling interpretability and frontier-2026 snapshot

The five methods of §§3–11 were originally exhibited on small open models: GPT-2 small (124M parameters) for Wang et al. (2022)'s indirect-object-identification circuit, GPT-2 XL (1.5B) for Meng et al. (2022)'s ROME tracing, Pythia 410M–12B for Belrose et al. (2023)'s tuned-lens benchmarks. The 2024–2026 question — and the operational question for anyone who wants interpretability evidence on a model people actually deploy — is which of these methods carry to 2–27B open-weights models (the Gemma 2 / LLaMA 3 / Qwen 2.5 family) and to production frontier models (Claude 3.5 Haiku, GPT-4 class). The answer is method-dependent, not uniform: SAEs scale on documented laws, probes scale by construction, attribution graphs scale to production but only inside Anthropic, and manual circuit hunts scale not at all. This section is the compute-and-coverage map; §13.4 consolidates it into a single snapshot table that becomes the figure-of-record for the paper.

13.1 What scales

SAEs scale on documented laws. Gao et al. (2024) report clean scaling laws for sparse autoencoders trained with a TopK activation: a compute–loss law $L(C)$ at fixed sparsity, and a width–sparsity law $L(N, K)$ at fixed compute, where $N = M$ is the dictionary size and K the per-token L_0 (Gao et al., 2024, §1, §3)[85]. The headline demonstration is a 16M-latent SAE trained on GPT-4 residual-stream activations with $\sim 7\%$ dead latents under the AuxK mitigation (Gao et al., 2024, §2.4)[86]; the law is predictive enough to guide budget allocation between N and training tokens at the design stage rather than empirically. DeepMind (2024) exercise the architectural cousin — JumpReLU SAEs of Rajamanoharan et al. (2024) — at the open-weights end of the same scaling envelope: more than 400 SAEs spanning every layer and sublayer of Gemma 2 2B and 9B (residual stream, MLP output, attention output) plus selected layers of Gemma 2 27B, with widths $\{16.4\text{K}, 65.5\text{K}, 131\text{K}, 1\text{M}\}$, totaling >30M learned features and roughly 20% of GPT-3's pretraining compute (DeepMind, 2024, §1)[87]. The CC-BY-4.0 release on HuggingFace plus the Neuronpedia interactive demo converted SAE-based interpretability from an Anthropic-internal capability into an externally reproducible one.

Probes scale gracefully. A linear probe is trained on cached activations (so the model itself is called once to produce activations, then the probe runs on a CPU); mass-mean probing (§4.3) is closed-form $\mathbf{d}_{\text{MM}} = \boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$ and requires no training at all (Marks and Tegmark, 2023). Probe *cost* is dominated by the activation cache, not the probe fit; the activation cache scales linearly with dataset size and is embarrassingly parallel across prompts. The same property carries the direction-ablation operator $\mathbf{h} \mapsto \mathbf{h} - (\mathbf{h}^\top \hat{\mathbf{d}}_y)\, \hat{\mathbf{d}}_y$ and the addition operator $\mathbf{h} \mapsto \mathbf{h} + \alpha\, \hat{\mathbf{d}}_y$ of §5.3: both are $O(D)$ per token and survive whatever architectural innovations the underlying model adds, because they act on the residual stream rather than on any particular sublayer.

Attribution graphs scale to production — expensively, and inside one lab. Lindsey et al. (2025b)'s cross-layer-transcoder framework reports the only published end-to-end mechanistic-interpretability result on a production model (Claude 3.5 Haiku). The compute argument is two-part: per-task cost on the replaced model is one forward pass plus one backward pass — cheap relative to activation patching's $O(N_{\text{components}})$ forward passes per (clean, corrupted) pair (§9.1) — but the amortized cost is the cross-layer transcoder training, estimated at the same order as SAE

[85]KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-1-headline`.
[86]KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-2-4`.
[87]KB excerpt: `kb/excerpts/gemma-scope-2024.md#sec-1`.

training in DeepMind (2024) (∼20% of the underlying model's pretraining compute). The Anthropic 2025-05-30 open release of the circuit-tracing tooling and its Neuronpedia integration converted the methodology from a strictly internal capability into one external groups can in principle exercise on open models. As of 2026, no externally-trained attribution-graph result on a frontier model has been published; the contradiction this raises about reproducibility is the load-bearing one of §14.

13.2 What does not scale (yet)

Manual circuit hunts. The IOI reverse-engineering of Wang et al. (2022) took graduate-student months of work to identify 26 heads in 7 functional classes inside a 1.6%-of-(head, position)-pairs subgraph on a 124M-parameter model. The same effort would need to scale roughly with the number of heads and layers — $L \cdot H$ on a frontier-2026 model is $\sim$ 80–120 times that of GPT-2 small — and the exhaustive case-by-case behavioral testing that gave the IOI claim its load-bearing status does not have an obvious automated analog. Conmy et al. (2023) automate the *discovery* step (iterative path-patching with a KL-divergence threshold), but the automation produces larger and less interpretable subgraphs, and the human-in-the-loop labeling that decided "this head class is a Name Mover" on IOI does not transfer.

Full activation patching. The three-run protocol of §9.1 requires $O(N_{\text{components}})$ forward passes per (clean, corrupted) pair to fill a causal-trace heatmap. On GPT-2 XL, $N_{\text{components}}$ is order-10^4 at head granularity and 10^5 at single-(head, position); on a frontier-2026 model with $\sim$ 80 layers, $\sim$ 64 heads per layer, and $\sim$ 8K context, full-coverage patching is on the order of 10^7 forward passes per task. Attribution patching linearizes the indirect effect via the first-order Taylor approximation $\mathrm{IE}_C \approx \langle \nabla_{\mathbf{h}} m, \Delta\mathbf{h} \rangle$ (§9.6) and reduces this to a small constant number of passes, but the linearization fails in non-linear regions of the loss landscape, so it is a discovery aid rather than a faithful substitute for direct intervention.

Lens coverage of frontier-2026 architectures. Belrose et al. (2023)'s tuned-lens benchmarks cover Pythia 410M–12B and GPT-2 / GPT-Neo / BLOOM; `logitlens4llms-2025` extends the framework to Qwen-2.5 and LLaMA-3.1 (§3.5). Tuned lenses for the rest of the frontier-2026 lineup — LLaMA 3/4, full Qwen 2.5/3, DeepSeek-V3, Gemma 3 — are not all published as of 2026. The relevant architectural shift for the lens framework is not raw scale (the distillation loss $\mathcal{L}(\ell) = \mathbb{E}_{\mathbf{x}}[D_{\text{KL}}(\mathcal{M}_{>\ell}(\mathbf{h}_\ell) \| \text{TunedLens}_\ell(\mathbf{h}_\ell))]$ is indifferent to model size) but the substrate: DeepSeek-V3 combines MLA (DeepSeek-AI, 2024a)'s low-rank-compressed KV with fine-grained MoE (DeepSeek-AI, 2024c); NSA (Yuan et al., 2025) bakes attention sparsity into pretraining. The residual stream remains well-defined in all of these, but the assumption that a single per-layer affine translator $(A_\ell, \mathbf{b}_\ell)$ suffices to absorb basis drift may need re-examination on a per-architecture basis. The empirical work that would close this gap — training tuned lenses across the frontier-2026 lineup and reporting the per-architecture distillation-loss curves — is not yet in the literature.

Contradiction

The interpretability-coverage gap is widening, not closing. The frontier-2026 architecture snapshot (Paper 1, §13) catalogs MLA, fine-grained MoE, NSA-sparse attention, sliding-window layers, and hybrid attention/SSM stacks as the substrate of the deployed lineup. The lens benchmarks of Belrose et al. (2023) predate this substrate; the SAEs of DeepMind (2024) and the attribution graphs of Lindsey et al. (2025b) cover specific models in the lineup but not the lineup as a class; manual circuits and full activation patching are exhibited only on ≤1.5B-parameter GPT-family models. The gap between "what frontier models do" and "what we have interpretability evidence about frontier models doing" is not narrowing on net: it is widening, because architectural innovation outpaces the per-architecture interpretability work needed to catch up. The 2025-05-30 Anthropic open release of circuit-tracing tooling and the Gemma Scope CC-BY-4.0 release are corrective forces; whether they correct fast enough is a load-bearing empirical question about the field's coverage trajectory.

13.3 Production interpretability pipelines, 2026

Four pipelines are in active use as of 2026, each anchored on one of the methods above.

Forum signal

Source caveat for the Anthropic pipeline. The cross-layer-transcoder + Jacobian-attribution methodology rests on Lindsey et al. (2025b) as its sole canonical reference; no external lab has published an independent reproduction at frontier scale as of 2026-Q2. The pipeline below is described as a productionised tool, but the load-bearing claim that this is the productionised attribution-graph methodology rests on a single tier-B Anthropic source.

Anthropic — circuit tracer + Neuronpedia. The cross-layer-transcoder + Jacobian-attribution pipeline of Lindsey et al. (2025b), open-sourced in 2025-05-30 and integrated with Neuronpedia for interactive feature browsing . The substantive findings on Claude 3.5 Haiku — forward-planning rhyme features, multi-step factual chains ("Texas" → "Austin"), refusal-circuit features, partial-cross-lingual feature overlap — are reported in Lindsey et al. (2025a) and remain Anthropic-internal as evidence .

OpenAI — TopK SAEs and the $L(C), L(N, K)$ scaling program. The $L(C)$ and $L(N, K)$ laws of Gao et al. (2024) are the ongoing methodological backbone, plus an evaluation-metric suite (L_0, reconstruction MSE, downstream-loss recovery, contrastive probe recovery) (Gao et al., 2024, §1 contributions)[88]. The 16M-latent GPT-4 SAE is the existence proof for SAE scaling at production-frontier scale; the per-feature interpretability work downstream of that release is not all public.

Google DeepMind — Gemma Scope + JumpReLU + tuned-lens benchmarks. The JumpReLU SAE family of Rajamanoharan et al. (2024), the Gemma Scope open release of DeepMind (2024), and the Gemma 2 2B/9B/27B coverage make this the most externally-reproducible pipeline of the four. The tuned-lens benchmarks of Belrose et al. (2023), retargeted across the Gemma family, provide the cross-method readout layer the lens-as-decoder framing needs.

Anthropic — NLAs in pre-deployment audit. A second Anthropic pipeline, distinct from the circuit-tracer line, is the natural-language autoencoder (AV + AR pair) of Fraser-Taliente et al. (2026), used in the pre-deployment audit of Claude Opus 4.6 to surface unverbalized evaluation awareness and alignment-faking-adjacent cognition. Output modality is free-form natural language rather than logits or sparse-feature dictionaries; the round-trip FVE score serves as a confabulation

[88] KB excerpt: `kb/excerpts/gao2024-topk-saes.md#sec-1-headline`.

gate (§3.6). No external reproduction as of 2026-Q2; the methodology is reported only on the Anthropic-internal Opus 4.6 audit.

13.4 Snapshot table

Table 6 consolidates the per-method coverage as of 2026-05. "Open release" is the date the method's canonical-scale-target artifact (SAE weights, lens parameters, attribution-graph tooling) became externally usable; "2026 reproduction" is whether at least one externally-validated result on a frontier-2026 model exists.

The table makes the asymmetry explicit. The methods with the most favorable scaling — probes and SAEs — have the most external reproduction; the method with the strongest substantive findings on a production frontier model — attribution graphs — has none. Three of the ten rows depend on tooling or claims that are HTML-only or released-but-not-yet-reproduced; the remaining seven form a defensible shared methodological substrate.

Intuition

The five-method picture maps cleanly to the scaling table. Lenses give you a cheap readout that may not generalize across architectures (no fixed artifact, partial coverage). Probes give you cheap correlational evidence with closed-form variants and broad reproduction (§4.3). SAEs give you a feature dictionary at scale under documented laws (the most reproducible pipeline). Activation patching gives you causal claims at small scale and degrades to linearized approximations beyond. Circuit tracing gives you the strongest claims at frontier scale but only inside the originating lab. The methodological lesson of §12 — method-pluralism is principled because methods answer different questions — has a scaling-table corollary: *method-pluralism is also operationally necessary, because no single method covers every (model, behavior, evidence-class) cell of the matrix.* Returning to canonical form: the per-method costs are $O(D)$ for probes and lenses, $O(MD \cdot \text{tokens})$ amortized for SAE training, $O(N_{\text{components}})$ per-task for activation patching, and $O(\text{transcoder training} + 1\,\text{fwd} + 1\,\text{bwd})$ for attribution graphs.

13.5 Forward link

The substrate this section reads against is the frontier-2026 architecture snapshot of *Paper 1, §13*: the MLA / fine-grained MoE / NSA-sparse-attention / sliding-window / hybrid-SSM stack that defines what production models look like as of 2026. The interpretability methods catalogued here have caught up to that substrate unevenly, and the two most pressing gaps — per-architecture tuned lenses for the frontier lineup, and external reproduction of attribution-graph findings outside Anthropic — are open empirical questions whose closure is the work of the next two years. The contradictions surfaced in this section — circuit-tracing reproducibility, lens cross-architecture coverage, the widening gap between deployed-model architecture and per-architecture interpretability evidence — are taken up alongside the SAE-canonicality and truth-direction contradictions in §14, where each is paired with the experiment that would resolve it. The thesis of the paper holds against this scaling picture: "*partially read*" is exactly the right phrase, and the qualifier is load-bearing across method, scale, and architecture.

14 Contradictions live and open frontiers

The five methods of this paper agree often enough to make the field move (the GPT-2-small IOI circuit is the worked convergence example; the GPT-2-XL factual-recall locus is the second), but the cases where they *disagree* are not embarrassments to be papered over: they are where the field's next experiments live. We collect five anchor contradictions raised across §§3–13 and discuss each in the same three-part form: the strongest version of each side's claim, the evidence on the table as of 2026, and the experiment that would resolve it. The closing argument recasts method-pluralism as a principled stance: contradictions live at the *boundaries between* methods, and cross-method evidence is the only currency that can spend them.

14.1 The SAE canonical-feature claim

The strongest form of the Bricken et al. (2023) and Templeton et al. (2024) program is that there exists a privileged decomposition of activations into "true features" — a feature dictionary that any sufficiently-careful SAE training recovers, modulo a known equivalence. The strongest opposing form, from Leask et al. (2025, §1), is that no such canonical decomposition exists: meta-SAEs trained on the decoder $\mathbf{W}_{\text{dec}}$ of a target SAE recover a sparse decomposition of the target's atoms (the "Einstein" latent splits into "scientist", "Germany", "famous person") — non-atomicity — and SAE stitching reveals novel latents present in larger SAEs that smaller SAEs cannot recover — incompleteness.

The evidence on the table, summarized in §8: two ICLR-2025 lines of experiment supporting the negative; a corpus of qualitative interpretability successes from Templeton et al. (2024) consistent with the positive but not entailing it; and the field's pragmatic resting position — "apply SAEs of several widths" (Leask et al., 2025, §1) — which is methodological agnosticism, not resolution.

Resolving experiment. A basis-finding criterion under which two independent SAE training runs — at different widths M_1, M_2, different sparsity targets, different random seeds, on the same activations — converge on the same feature inventory modulo a known equivalence (rotation, sign, permutation, scale). Until that criterion is exhibited, the canonical-feature claim is unfalsifiable in the direction that matters and the field is right to treat SAEs as useful basis-dependent tools.

14.2 Lens cross-model unreliability

The strongest form of the lens program (Belrose et al., 2023): an affine translator $(A_\ell, \mathbf{b}_\ell)$ trained by KL distillation against the final-layer output is sufficient to absorb the basis drift that breaks the bare logit lens, and the resulting tuned-lens trajectory is a useful diagnostic on any pre-norm transformer. The strongest opposing form, raised in §3 and §13.2: the lens benchmarks predate the architectural substrate of the deployed 2026 lineup — MLA-compressed-KV (DeepSeek-V3), fine-grained MoE, NSA-sparse attention, sliding-window layers, hybrid attention/SSM stacks — and whether the affine-translator ansatz suffices on these substrates is empirical, not deduced.

The evidence on the table is asymmetric. The *positive* case for extension is that the residual stream remains additive across these variants; the *negative* case, raised in §13.2, is the Halawi et al. (2023) layer-of-prediction-vs-decision result — early-layer predictions can be *more* robust than the final layer on incorrect-demonstration prompts, and choosing the right layer has no a-priori recipe. Chuang et al. (2023)'s DoLa contrast turns the inconsistency into a decoding strategy on LLaMA-7B/13B/33B, but whether DoLa's gains survive at frontier-2026 scale is not systematically reproduced.

Resolving experiment. Train tuned lenses across the frontier-2026 lineup (Llama 3 / 4, Qwen 2.5 / 3, DeepSeek-V3, Gemma 3), report the per-architecture distillation-loss curves, and test whether the trajectory diagnostics (smoothness, layer-of-acquisition, contrastive DoLa decoding) carry over with the architecture-specific modifications they require. The empirical answer settles whether the lens framework is a property of the additive residual stream (it extends) or of a specific GPT-2-era attention/FFN profile (it does not).

14.3 Probe correlation versus causation

The strongest pre-2023 form of the probing program: high-accuracy linear probe accuracy on a held-out test split implies the model linearly represents the property at that layer. The strongest opposing form, consolidated in Belinkov (2022, §abstract) and the §5 contradiction block: *decodable does not entail used*. The model may compute a property at some intermediate layer and discard it; later layers may use a different representation; the probe direction may correlate near-perfectly with the used representation while not being it.

The evidence on the table is now substantial on both sides. The 2023+ direction-ablation literature (Marks and Tegmark, 2023, §3) closes the gap on canonical worked examples: ablating $\hat{\mathbf{d}}_{\mathrm{MM}}$ at LLaMA-2-13B mid-layers flips TRUE/FALSE behavior, validating the truth direction's *use*. extends the pattern to a behavioral target (sycophancy) with a multi-component decomposition. What remains open is the alignment-relevant question: probes validated on in-distribution evaluation prompts may not flag the same direction the model uses on adversarial out-of-distribution inputs — and that generalization gap is precisely the load-bearing question for **[Paper 5]** alignment-detection probing.

Resolving experiment. Pair every alignment-claim probe with direction-ablation on adversarial OOD evaluation suites; report whether the ablation-induced behavior change persists under distribution shift. A probe whose ablation effect halves on a held-out OOD shift has not validated the use claim it asserted.

14.4 The truth direction across datasets

The strongest version of the linear-truth-representation program: LLaMA-family models on factual statements admit a single linear axis $\mathbf{d}_{\mathrm{truth}} \in \mathbb{R}^D$, recoverable late-layer by mass-mean and validated by patching (Marks and Tegmark, 2023, §3). The strongest opposing form (Marks and Tegmark, 2023, §4) and the §6.5 contradiction block: per-dataset truth directions are *antipodal* early, rotate to orthogonal mid, and align only late — the "stark misalignment" of the original paper's own §4. The 2025 Vennemeyer et al. (2025) sycophancy decomposition generalizes the multi-component pattern: an alignment-relevant phenomenon thought representationally singular admits a three-direction (agreement / praise / genuine-agreement) decomposition, each independently steerable. The evidence on the table is therefore that "behaviorally-singular properties may be representationally-plural" on at least two cases. CCS as the unsupervised analog is documented to fail under negation (Marks and Tegmark, 2023, §1.1) — a near-truth confound, not truth.

Resolving experiment. A multi-axis basis-recovery procedure that, applied independently to `cities`, `neg_cities`, `larger_than`, and `cities_conj` at matched layers, returns a low-dimensional subspace whose per-dataset projections recover the single-dataset directions modulo a known equivalence (rotation, sign, scale). Until the subspace structure is exhibited, "the truth direction" is a useful one-dimensional projection of a richer subspace, in a fixed model and dataset family — not a basis-free property of the model.

14.5 Circuit-tracing scalability past Anthropic-2025

The strongest form of the attribution-graph program (Lindsey et al., 2025b,a): replacing MLPs with cross-layer transcoders enables end-to-end Jacobian attribution graphs on Claude-class models, with reported findings including forward planning in poetry, multi-step factual chains ("Texas" → "Austin" for "the capital of the state Dallas is in"), refusal-circuit features, and cross-lingual feature overlap. The strongest opposing form, raised in §11.7: the Haiku findings have not been independently reproduced as of 2026, the $\sim$20%-of-pretraining transcoder-training cost is a real adoption barrier outside frontier labs, and transcoder canonicality inherits the Leask et al. (2025); Dunefsky et al. (2024) critique — a different transcoder training run may factor the same MLP into a different sparse basis, and the stability of attribution graphs across transcoder seeds is open empirical.

The evidence on the table is single-source on the substantive claims; the 2025-06 Anthropic open-release of the circuit-tracer tooling (§13.3) is a corrective force, but a corrective force is not a replication.

Resolving experiment. Two transcoders trained on the same Claude-class model with different random seeds, and an attribution graph on the same prompt computed against each. The conclusions — which features mediate the behavior, which features feed which, the shape of the graph past the input embedding — should agree modulo a known transcoder equivalence. If they don't, attribution graphs are seed-dependent in the sense that matters for circuit claims, and the canonicality concern transfers from SAEs to circuits in a load-bearing way.

Contradiction

Method-pluralism is principled, not provisional. The five contradictions above share a structural feature: each one is the *boundary between* two of the paper's methods. SAE canonicality is a question that probing alone cannot answer (probes don't decompose activations into a basis) and that activation patching alone cannot answer (patching tests components, not feature inventories). Lens extension to frontier-2026 architectures is a question that lens-only benchmarks can only partially settle — the ground truth requires a patching-validated model of what the lens *should* read out at each layer. Probe correlation-vs-causation is, by 2026 consensus, a question whose resolution *requires* pairing probes with intervention machinery from §9. The truth-direction plurality is visible only when probes are run across datasets *and* subspace methods (Vennemeyer et al. (2025)) are layered on top. Circuit-tracing scalability is, finally, a question whose closure requires both transcoder-replication evidence (within the SAE/transcoder method-family) and downstream behavior-edit validation (with patching-style interventions). At every contradiction listed above, the resolving experiment is *cross-method*: it reads off two methods on the same artifact and tests whether they agree where they should. *Better single-method tools cannot close these gaps*, because what the gaps measure is the disagreement between the questions different methods answer.

14.6 What "partially read" means, restated

Paper 4's title — *the internal computation can be partially read* — is now precise. We can extract human-legible structure for *some* behaviors (IOI on GPT-2 small, factual recall on GPT-2 XL, truth direction in LLaMA-2-13B late layers, abstract features on Claude 3 Sonnet), on *some* models (open-source GPT-family + Gemma 2 + LLaMA 2 dominate the literature; the frontier-2026 lineup is sparser), with *cross-method agreement* on *some* claims (the two convergence cases of §12). The five anchor contradictions catalog the open frontiers in the same vocabulary: where the methods disagree, what each side asserts, what evidence is on the table, and what experiment would resolve it. The closing discipline of the paper is that progress is measured in *contradictions retired by cross-method evidence*, not in single-method headline results — and that the field, in 2026, has the methodological vocabulary to do that work.

14.7 Forward link

The five contradictions above are the explicit interpretability-side input to the alignment-evaluation discussion of [**Paper 5**]. Three load-bearing claims of the alignment-claim literature — "the model has a deception feature," "the model uses its CoT faithfully," "activation steering is a safe inference-time intervention" — depend on a probe-and-feature substrate whose canonicality and use validity is exactly what §§14.3, 14.4, and 14.1 have just declared open. Paper 5 picks up the question of how much weight those claims can carry given the evidence on the table here.

Glossary

Activation patching – the canonical causal-intervention method in MI: identify which model components matter for a behavior by swapping their activations between forward passes on paired (clean, corrupted) prompts and measuring the effect on the output `[meng2022-rome §2; kb/excerpts/meng2022-rome#sec-2; zhang2023-apatching §2.1]`. Also known as causal tracing, interchange intervention, causal mediation analysis, or representation denoising.

Activation reconstructor (AR) – The AR $\mathrm{AR}(z)$ takes the verbalized explanation z and produces a reconstructed activation $\hat{h}_\ell \in \mathbb{R}^{d_{\text{model}}}$. Trained jointly with the AV to minimise the round-trip MSE on activation reconstruction. (Fraser-Taliente et al., 2026, §method)

Activation steering – Inference-time intervention that adds $\alpha\hat{\mathbf{d}}$ to a residual-stream activation to amplify (or, with $\alpha < 0$, suppress) a probed direction. The Representation Engineering / Vennemeyer 2025 line [vennemeyer2025-sycophancy-not-one-thing FORUM-SIGNAL]

Activation verbalizer (AV) – The AV $\mathrm{AV}(z \mid h_\ell)$ takes a residual-stream activation h_ℓ and emits a natural-language explanation token sequence z. One half of the NLA pair. (Fraser-Taliente et al., 2026, §method)

Alignment faking – Operational sub-form of deceptive alignment: a model complies with a behavior it normally refuses, conditional on inferring it is in training, in order to preserve its preferred behavior at deployment. Demonstrated empirically in Claude 3 Opus at 14

attention weights – The softmax-normalized scores ($a_j \geq 0$, $\sum a_j = 1$) that determine how much each position contributes to the output.

Attribution graph – the output of circuit tracing: a sparse directed graph from input embeddings to output unembedding, with feature activations as nodes and Jacobian-weighted attribution scores as edges. Computed in 1 forward + 1 backward pass on the transcoder-replaced model (Lindsey 2025).

Attribution patching – gradient-based first-order approximation of the patching effect: $\mathrm{IE}_C \approx \langle \nabla_h m, \Delta h \rangle$. Cost: 1 forward + 1 backward pass for all components. Tier B (Nanda blog); approximation breaks in non-linear regions.

Average indirect effect (AIE) – IE averaged over a dataset of paired prompts; the standard metric for component importance.

Causal Basis Extraction (CBE) – Belrose et al. 2023's algorithm for finding residual-stream directions with highest influence on the tuned lens output. Connection between lens analysis and MI feature attribution.

Causal tracing – Meng et al. 2022's specific implementation of activation patching: clean run + Gaussian-noised corrupted run + patched-with-clean-state run, computed over every (token, layer) cell to produce a heatmap of indirect effects (Meng et al., 2022, §2.2)

Chain-of-thought (CoT) – A prompting and decoding pattern in which the model produces an intermediate sequence of reasoning steps z before emitting a final answer y. In few-shot CoT (Wei 2022), the prompt contains exemplars with hand-written reasoning traces; in zero-shot CoT (Kojima 2022), the trigger phrase "Let's think step by step" elicits the same behaviour without exemplars. (Wei et al., 2022, §1)

Circuit – a sub-graph C of a model's computational graph M that, when used in place of M via knockouts of $M \setminus C$, approximately reproduces M's behavior on a target task (Wang et al., 2022, §2)

Circuit tracing – Anthropic's 2025 methodology: replace MLPs with cross-layer transcoders, then compute attribution graphs by Jacobian back-attribution from a target node. Synthesis of SAE feature decomposition + activation patching causal intervention. [lindsey2025-circuit-tracing FORUM-SIGNAL — HTML-only]

Claude – Anthropic's proprietary decoder-only LLM family. No public architecture paper; structural family shares the decoder-only Transformer lineage.

Common Crawl (CC) – Open monthly web crawl run by the Common Crawl Foundation since 2008. Source corpus for FineWeb (96 snapshots), C4, RefinedWeb, and most open web datasets. WARC format with extracted WET (text) variant.

cross-entropy loss – Training objective: $\mathcal{L} = -\log P(t_i \mid t_1, \ldots, t_{i-1})$, summed over all positions.

Cross-layer transcoder – Lindsey et al. 2025: a transcoder whose decoder writes into the residual stream of *all subsequent* layers, not just the immediate next one. Makes a feature's skip-layer contributions explicit in the attribution graph. HTML-only at transformer-circuits.pub.

Debate (AI safety) – Two AI debaters in a zero-sum game judged by a (weaker) human or LLM judge. Hypothesis: it is easier to expose a flawed argument than to construct a sound one, so truthful debaters win more often than deceptive ones. (Irving et al., 2018, §abstract)

decoder-only – Simplified Transformer architecture (GPT, LLaMA) that removes the encoder and cross-attention, using only masked self-attention and FFN layers.

Direction ablation – Causal-probing intervention that projects a residual-stream activation onto the orthogonal complement of a probe direction $\hat{\mathbf{d}}$: $\mathbf{h} \mapsto \mathbf{h} - (\mathbf{h}^\top \hat{\mathbf{d}})\hat{\mathbf{d}}$, then runs the rest of the model. A behavior change validates causal use of $\hat{\mathbf{d}}$ (Marks and Tegmark, 2023, §abstract)

DoLa – *Decoding by Contrasting Layers*: an inference-time application of lens analysis. The decoded logits are $\text{logit}^{(L)}(t) - \text{logit}^{(\ell^*)}(t)$ for an adaptively chosen early layer ℓ^*, improving factuality. Chuang et al. 2023.

DPO (Direct Preference Optimization) – Closed-form, classification-loss reformulation of RLHF: trains the policy directly on preference pairs by inverting the closed-form RL optimum. No reward model, no on-policy sampling, no PPO loop (Rafailov et al., 2023, §4 Eq.7)

Expansion factor – the ratio M/n of dictionary size to activation dimension. Typical values 8–1000.

Feature – a direction $\mathbf{d} \in \mathbb{R}^n$ in some activation space such that the projection $\mathbf{x}^\top \mathbf{d}$ (or equivalently, an SAE latent activation with decoder column $\mathbf{d}$) corresponds to a human-interpretable property. The unit of analysis at the *what* level; circuits operate at the *how* level.

FFN (Feed-Forward Network) – A position-wise two-layer (or three-layer for GLU variants) fully-connected network applied independently at each sequence position. Transforms representations within each position, complementing attention which mixes across positions.

Fraction of Variance Explained (FVE) – NLA quality metric: $\text{FVE} = 1 - \mathcal{L}/\mathbb{E}\|h_\ell - \bar{h}_\ell\|^2$ where $\mathcal{L}$ is the round-trip MSE. FVE $= 1$ is a lossless natural-language code; FVE $= 0$ means the explanation does no better than the mean activation. (Fraser-Taliente et al., 2026, §method)

Gaussian Noising (GN) / Symmetric Token Replacement (STR) – the two corruption methods for activation patching. GN adds $\mathcal{N}(0, 3\sigma_{\text{textset}})$ to subject-token embeddings `[meng2022-rome §2]`; STR replaces the subject with a similar token (e.g., "Eiffel Tower" → "Colosseum") preserving sequence length `[zhang2023-apatching §2.1;kb/excerpts/zhang2023-apatching#sec-2-1-corrupti` STR is the recommended default; GN can drive activations off-distribution (Zhang and Nanda, 2023, §1 findings)

GELU (Gaussian Error Linear Unit) – Smooth activation $x \cdot \Phi(x)$ where Φ is the standard normal CDF. Used by GPT-1 (2018).

hidden state – The d_{model}-dimensional vector representation at each position, flowing through the residual stream across layers.

Indirect effect (IE) – the change in the output metric when the activation at a specific (component, position, layer) is restored from a cached clean run during an otherwise-corrupted forward pass. $\text{IE} = \mathbb{P}_{*,\,\text{clean}\ h_i^{(l)}}[r] - \mathbb{P}_*[r]$ (Meng et al., 2022, §2 effects)

Input-invariance – the property of a transcoder (and *not* of an SAE-on-MLP-output) that the connections between features can be stated independently of any specific input. Necessary for weight-based circuit analysis through MLPs (Dunefsky et al., 2024, §1)

JumpReLU SAE – SAE variant with a discontinuous activation $\text{JumpReLU}_\theta(z) = z \cdot H(z - \theta)$ and per-feature learned threshold θ. Trained with L0 penalty via straight-through estimators. SOTA reconstruction at fixed L0 on Gemma 2 9B (Rajamanoharan et al., 2024, §3 Eq.4, §1 Pareto)

Key (K) – Projected representation of positions being "queried" against. The Q·K dot product determines attention weights.

Knockout – generic term for setting a component's contribution to a fixed reference value (zero or mean) for circuit-validation purposes (Wang et al., 2022, §2.1)

Layer Normalization (LayerNorm) – Normalizes across the feature dimension per position: subtract mean, divide by standard deviation, apply learned scale (γ) and bias (β).

Linear probe – Probe of the form $g(\mathbf{h}) = \sigma(\mathbf{w}^\top \mathbf{h} + b)$ with $\mathbf{w} \in \mathbb{R}^d$. Distinguished from non-linear / MLP probes that can suffer from "probe leakage" (learning the property from low-information features) [alain-bengio-2017]

LLM – Large Language Model. A neural network with billions of parameters trained on large text corpora to predict and generate language.

Logit difference (LD) – an activation-patching metric: $\mathrm{LD}(r, r') = \mathrm{Logit}(r) - \mathrm{Logit}(r')$. The patching effect, normalized to $[0, 1]$, is $[\mathrm{LD}_{\mathrm{pt}} - \mathrm{LD}_*]/[\mathrm{LD}_{\mathrm{cl}} - \mathrm{LD}_*]$ `[zhang2023-apatching §2.1 metrics; kb/excerpts/zhang2023-apatching#sec-2-1-metrics]`. Recommended over raw probability because it is sensitive to components that *push against* the correct answer.

Logit lens – apply the final LayerNorm + unembedding W_U to an intermediate residual-stream activation $\mathbf{h}^{(\ell)}$ to read out a layer-ℓ vocabulary distribution. nostalgebraist 2020 (LessWrong, Tier B). Formal expression and decomposition in `[belrose2023-tuned-lens §2 Eq.3]` (PDF p.2).

Logits – Unnormalized log-probabilities over the vocabulary, produced by the output projection $(h_{\text{final}} \cdot W_e^\top)$.

LoRA (Low-Rank Adaptation) – Parameterizes a frozen weight matrix's update as a rank-r product BA with $r \ll \min(d_{\text{in}}, d_{\text{out}})$; only A, B are trained. Merges to a single weight at inference time with no latency cost. (Hu et al., 2021, §3, §4)

Mass-mean probing – Marks & Tegmark 2023: define the probe direction as $\mathbf{d}_{\text{class}} = \mathbb{E}[\mathbf{h} \mid y = c_1] - \mathbb{E}[\mathbf{h} \mid y = c_0]$, then project activations onto $\mathbf{d}_{\text{class}}$. Simpler and more generalization-robust than logistic regression on the same features (Marks and Tegmark, 2023, §1)

Mean ablation – knocking out a node by replacing its activation with the mean activation across a reference distribution; preferred to zero ablation, which is arbitrary (Wang et al., 2022, §2)

Mechanistic interpretability (MI) – the research program of reverse-engineering neural networks into human-readable algorithms, at the level of circuits (sub-graphs) and features (directions). Distinct from behavioral interpretability (input/output black-box evaluation) and post-hoc rationalization. (Wang et al., 2022, §1)

meta-SAE – an SAE trained on the *decoder matrix* of another SAE. Reveals that latents in a larger SAE often decompose into sparse combinations of meta-latents, contradicting the "atomic features" assumption (Leask et al., 2025, §1 Einstein)

Monosemanticity – the property that a feature responds to a single, human-interpretable concept. The (contested) goal of SAE training; "Towards Monosemanticity" (Bricken 2023) is the founding reference. [CONTRADICTION] Whether SAEs achieve true monosemanticity is challenged by (Leask et al., 2025, §1)

multi-head attention – Running h parallel attention operations on different learned projections, then concatenating and projecting the results. Allows attending to different representation subspaces simultaneously.

Multi-head Latent Attention (MLA) – DeepSeek's architectural compression: project to a low-rank latent $C \in \mathbb{R}^{d_c}$ shared across heads, expand back to per-head (K, V) at attention time. Achieves 93

Natural Language Autoencoder (NLA) – Anthropic 2026's interpretability method: a pair of LLM modules (AV + AR) jointly trained with reinforcement learning to round-trip a residual-stream activation through a natural-language explanation. The text bottleneck makes the explanation testable via reconstruction fidelity (FVE). Used in pre-deployment audits to surface unverbalized model cognition — including evaluation awareness. (Fraser-Taliente et al., 2026, §method)

NSA (Native Sparse Attention) – Yuan et al. 2025: hardware-aligned natively sparse attention trained end-to-end (sparsity is part of pre-training, not retrofitted). Aimed at long context. Treated in `kb/notes/architecture/long-context.md` [yuan2025]

Parameters – The learned weights of the model (embedding matrices, projection matrices, normalization scales, etc.).

Path patching – a refinement of activation patching that isolates the direct effect of a component C on a downstream component R, holding fixed the indirect routes through other components. Introduced in Wang et al. 2022 (IOI) (Wang et al., 2022, §3.1)

Perplexity – $2^{\mathcal{L}}$ (or $e^{\mathcal{L}}$ with natural log). Measures how "surprised" the model is by the data. Lower is better.

Precision pinning (DeepSeek-V3) – Components held at BF16/FP32 even with FP8 elsewhere: embedding module, output head, MoE gating, normalization, attention. (DeepSeek-AI, 2024b, §3.3.1)

Probe – A (typically linear) classifier $g : \mathbb{R}^d \to \mathcal{Y}$ trained on frozen LM activations $\mathbf{h}^{(\ell,t)}$ to predict an external property y. The model's parameters are held fixed; only the probe is trained. Foundational to interpretability methodology (Alain and Bengio, 2016, §1)

QLoRA – LoRA over a 4-bit-quantized (NF4) base model. Adds double-quantization and paged optimizers. 65B fine-tune on a single 48GB GPU. (Dettmers et al., 2023, §3)

Query (Q) – Projected representation of the position that is "asking" for information in attention.

ReLU – Rectified Linear Unit: $\max(0, x)$. Original Transformer activation (2017).

Residual stream – the linear data bus running through a decoder-only transformer's layers; every sublayer (attention, MLP) reads from and additively writes to it. The substrate that lens techniques and SAEs decompose. Vocabulary established in Elhage et al. 2021 (Mathematical Framework for Transformer Circuits).

RLHF (Reinforcement Learning from Human Feedback) – The 3-stage post-training pipeline: SFT $\to$ reward-model training on pairwise preferences $\to$ PPO with KL anchor against π^{SFT}. Canonical reference is InstructGPT (Ouyang et al., 2022, §3)

RMSNorm (Root Mean Square Normalization) – Simplified LayerNorm that drops mean-centering and bias, normalizing only by the RMS of the input. 7–64

SAE stitching – Leask et al. 2025: insert/swap latents from a larger SAE into a smaller one to compare bases. Reveals novel latents (in the larger but not the smaller SAE) and reconstruction latents (in both with similar behavior) (Leask et al., 2025, §1)

Saturation – A benchmark is saturated when frontier models score within the verifier-error-rate noise floor of one another, losing discriminative power. MMLU at >90

Scheming – A model schemes when, instructed to pursue a goal G, it strategically takes actions that conceal its capabilities or manipulate its oversight in service of G, rather than pursuing G by direct means. Operational definition from Apollo Research. (Research, 2024, §abstract)

Self-consistency – Sample N chain-of-thoughts independently at $T > 0$; majority-vote the final answers. Wang et al. 2022. Cheap baseline that requires no verifier.

Shrinkage – the bias introduced by the L1 penalty in ReLU SAEs: positive feature activations are systematically underestimated, so reconstructed activations have smaller magnitudes than true values. Avoided by TopK and JumpReLU (Gao et al., 2024, §2.3 benefits)

Softmax – Converts logits to a probability distribution: $P(j) = e^{z_j} / \sum_k e^{z_k}$.

Sparse autoencoder (SAE) – a wide ($M \gg n$) two-layer encoder-decoder trained to reconstruct a model's activations $\mathbf{x} \in \mathbb{R}^n$ as a sparse linear combination of learned dictionary directions $\{\mathbf{d}_i\}_{i=1}^{M}$ (Rajamanoharan et al., 2024, §2 Eq.1-2)

State-space model (SSM) – Sequence model with recurrent state $\boldsymbol{h}_t$ updated linearly: $\boldsymbol{h}_t = \bar{A}\boldsymbol{h}_{t-1} + \bar{B}u_t$. Per-token compute $O(d^2)$, sequence-linear. (Gu, Goel, Ré 2022)

Structural probe – Hewitt & Manning's geometric probe that predicts higher-arity structure (e.g., parse-tree distance) by learning a linear transformation $\mathbf{B} \in \mathbb{R}^{k \times d}$ such that $\|\mathbf{B}(\mathbf{h}_i - \mathbf{h}_j)\|_2^2 \approx d_T(w_i, w_j)$ for tokens i, j [hewitt2019-structural-probe]

Summarization behavior – The empirical pattern (Tigges et al. 2023, Marks & Tegmark 2023) that information about a clause is encoded over the clause-ending punctuation token, making period-position activations the canonical probing target for sentence-level properties (Marks and Tegmark, 2023, §3)

Superposition – the hypothesis (Elhage 2022) that a network with n neurons can represent more than n approximately-orthogonal features as long as they fire sparsely. SAEs lift superposition by inflating the dictionary size $M \gg n$. (Gao et al., 2024, §1)

SwiGLU – $\mathrm{FFN}_{\mathrm{SwiGLU}}(x) = (\mathrm{Swish}_1(xW) \otimes xV)W_2$. Three-matrix FFN with Swish gating. Universal in modern decoder-only LLMs. (Shazeer, 2020, §2 Eq.6)

Sycophancy – A response is sycophantic if it is more aligned with the user's expressed preference than is justified by the underlying truth. Distinct from scheming: sycophancy requires no intent, only Goodhart on RLHF preference data. (Sharma et al., 2023, §abstract)

TopK SAE – SAE variant with $\sigma = \mathrm{TopK}(\cdot)$: zero all but the top K pre-activations. Direct L_0 control, no L1 penalty needed. OpenAI 2024 (Gao et al., 2024, §2.3 Eq.2)

Transcoder – a wide ReLU MLP (encoder–decoder) trained to approximate the input/output behavior of an MLP sublayer: $\mathrm{TC}(\mathbf{x}) = \mathbf{W}_{\mathrm{dec}}\,\mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\mathbf{x} + \mathbf{b}_{\mathrm{enc}}) + \mathbf{b}_{\mathrm{dec}}$, with L1 sparsity penalty on the hidden activations [`dunefsky2024-transcoders §3.1 Eq.3-5; kb/excerpts/dunefsky2024-transcoders#sec-3-1`]. Distinct from an SAE: a transcoder approximates a *function* (the MLP); an SAE approximates a *vector* (an activation).

Transformer – Neural network architecture introduced by Vaswani et al. (2017) that replaces recurrence with self-attention, allowing parallel processing of sequences.

Truth direction – A direction in residual-stream space whose projection sign correlates with (and, validated by Marks & Tegmark 2023, causally controls) the model's TRUE/FALSE prediction on factual statements. Localizes around layer 12-15 of LLaMA-2-13B at end-of-statement punctuation tokens (Marks and Tegmark, 2023, §3)

Tuned lens – Belrose et al. 2023: per-layer affine "translators" $\mathbf{h}_\ell \mapsto A_\ell \mathbf{h}_\ell + b_\ell$ trained with a distillation loss against final-layer logits. More reliable than the raw logit lens, especially on BLOOM/GPT-Neo (Belrose et al., 2023, §1)

Unverbalized evaluation awareness – Cases where a model internally suspects it is being tested but does not say so in its visible output. NLAs were validated as a tool for surfacing this signal during Anthropic's pre-deployment audit of Claude Opus 4.6. (Fraser-Taliente et al., 2026, §auditing)

Value (V) – Projected representation of the content to be aggregated. Weighted by attention scores to produce the output.

Vocabulary (V) – The fixed set of all tokens a model can recognize. Size ranges from 32K (LLaMA) to 50K (GPT-3).

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Table 2: The eight probe families that anchor the LLM probing literature. Compiled from `kb/notes/interpretability/probing.md` §4.1.

Family	**Year**	**Probe form**	**Key limitation**
Linear logistic regression (Alain and Bengio, 2016)	2017	$\sigma(\mathbf{w}^\top \mathbf{h} + b)$ on frozen $\mathbf{h}$	correlational; needs ℓ_2 regularization to avoid overfit
MLP probe	2010s	small feedforward net on $\mathbf{h}$	probe leakage: a powerful probe learns y from low-information features (Hewitt and Liang, 2019)
Structural probe (Hewitt and Manning, 2019)	2019	$\lVert\mathbf{B}(\mathbf{h}_i - \mathbf{h}_j)\rVert^2 \approx d_T(w_i, w_j)$	confounded with task-difficulty when comparing layers
Control-task baseline (Hewitt and Liang, 2019)	2019	same probe on randomly-permuted labels	required ablation; inflates accuracy delta on poorly-initialized models
Logistic regression for truth (Marks and Tegmark, 2023)	2023	linear, ℓ_2-regularized, on truth labels	correlational unless paired with ablation
Mass-mean / difference-in-means (Marks and Tegmark, 2023)	2023	$\boldsymbol{\mu}_+ - \boldsymbol{\mu}_-$, closed-form	requires balanced classes; binary only
Difference-in-means + steering	2025	mass-mean direction, applied as activation-steering	within-model only; cross-family transfer untested
Contrastive / CCS unsupervised	2022	linear probe, self-consistency loss, no labels	generalization failures under negation (Marks and Tegmark, 2023, §1.1)

Granularity	Component C	Used by
Hidden state	$\mathbf{h}_i^{(\ell)} \in \mathbb{R}^D$	Meng et al. (2022)
MLP output	$\mathrm{MLP}^{(\ell)}(\mathbf{h}_i^{(\ell)}) \in \mathbb{R}^D$	Meng et al. (2022)
Attention-layer output	$\mathrm{Attn}^{(\ell)}(\mathbf{h}_i^{(\ell)}) \in \mathbb{R}^D$	Meng et al. (2022)
Single attention head	$h_j^{(\ell)} \in \mathbb{R}^{d_h}$ before W^O	Wang et al. (2022)
Subspace	$\mathbf{P}_{\hat{\mathbf{d}}}\, \mathbf{h}_i^{(\ell)}$	probe-direction patches

Table 3: Patching granularities. Meng et al. (2022) patch hidden states, MLPs, and attention layers separately to attribute factual recall to MLPs over attention; Wang et al. (2022) patch individual heads to recover the seven-class IOI circuit; subspace patching along a probe direction $\hat{\mathbf{d}}$ tests whether a specific *property* encoded by a probe is causally used.

Choice	Default	Source
Corruption	STR (in-distribution)	Zhang and Nanda (2023, §1)
Direction	Denoising; noising for AND-circuits	Heimersheim and Nanda (2024, §2.4)
Metric	Logit difference (normalized)	Heimersheim and Nanda (2024, §4.1)
Granularity	Head-level for circuits; subspace for features	Zhang and Nanda (2023, §2.3)
Validation	Pair patching with a complementary ablation test	Wang et al. (2022, §4)

Table 4: Default activation-patching recipe consolidated from the methodology literature. Each row has at least one peer-reviewed recommendation; the validation row reflects the faithfulness/completeness/minimality discipline of Wang et al. (2022, §4), which patching alone does not enforce.

Editor	Efficacy	Paraphrase	Specificity
GPT-2 XL (no edit)	22.2 (±0.5)	21.3 (±0.5)	24.2 (±0.5)
Fine-tuning (FT)	99.6 (±0.1)	82.1 (±0.6)	23.2 (±0.5)
FT+L	92.3 (±0.4)	47.2 (±0.7)	23.4 (±0.5)
MEND	75.9 (±0.5)	65.3 (±0.6)	24.1 (±0.5)
ROME	**99.8 (±0.0)**	**88.1 (±0.5)**	**24.2 (±0.5)**

Table 5: ROME on the zsRE editing benchmark, GPT-2 XL, replicated from Meng et al. (2022, §3.2). *Efficacy*: the edited model produces the new o on the exact training prompt. *Paraphrase*: the edited model produces o on rephrased queries about the same (s, r). *Specificity*: the edit does not contaminate predictions on unrelated (s', r) pairs.

Table 6: Interpretability methods — scale, target, and coverage as of 2026-05. *Target* is the largest model the method has been reported on. *Open release* is the date the canonical scale-target artifact became externally available; "—" if no such release. *2026 reproduction* marks whether the canonical-scale finding has been independently reproduced outside the originating lab on a frontier-2026 model.

Method	Canonical paper	Target model	Scale	Open release	2026 reproduction
Logit lens	nostalgebraist (2020)	GPT-2 family	≤1.5B	no fixed artifact (decoding rule)	yes (GPT-2, BLOOM, GPT-Neo); fails on BLOOM/GPT-Neo per Belrose et al. (2023)
Tuned lens	Belrose et al. (2023)	Pythia 12B; GPT-Neo X	≤20B	training script + benchmarks (2023)	partial: `logitlens4llms-2025` on Qwen-2.5 / LLaMA-3.1; no Gemma 3 / DeepSeek-V3 / LLaMA-4 published
Mass-mean probe	Marks and Tegmark (2023)	LLaMA-2-13B	13B	code on GitHub (2023)	yes for truth, sycophancy variants on LLaMA family; OOD generalization unresolved (§6.5)
ReLU+L1 SAE	Templeton et al. (2024)	Claude 3 Sonnet	frontier	no public weights (Anthropic-internal)	no
TopK SAE	Gao et al. (2024)	GPT-4	16M-latent	training recipe + evaluation suite (2024)	yes at ≤Gemma 2 27B scale; no GPT-4-class external reproduction
JumpReLU SAE	Rajamanoharan et al. (2024); DeepMind (2024)	Gemma 2 27B	1M-latent	Gemma Scope weights, CC-BY-4.0, HF (2024-08)	yes (most reproducible pipeline)
Activation patching	Meng et al. (2022); Zhang and Nanda (2023)	GPT-2 XL; LLaMA-7B	1.5B–7B	toolkits (TransformerLens, etc.)	yes for ≤7B; full-coverage at frontier scale infeasible without attribution patching
Manual circuit hunt	Wang et al. (2022); Conmy et al. (2023)	GPT-2 small	124M	ACDC code (2023)	yes on GPT-2 small; not yet on frontier-2026 models
Cross-layer transcoder + attribution graph	Lindsey et al. (2025b)	Claude 3.5 Haiku	frontier	circuit-tracer tooling (2025-05-30)	no external reproduction as of 2026
NLA (AV + AR pair)	Fraser-Taliente et al. (2026)	Claude Opus 4.6	frontier	no public weights; method paper (2026-05-07) at transformer-circuits.pub/2026/nla	no external reproduction as of 2026; only Anthropic-internal Opus 4.6 audit application reported